

King Faisal University  
School of Business Administration  
Management Information System



# HackSynk

*Project submitted.*

*in partial fulfillment of the requirements for the degree*

*of*

*Bachelor of Management Information Systems*

**by**

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## UNDERTAKING

This is to declare that the project entitled “HackSynk” is an original work done by undersigned, in partial fulfillment of the requirements for the degree of “Bachelors of Business Administration” at Management Information Systems Department, School of Business, King Faisal University.

Further, it is hereby declared that the entire project work including the analysis, design and system development has been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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## **ABSTRACT**

This report explores the HackSynk platform, a comprehensive web-based system designed to serve as a central hub for managing hackathon events efficiently. HackSynk provides an integrated environment where organizers can create and manage hackathons, participants can form or join teams and submit projects, and judges can evaluate submissions through a structured and transparent process. The platform brings together individuals with diverse skills and backgrounds, enabling effective collaboration and innovation within a single digital space.

HackSynk offers organizers the ability to oversee event timelines, assign judges, track project submissions, and announce results seamlessly. Participants can explore available hackathons, collaborate through built-in communication tools, and showcase their projects in an organized manner. Judges benefit from a standardized evaluation system that ensures fairness, consistency, and meaningful feedback.

By centralizing all hackathon-related activities, HackSynk enhances user experience, reduces administrative complexity, and improves coordination among all stakeholders. Beyond its technical functionality, the platform acts as a catalyst for creativity, teamwork, and knowledge exchange. HackSynk contributes to a more structured and engaging hackathon ecosystem by providing a digital space where ideas can grow, collaboration can thrive, and innovation can be effectively recognized.

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## **1. PRELIMINARY INVESTIGATION**

### **1.1 Introduction**

Hackathons have proved to be the best tool for nurturing creativity, teamwork, and expertise. But manual handling or the use of simple tools for the handling of the activities creates inefficiencies. Team formation, communication, and submission of projects can cause great hurdles for the contestants, whereas the organizers find it hard to manage the activities, track progress, and maintain the judging process seamlessly. An efficient online solution is needed for automating the processes and delivering the best hackathon experience.

This project establishes a Hackathon Management System for one-stop-shop access for the organizers and the participants. With the addition of the features required like formal evaluation, communication tools, project tracking, and team collaboration, the system will make the process easier and efficient for all the stakeholders for the execution of the hackathon.

### **1.2 Problem statement**

Hackathons are widely recognized as effective platforms for fostering innovation, collaboration, and skill development. However, many challenges arise when managing hackathons manually or using basic tools. Participants often struggle with team formation, as there is no structured way to find or join suitable teams, leading to an imbalance in skill distribution and a lack of coordination. Communication between team members is often fragmented, as they rely on multiple external tools, making collaboration inefficient. Additionally, project submission and tracking can become disorganized, leading to confusion and last-minute errors.

For organizers, handling hackathons without an integrated system presents significant difficulties. They need to manage participant registration, team formation, scheduling, and judging, often using spreadsheets and email chains, which can be time-consuming and prone to errors. Judges face challenges in accessing and evaluating projects, as there may be no centralized system to streamline scoring, feedback, and ranking. Moreover, the lack of a structured reward system makes it difficult to ensure fairness and transparency in awarding winners.

Without an efficient hackathon management system, these inefficiencies can lead to lower participant engagement, delayed evaluations, and a frustrating experience for both participants and organizers. There is a need for an all-in-one digital solution that simplifies the entire hackathon

process, from team formation to final project evaluation, ensuring a smooth, fair, and engaging competition for all stakeholders.

### **1.3 Solution**

HackSynk is a comprehensive digital platform designed to streamline the management of hackathons by addressing the inefficiencies faced by participants, organizers, and judges. The platform simplifies team formation by allowing participants to search for and join teams based on skills and interests, ensuring balanced and well-matched groups. It also integrates communication tools that facilitate seamless collaboration within teams, reducing reliance on external messaging platforms. Organizers can efficiently manage event creation, participant registration, scheduling, and rule-setting, all from a single interface, minimizing administrative burdens. HackSynk also enhances the project submission and evaluation process by providing a structured system where teams can submit their projects, and judges can review, score, and provide feedback through a standardized evaluation framework. Additionally, the platform automates award allocation and winner announcements, ensuring a transparent and fair judging process. By consolidating all essential hackathon operations into one streamlined system, HackSynk improves efficiency, enhances user experience, and creates an organized and engaging competition environment for all stakeholders.

### **1.4 Scope of Project**

The Hackathon Website Project aims to create a digital platform that helps individuals without a team find suitable groups to participate in hackathons. The platform also allows the complete organization and management of hackathons within the website itself, without any external intervention, making it easier for participants and organizers to handle all event stages in one place.

#### ***1.4.1 What is Included in the System***

User Registration

Team Search

Team Creation



Hackathon Creation

Hackathon Participation

Hackathon Management

Member Communication

Project Evaluation System

Rewards System

#### ***1.4.2 What the System Might Not Cover***

Financial or Logistical Support

Integration with External Tools

Training or Consultation

Team Success Guarantee

External Rewards Management

### **1.5 Feasibility Study**

The feasibility study is an analytical process aimed at evaluating the project's viability from multiple perspectives. Here, we will focus on the operational and technical aspects.

#### ***1.5.1 Operational Feasibility***

The project is operationally feasible as it addresses key challenges in hackathon management. It helps by:

- Automatically forming teams based on skills.
- Providing an effective communication platform for organizers and participants.
- Enabling electronic project submission and evaluation, saving time and effort.

#### ***1.5.2 Technical Feasibility***

The project is technically feasible as it relies on stable, well-established, and widely supported technologies:

- **Vanilla JavaScript** for building a responsive and interactive user interface without requiring heavy frontend frameworks.
- **PHP** for handling server-side logic, data processing, and application operations.
- **MySQL** as a reliable relational database management system for structured and efficient data storage.

#### 1.5.2.1 Hardware

Table 1.1 provides an overview of the hardware and network requirements for the system designed to handle hacksync business. The table outlines the essential devices, processing capabilities, memory specifications, and network infrastructure necessary for optimal performance. A laptop or specialized machine serves as the primary device for executing the conversion process, powered by an Intel Core i7 processor. The system requires a minimum of 8 GB RAM and 1 TB of internal memory to handle data-intensive operations efficiently. For connectivity, the setup includes a router supported by high-speed internet access, utilizing both 5G and Fiber Internet technologies to ensure seamless communication and data exchange.

Table 1:Hardware Requirements

| Hardware               | The requirements                   |
|------------------------|------------------------------------|
| Devices                | Laptop / PC                        |
| Processor              | Intel Core i7                      |
| Memory                 | -8 ram<br>- 1 tera internal memory |
| Network infrastructure | Router                             |
| Internet access        | 5G - Fiber internet.               |

### 1.5.2.2 Software

Table 1.2 provides an overview of the software requirements for the project. It lists the essential software and tools needed to implement and operate the system. The table includes the following key components:

Table 2:Software Requirements

| Software             | The requirements                                                                                                 |
|----------------------|------------------------------------------------------------------------------------------------------------------|
| Operating system     | Microsoft Windows, MacOS, Android and Apple Using these four operating systems will facilitate using the system. |
| Web server           | A hosting environment or cloud server capable of running PHP applications.                                       |
| Database             | MySQL for structured and reliable data storage.                                                                  |
| Programing languages | HTML, CSS, JavaScript (Vanilla JS), and PHP.                                                                     |
| Design tools         | CSS for styling and Figma for user interface design and prototypes.                                              |
| GPS and map software | Google and Apple maps                                                                                            |

### 1.5.3 Economic Feasibility

HackSynk project presents an attractive case for economic feasibility through its defined benefits and manageable costs. Its aims to enhance participation in hackathons through an easy-to-use platform that attracts a larger audience. This platform improves user experience and facilitates interaction, leading to increased satisfaction. It also simplifies team formation among participants, fostering collaboration and creativity. Additionally, automating processes save time and resources, while data collection helps in informed decision-making, Project costs include

development expenses such as programming and interface design, along with operational costs like website hosting and system maintenance. Overall, HackSynk is strategically positioned to create a thriving environment for hackathons while maintaining economic viability.

## **1.6 Functional Requirements**

### *1.6.1 User Registration and Authentication*

- **Role-Based Registration:** The system should provide separate registration flows for Participants, Organizers, and Judges, each with role-specific profile information.
- **Secure Authentication:** The system should provide secure login/logout functionality with session management for all user roles.
- **Profile Management:** The system should allow users to create and manage profiles with skills, education, experience, and contact information.

### *1.6.2 Team Management*

- **Team Creation:** The system should enable participants to create teams for hackathons with configurable size limits and required skills.
- **Team Joining:** The system should support direct join (open teams) and request-based join (requires team leader approval).
- **Team Member Management:** The system should allow team leaders to approve/reject join requests and manage team membership.
- **Team Discovery:** The system should allow participants to browse and search teams by name, required skills, and member count.

### *1.6.3 Project Submission*

- **Project Submission:** The system should enable teams to submit projects with title, description, demo link (optional), and PDF documentation files.
- **Submission Timeline:** The system should enforce submission deadlines based on hackathon end dates.
- **Project Management:** The system should allow teams to view, update, and manage submitted projects within allowed timeframes.

#### *1.6.4 Search and Discovery*

- **Hackathon Discovery:** The system should allow users to browse and search hackathons by name, description, location, dates, and themes.
- **Team Search:** The system should enable participants to search teams within hackathons by name, skills, and keywords.
- **Project Search:** The system should allow judges and organizers to search projects by title, team name, or hackathon name.
- **Filtering and Sorting:** The system should provide filtering and sorting for hackathons, teams, and projects by dates, status, and relevance.

#### *1.6.5 Communication Tools*

- **Direct Messaging:** The system should provide real-time one-to-one messaging between participants.
- **Team Group Chat:** The system should enable team-based group chat rooms to support real-time collaboration.
- **Notification System:** The system should send notifications for join requests, new messages, hackathon updates, judge invitations, evaluation progress, and result announcements.
- **Notification Management:** The system should allow users to view, mark as read, organize, and manage all received notifications.

#### *1.6.6 Event Management (Organizer Features)*

- **Hackathon Creation:** The system should allow organizers to create new hackathons with details such as name, description, theme, dates, judging period, rules, prizes, team limits, location, and event image.
- **Hackathon Management:** The system should enable organizers to edit, update, and manage active or upcoming hackathons.
- **Judge Assignment:** The system should allow organizers to invite judges, while judges can accept or reject invitations.

- **Team Assignment:** The system should enable organizers to assign specific teams to judges for evaluation.
- **Event Analytics:** The system should provide analytics on number of participants, teams, submitted projects, judge assignments, and evaluation progress.

#### *1.6.7 Judging and Evaluation*

- **Judge Invitation Management:** The system should allow judges to view, accept, or reject hackathon invitations.
- **Project Evaluation:** The system should support a 10-criteria evaluation system (Innovation, Technical Implementation, Usability/Design, Impact, Problem Relevance, Functionality, Feasibility, Collaboration/Presentation, Technical Complexity, and Testing/Reliability).
- **Scoring System:** The system should allow judges to score each criterion on a 1–10 scale, with automatic total score calculation.
- **Feedback System:** The system should allow judges to provide qualitative written feedback for each project.
- **Evaluation Timeline Enforcement:** The system should restrict evaluations to designated judging periods.
- **Evaluation Tracking:** The system should track evaluation status (pending, in-progress, completed) per project-judge pair.
- **File Access:** The system should allow judges to view and download project documentation files.

#### *1.6.8 Results and Rankings Management*

- **Results Compilation:** The system should aggregate and calculate final results based on judges' scores, producing average scores per project.
- **Project Rankings:** The system should allow organizers to define and manage project rankings, including positions and status.

- **Results Announcement:** The system should allow organizers to publish results and make rankings visible to participants.
- **Results Viewing:** The system should allow participants to view final scores, rankings, and judge feedback after the announcement.

#### *1.6.9 Reporting and Analytics*

- **Participant Statistics:** The system should generate reports on registered participants, team formation, and submission rates per hackathon.
- **Team Activity Reports:** The system should provide analytics on team creation, team size averages, and join request patterns.
- **Project Submission Reports:** The system should generate reports on total project submissions, completion rates, and evaluation status.
- **Evaluation Reports:** The system should offer analytics on completed vs. pending evaluations, average scores, judge activity, and score distribution.
- **Success Metrics:** The system should provide aggregated event performance indicators, including engagement, participation, and overall project quality.

#### *1.6.10 Administrative Features*

- **User Management:** The system should allow administrators to manage users, roles, and permissions.
- **System Overview:** The system should provide administrators with a dashboard to monitor platform activity, user counts, and system health.
- **Content Moderation:** The system should enable moderators to oversee hackathons, projects, chats, and other user-generated content.

## 1.7 List of staff functions

### 1.7.1 Organizer

- **Hackathon Creation and Management:** Organizers can create, edit, update, and delete hackathon events, providing full event details including name, description, dates, location, rules, prizes, and images.
- **Judge Assignment:** Organizers can assign judges to hackathons, manage invitations, handle judge responses, and distribute teams to judges for evaluation manually or automatically.
- **Event Configuration:** Organizers can configure event timelines including start and end dates, judging periods, and submission deadlines, as well as manage event statuses (upcoming, active, or completed).
- **Content Management:** Organizers can upload hackathon images, define rules and guidelines, specify prizes, and update event descriptions and requirements.
- **Results Management:** Organizers can review project evaluations, determine final rankings based on judges' scores, and announce results for participants.
- **Analytics and Statistics:** Organizers can monitor participant engagement, project submissions, evaluation progress, and overall event performance metrics.
- **Team Distribution:** Organizers can assign teams to judges manually or automatically to ensure balanced workload distribution during evaluation.

### 1.7.2 Participant

- **Registration and Profile Management:** Participants can register securely, create and update profiles containing skills, education, experience, qualifications, and certificates.
- **Hackathon Discovery:** Participants can browse, filter, and search hackathons by various criteria and join events of interest.
- **Team Formation:** Participants can create teams, join existing ones, send or receive join requests, manage team membership, and view team details.
- **Project Submission:** Participants can submit projects with titles, descriptions, demo links, and PDF file uploads, and update or delete submissions before the deadline.



- **Real-Time Communication:**
  - Engage in private one-to-one messaging with other participants.
  - Collaborate through team-based group chats for coordination.
  - Receive and send notifications regarding team activities, invitations, and messages.
- **Results Viewing:** Participants can view project results, scores, rankings, and feedback after official announcement.
- **Collaboration:** Participants can interact, share ideas, and work together on projects through the platform's communication tools.

### 1.7.3 Judge

- **Project Evaluation:** Judges evaluate projects using a 10-criteria scoring system covering:  
Innovation and Creativity, Technical Implementation Quality, Design and Usability, Impact and Relevance, Problem Relevance, Functionality and Completeness, Feasibility and Scalability, Team Collaboration and Presentation, Technical Complexity, and Testing and Reliability.
- **Scoring System:** Judges provide scores on a scale of 1–10 for each criterion, with automatic total and average score calculations.
- **Feedback Provision:** Judges write detailed feedback highlighting project strengths and areas for improvement.
- **Assignment Management:** Judges can view assigned hackathons, accept or decline invitations, and access assigned teams or projects for evaluation.
- **Evaluation Timeline:** Judges must submit evaluations within the designated judging period defined by organizers.
- **Project Review:** Judges can access all project details, including titles, descriptions, demo links, uploaded files, and team information.
- **Evaluation Status Tracking:** Judges can track evaluation progress (evaluated, pending) and review completed evaluation history.

#### *1.7.4 Project*

- **Project Submission:** Teams submit hackathon projects including title, description, demo links, PDF documentation (up to 10MB), and submission timestamps.
- **Evaluation Requirements:** Projects must be submitted before the deadline to qualify for evaluation.
- **Multi-file Support:** Teams can upload multiple PDF files to provide complete project documentation.
- **Team Association:** Each project is linked to a specific team and hackathon for proper organization and evaluation tracking.
- **Evaluation Tracking:** Projects maintain records of evaluation status, judges' scores, average scores, rankings, and written feedback.
- **Results Visibility:** Project results and rankings become visible to participants only after official announcement by the organizer.

#### *1.7.5 Award/Ranking*

- **Ranking System:** The system ranks projects based on aggregated judge scores, assigning positions according to average scores.
- **Ranking Management:** Organizers can manually set, adjust, or confirm rankings, assign positions (1st, 2nd, 3rd, etc.), and add notes or status information.
- **Results Announcement:** Organizers control when to make results public, allowing participants to view their scores, rankings, and feedback.
- **Award Recognition:** Top-ranked projects receive recognition or awards visible to all participants after results are published.
- **Evaluation Aggregation:** The system automatically computes average scores from all judges to determine project rankings.
- **Ranking Status:** Each project can have a defined ranking status (e.g., pending, confirmed) with optional organizer notes for context or award details.

## 1.8 Data Collection Techniques

To have a good information gathering, we decided to use questionnaires and interviews for people who most likely to interact with our system. The gathered data will help us to shape the development of our hackathon management system, by making sure all user can have the best experience and feel satisfied about the system.

### 1.8.1 Surveys Questions

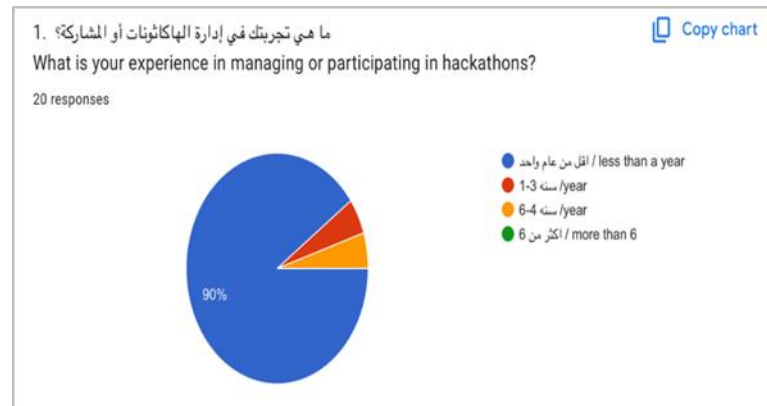


Figure 1 Responses of Hackathon System

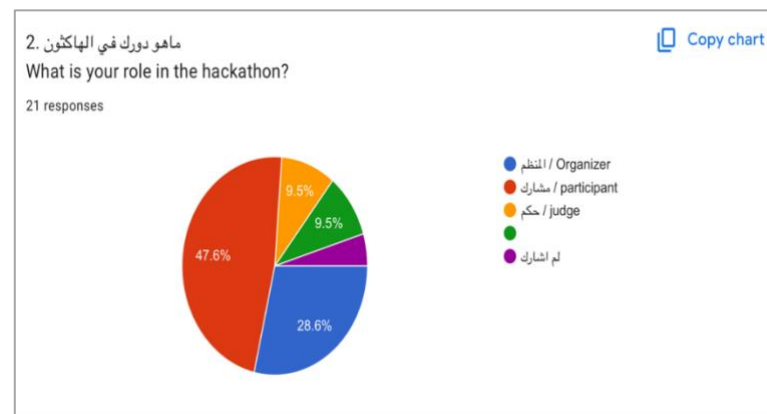


Figure 2 Responses of Hackathon System



Figure 3Responses of Hackathon System



Figure 4Responses of Hackathon System



Figure 5Responses of Hackathon System



Figure 6 Responses of Hackathon System

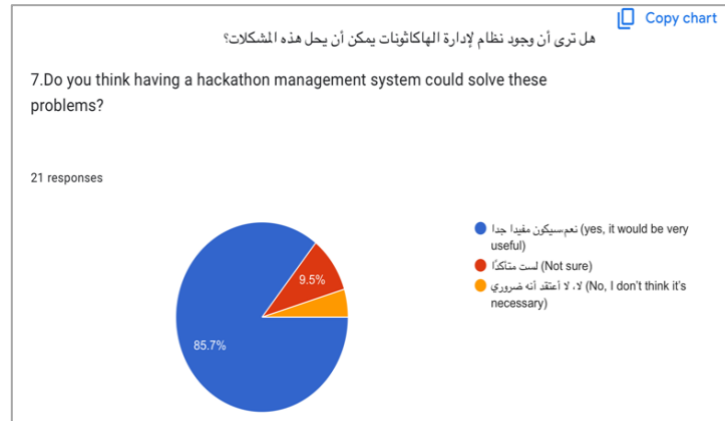


Figure 7 Responses of Hackathon System



Figure 8 Responses of Hackathon System

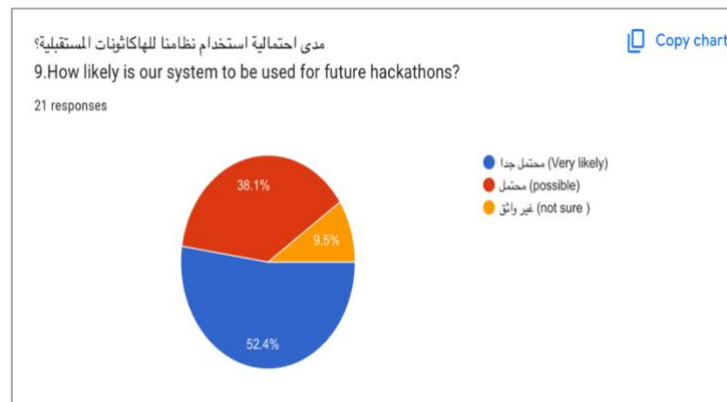


Figure 9 Responses of Hackathon System



Figure 10 Responses of Hackathon System

### 1.8.2 Interview Questions

1. What challenges have you faced in previous hackathons, and how do you think a digital system like HackSynk can improve the experience?
  - Finding a team is hard, and communication is often messy. A system that helps with team matching and real-time updates would be great.
  - Submitting projects can be confusing, especially with different formats. A structured submission system would make it easier.
  - Getting feedback is slow, and sometimes we don't even receive any. A system that ensures judges provide feedback would help us improve.
2. How important is real-time communication in a hackathon, and what tools do you currently use?
  - Very important! We use WhatsApp, but having a built-in chat in the system would be more convenient.
  - Communication is key, especially for remote teams. Right now, we use Discord, but an integrated chat would help a lot.
  - We mostly rely on Telegram, but it's hard to track important messages. A system with notifications would be better.
3. What features do you think are most important in a Hackathon Management System?
  - A team-matching tool and an easy way to submit projects.
  - Clear deadlines, project tracking, and real-time notifications.
  - An evaluation system that lets judges give structured feedback.
4. How do you think project evaluations can be made more efficient in an online hackathon system?
  - Judges should have a clear scoring system to avoid confusion.
  - A system that automatically organizes and calculates scores would save time.
  - Judges should be able to leave comments so we understand our mistakes.
5. What additional features would you like to see in a hackathon management system?
  - A mentorship feature where experienced participants can help beginners.
  - A project showcase where we can see past winning projects for inspiration.

Integration with GitHub or Figma for easy project submission.

## **1.9 TECHNOLOGY**

### **1. Identifying Problems, Opportunities, and Objectives**

- SWOT analysis was utilized to:
- Identify the issues the platform aims to solve.
- Determine the opportunities that can be improved.
- Define the objectives the system is designed to achieve.
- Reliable sources such as academic research and technical articles were used to understand the challenges faced by hackathon participants and the potential opportunities the platform can provide.
- Social media platforms like Twitter and LinkedIn offer valuable insights into the primary challenges faced by participants, as well as opportunities for improvement by collecting and analyzing user feedback.

### **2. Determining Human Requirements**

- Surveys and interviews were conducted with participants and organizers to understand user needs and design a user-friendly and secure system.
- The tools used include:
- Google Forms for creating digital surveys and analyzing user responses.
- Paper-based surveys distributed during hackathon events to gather feedback from attendees about their experience with existing tools.

### **3. Analyzing the System's Needs**

- To understand and structure the system's requirements, the following analytical tools were used:
- Data Flow Diagrams (DFD) to map out the system's inputs, processes, and outputs.
- Context Diagrams to provide a high-level view of the relationships between the platform and external entities.
- Entity-Relationship Diagrams (ERD) to illustrate the data model and relationships between different system elements, such as teams, projects, and awards.

### **4. Designing**



- The design phase focuses on developing the platform's functionalities and managing the backend system. This phase includes:
- UML for creating process flow diagrams.
- Diagramming tools such as StarUML and Draw.io for visual representations.

#### 5. Developing and Documenting Software

- The following tools were employed for platform development and documentation:
- Server development tools such as XAMPP (including Apache, MySQL, and PhpMyAdmin).
- Programming development tools like Visual Studio Code and HTML Editors.
- Web page formatting using CSS and frameworks like Bootstrap and Tailwind.
- Designing logos, icons, and graphics using tools such as Figma, Adobe XD, Freepik, and Bootstrap Icons.
- Additionally, Code Walkthroughs were provided to explain complex sections of the system in a clear manner.

#### 6. Testing and Maintaining the System

- To ensure stability and efficiency, the system underwent thorough testing and validation using:
- Validator.w3.org to detect errors and validate the code.
- Sample and real-world data to ensure stable performance.
- Program updates uploaded to the hosting server for ongoing maintenance.

#### 7. Implementing the System

- The final phase involves deploying the platform and making it available for use. The system will be optimized for multiple web browsers such as:
- Google Chrome
- Safari
- Microsoft Edge
- This ensures compatibility and accessibility for all users.

## 1.10 project Timeline

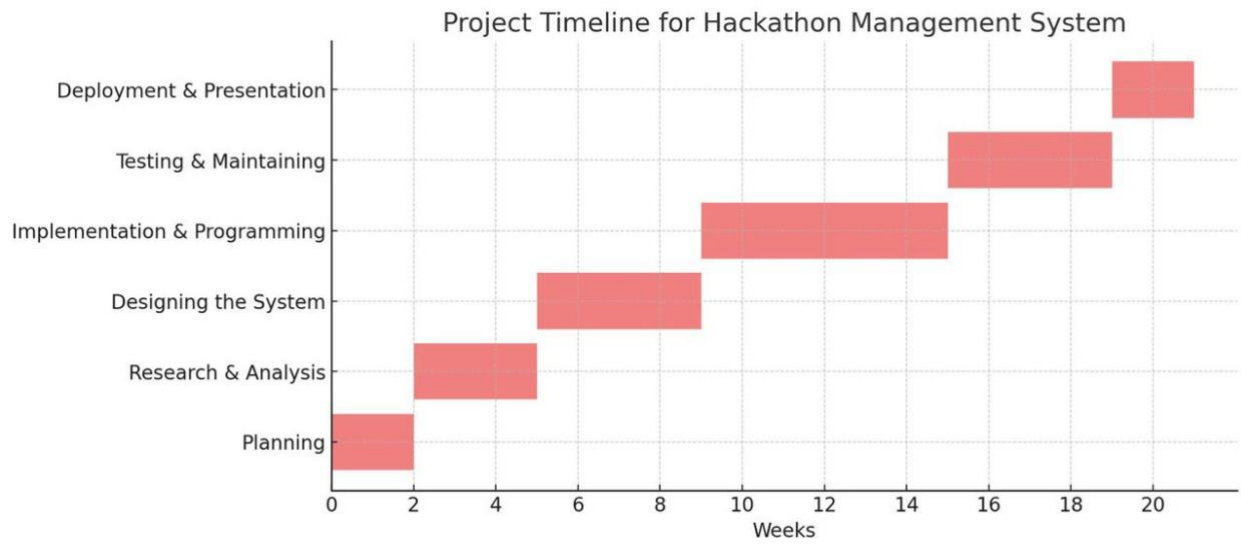


Figure 11 project timeline

## **2. SYSTEM ANALYSIS**

### **2.1 Overview**

This chapter provides a detailed explanation of our system HackSynk and its interaction with external entities. It introduces the core functionalities of the system, the roles of different users, and how data flows between them. The main objective of this chapter is to define the system's boundaries and scope by presenting a Context Diagram, which visually represents how the system communicates with external entities the users, teams, judges, hackathons, projects, and awards.

The chapter discusses each external entity's role in the system, detailing the input they provide and the output they receive. This helps in understanding how various stakeholders interact with the HackSynk, ensuring an organized and efficient hackathon management process. Additionally, this chapter sets the foundation for further detailed system design by breaking down high-level interactions.

## 2.2 Context Diagram

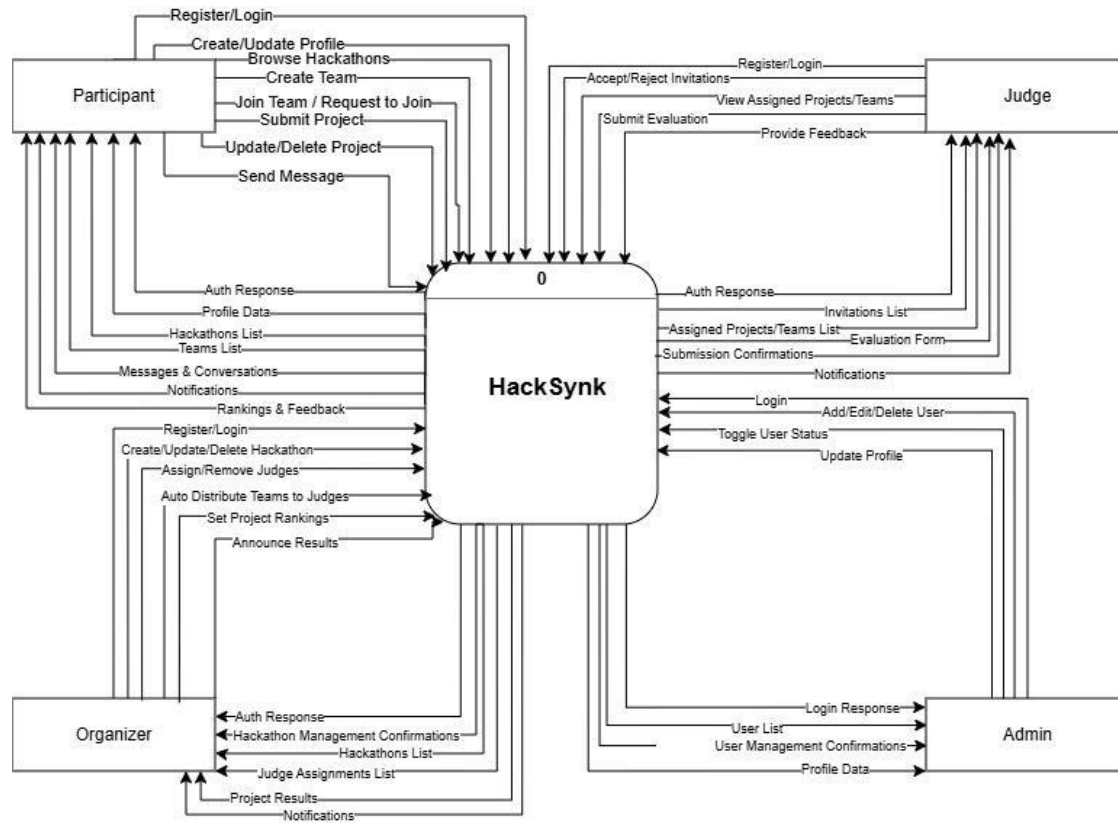


Figure 11 Context Diagram

Context diagrams provide clarity, aid communication, define scope, and assist in requirements gathering. This diagram illustrates how four user types interact with HackSynk.

### **Participant:**

After registration, participants can browse hackathons, form or join teams, submit projects with files, communicate via direct and team chat, and view evaluation results after announcement.

### **Organizer:**

Organizers create and manage hackathons, assign judges, distribute teams for evaluation, set project rankings, and announce results. The system provides analytics and tracks submissions throughout the event lifecycle.

### **Judge:**

Judges receive project assignments, evaluate submissions using a 10-criteria scoring system, and provide feedback. The system ensures evaluations occur during designated judging periods.

## Admin:

Admins manage all user accounts (participants, organizers, judges), control user status, and maintain system stability and security.

## 2.3 Functional Decomposition Diagram

A Functional Decomposition Diagram is a visual representation that breaks down a system or process into its smaller, more manageable components or functions. It shows the hierarchy of system functions, starting from the high-level objectives and progressively dividing them into more detailed sub-functions or tasks. This diagram helps to clarify the structure and flow of operations within the system. The detailed Functional Decomposition Diagram is provided in Figure 13.

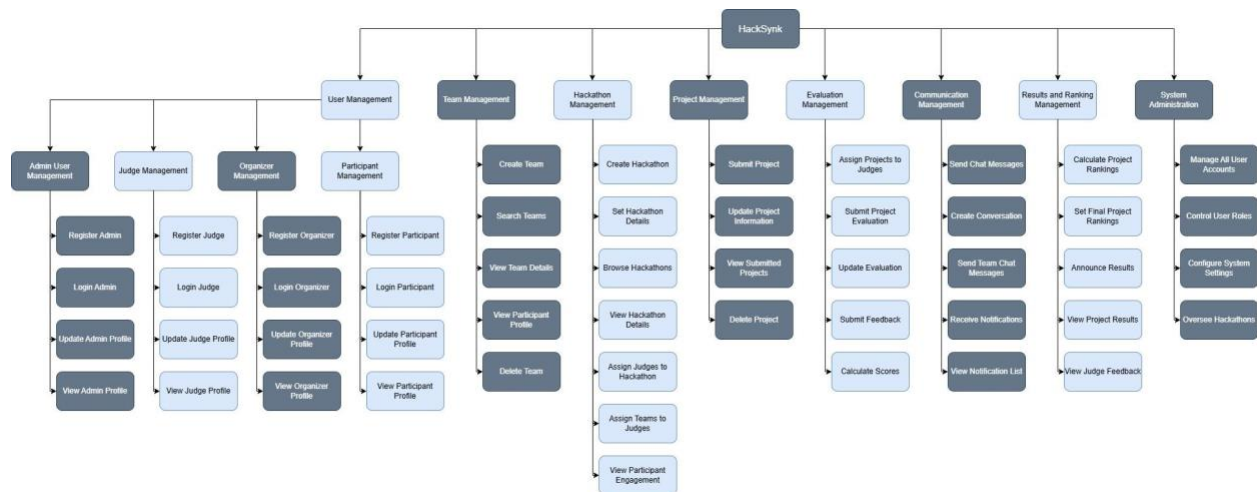


Figure 12Functional Decomposition Diagram

## 2.4 Data Flow Diagram

The Data Flow Diagram (DFD) for HackSynk illustrates the comprehensive data flow architecture of the hackathon management platform through a two-level decomposition. The Level 0 context diagram identifies four external entities (Participant, Organizer, Judge, and Admin) that interact with the system, while the Level 1 diagram decomposes the system into seven core processes: User Management, Team Management, Hackathon Management, Project

Management, Evaluation Management, Communication Management, and Rewards & Notifications Management. The DFD documents fourteen data stores that persist user profiles, team information, hackathon details, project submissions, evaluations, chat messages, rankings, and system notifications. Each process is detailed with its specific data flows, showing how information moves between external entities and processes, and how processes interact with data stores and each other. This diagram serves as a critical documentation tool that visualizes the complete system architecture, enabling stakeholders to understand data movement patterns, process dependencies, and system boundaries, while providing developers with a clear reference for implementation and maintenance of the platform.

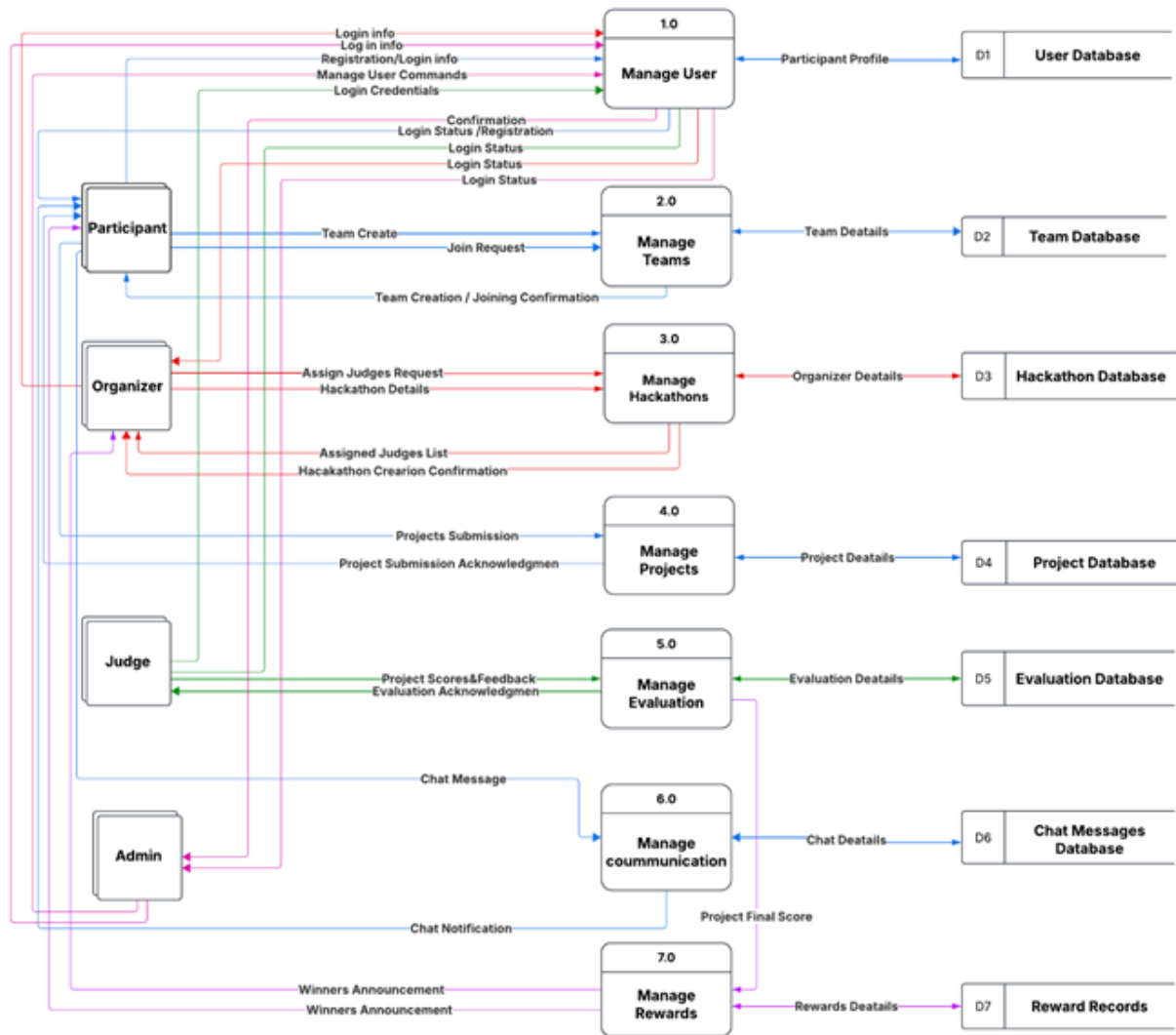


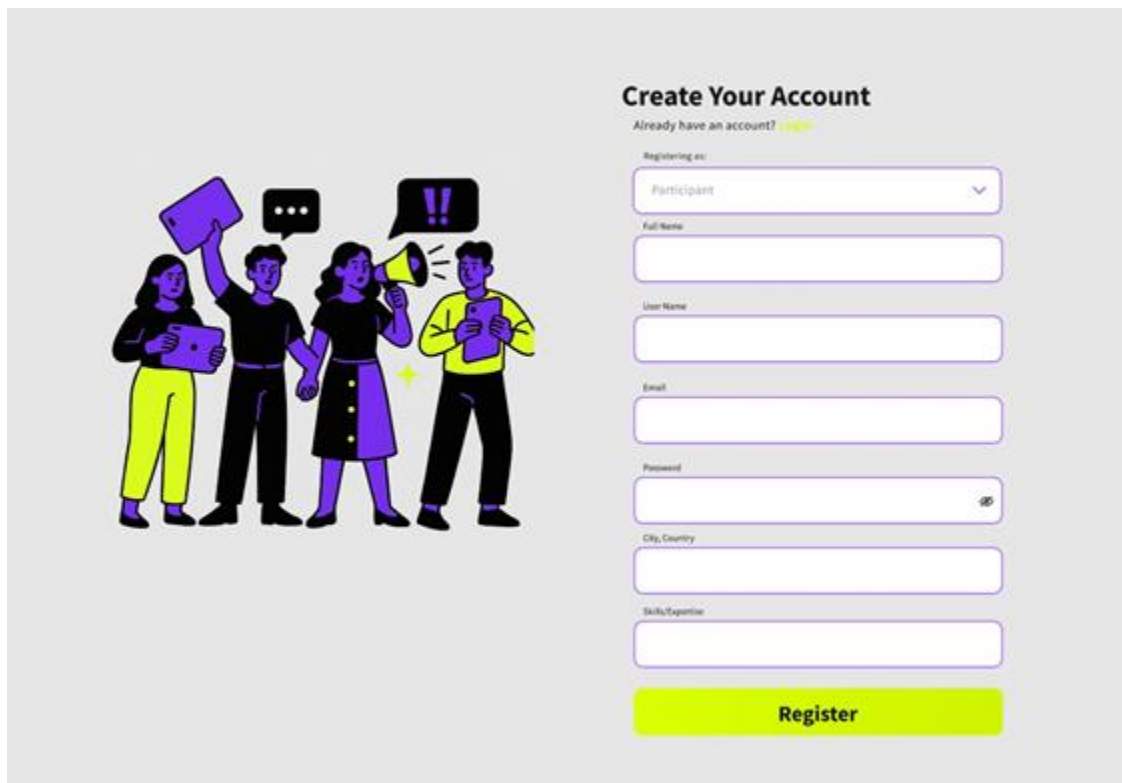
Figure 13DFD

### 3. SYSTEM DESIGN

#### 3.1 Overview

In this chapter focuses on the design aspects of the HackSynk System, covering input and output forms design for user interaction, prototype design to visualize functionality, and the Entity Relationship Diagram to illustrate data relationships. It also includes normalization to reduce redundancy, logical database design for organizing data access, and physical database design for implementation details. Together, these elements provide a cohesive framework that enhances the system's effectiveness and usability.

#### 3.2 Prototype Design



**Create Your Account**

Already have an account? [Click here](#)

Registering as:

Participant

Full Name

User Name

Email

Password

City, Country

Skills/Expertise

**Register**

Figure 14 Participant Sign Up





## Create Your Account

Already have an account? [Log In](#)

Registering as:

Organizer

Full Name

User Name

Email

Password

Organization Name

Job Title/Position

Register

Figure 15 Organizer Sign Up



### Create Your Account

Already have an account? [Log in!](#)

Registering as:

Full Name

User Name

Email

Password

Professional Title

**Register**

Figure 16 Judge Sign up



### Log in!

Dose not have an account? [Register](#)

User Name

Password

**Register**

Figure 17 Log in Forms

The screenshot shows the HackSynk website's 'Health Innovation Hackathon' registration page. The header includes the HackSynk logo, navigation links for Home, Hackathons, and About Us, and a search icon with 'Log in' and 'Sign up' buttons. The main content area features a large hexagonal logo on the left and a registration form on the right. The form includes a 'Short Description' section, a 'Start Date' of May 28, 2025, and an 'End Date' of May 31, 2025. The 'Location' is listed as Riyadh, Saudi Arabia. The 'Rules and Guidelines' section lists four rules: teams must consist of 3-5 members, participants must register before the deadline, projects should focus on innovative healthcare solutions, and presentations will be judged based on creativity, feasibility, and impact. The 'Prizes' section lists two prizes: 1st Place (\$5,000 and mentorship opportunities) and 2nd Place (\$3,000). A 'Register now' button is located at the bottom right of the form.

**HackSynk** Home Hackathons About Us

**Health Innovation Hackathon**

**Short Description:**  
Join us for the Health Innovation Hackathon, where teams will develop creative solutions to improve healthcare access and efficiency. Collaborate with like-minded individuals and experts to make a difference in the health sector!

**Start Date:** May 28, 2025 **End Date:** May 31, 2025

**Location:** Riyadh, Saudi Arabia

**Rules and Guidelines:**

- Teams must consist of 3-5 members.
- Participants must register before the deadline.
- Projects should focus on innovative healthcare solutions.
- Presentations will be judged based on creativity, feasibility, and impact.

**Prizes:**

- 1st Place: \$5,000 and mentorship opportunities
- 2nd Place: \$3,000

**Register now**

Figure 18Hackathon Register

The screenshot shows the HackSynk website's 'Join Hackathon' registration form. The header includes the HackSynk logo, navigation links for Home, Hackathons, and About Us, and a search icon with a user profile icon. The main content area features a 'Join Hackathon' section with a form. The form includes a 'Project Title' field, a 'Leader Name' field with the value 'Abdullah Hamad', an 'Upload File' section with a 'Choose a file' button and a 'PDF file only' note, and a 'Group Members Number' field with the value '3'. A 'Register now' button is located at the bottom right of the form.

**HackSynk** Home Hackathons About Us

**Join Hackathon**

**Project Title:**

**Leader Name:**  
Abdullah Hamad

**Upload File**

**Choose a file** PDF file only

**Group Members Number:**  
3

**Register now**

Figure 19Leader Register Form

**HackSynk** Home Hackathons About Us

### Join Hackathon

Project Title:

Leader Name:

Upload File

Chose a file PDF File only

Group Members Number:

\*Can Not join As You Don't Have Team Yet

Register now

Figure 20 Not a Leader Register Form

**HackSynk**

Search

Profile

Create/Manage Hackathons

Assign Judges

Log Out

### Create New Hackathon

Hackathon Name (Required)

Short Description (Required)

Start Date (Required) End Date (Required)

Location (Optional)

Rules and Guidelines

Prizes (Optional)

Upload Hackathon Image (Optional)

Chose a file

Create Hackathon

Figure 21 Create New Hackathon

**HackSynk**

Search

**Edit Hackathon**

Hackathon Name (Required)  
Saudi TechHackathon

Short Description (Required)  
The Saudi Tech Challenge invites innovators and tech enthusiasts to develop solutions addressing local challenges in .....

Start Date (Required) 2025-12-01 End Date (Required) 2025-12-03

Location (Optional)  
King Abdulaziz University, Jeddah

Rules and Guidelines

Prizes (Optional)

Upload Hackathon Image (Optional)  
Choose a file

**Edit Hackathon**

**Log Out**

Figure 22Edit Hackathon

**HackSynk**

Search

**Assign New Judges**

Judge Name:

Judge Email:

Select Hackathon:  
Select Hackathon

**Send Invite**

**Log Out**

Figure 23Assign New Judge

**HackSynk**

Search

**Hackathon Management**

هـاڪٿون  
ٽڪنولاجيا  
الازياء

**Announcement**

Title of Announcement (Required)

Announcement Content (Required)

Send Announcement

Log Out

Figure 24 Hackathon Management

**HackSynk**

Search

**Create a Team**

Team Name (Required)

Team Description (Optional)

Select Hackathon (Required)

Select Hackathon

Max Participants

3

Create

Log Out

Figure 25 Create Team



Profile

Create Team

Join Team

Submit Project

Send Chat Message

Log Out

Search

Submit Project

Project Title:

Leader Name:

Abdullah Hamad

Upload File

Chose a file

PDF File only

Group Members Number:

3

Hackathon:

Health Innovation Hackathon

Submit

Figure 26 Submit Project



Profile

Invitations

Project Evaluation/Feedback

Log Out

Search

Project Evaluation-Feedback



X Team

View Project

Project description

| Evaluation Criteria           | Bad                   | Normal                | Excellent             |
|-------------------------------|-----------------------|-----------------------|-----------------------|
| 1. Innovation and Creativity: | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 2. Technical Implementation:  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 3. Usability and Design:      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 4. Impact and Relevance:      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| 5. Presentation:              | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Write a Feedback.....

Send

Figure 27 Project Evaluation

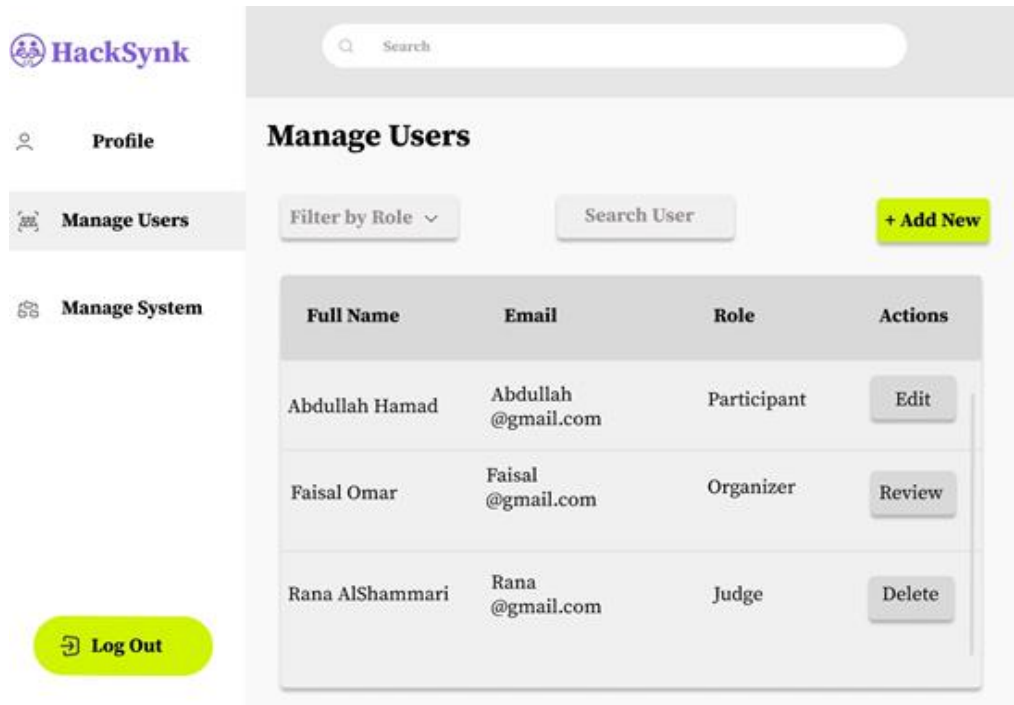


Figure 28Manage Users

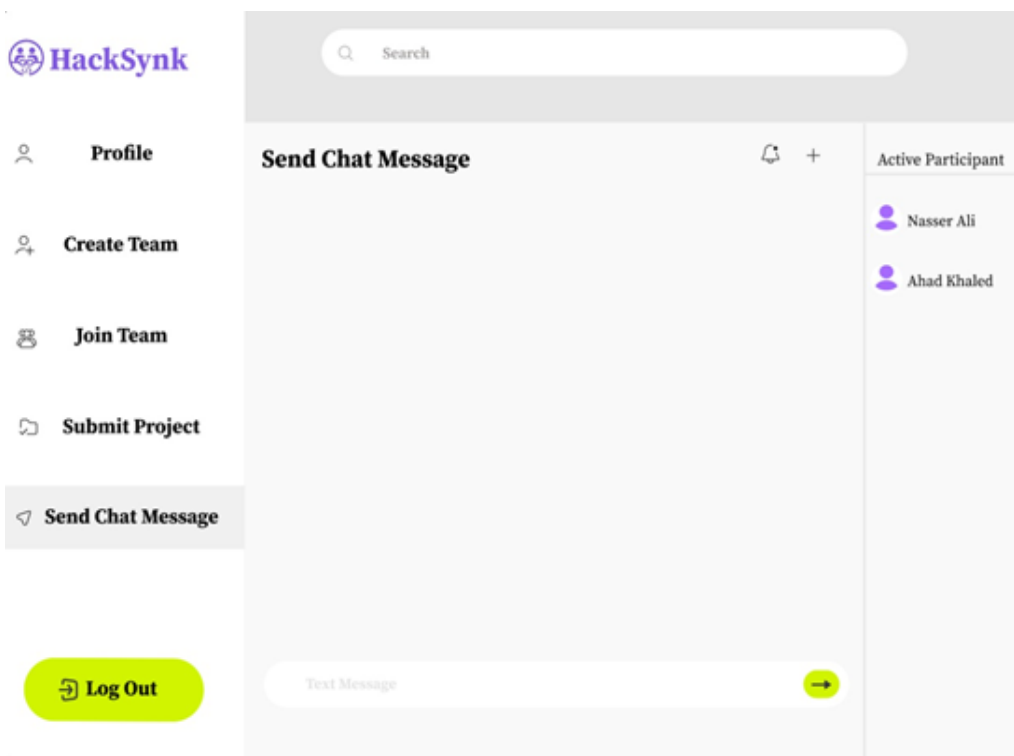


Figure 29Chat Message



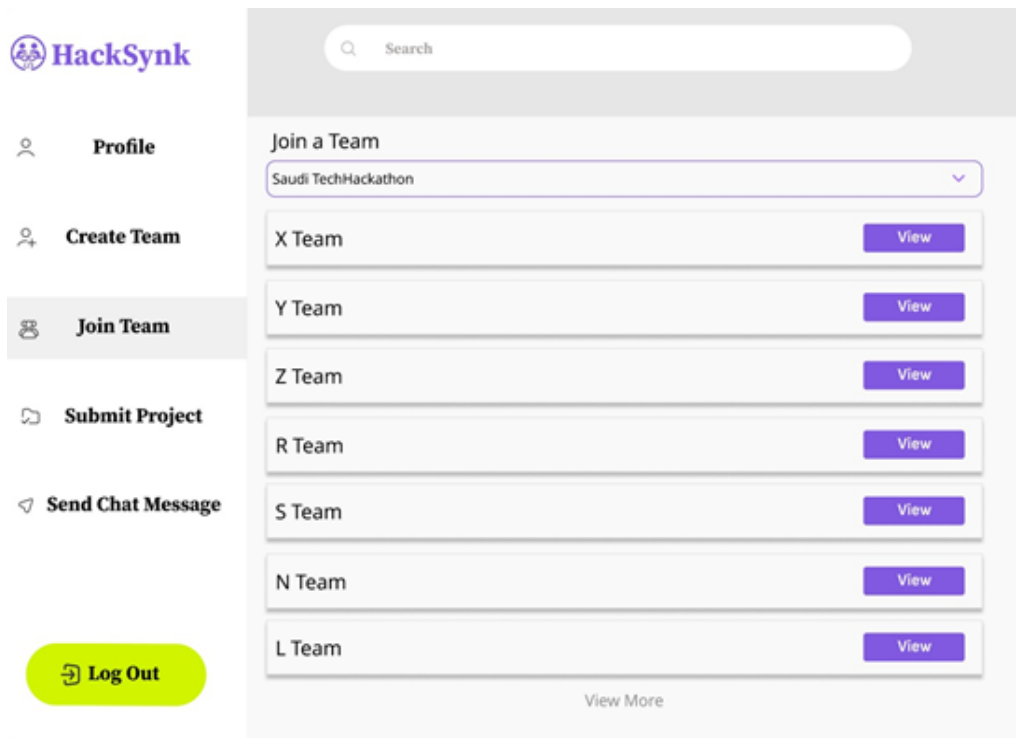


Figure 30Join Team

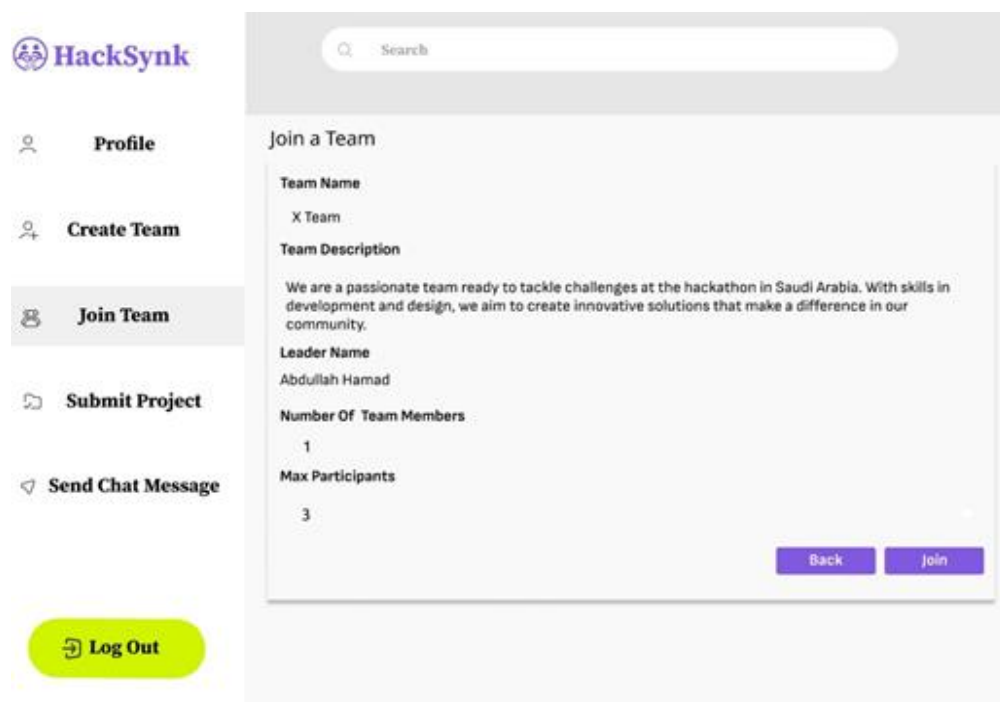


Figure 31Join Team 2

The screenshot shows the 'Join a Team' form in the HackSynk application. The left sidebar contains navigation links: Profile, Create Team, Join Team (highlighted), Submit Project, and Send Chat Message, along with a Log Out button. The main content area has a search bar at the top. The form fields are as follows:

| Field                  | Value                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Team Name              | X Team                                                                                                                                                                                                      |
| Team Description       | We are a passionate team ready to tackle challenges at the hackathon in Saudi Arabia. With skills in development and design, we aim to create innovative solutions that make a difference in our community. |
| Leader Name            | Abdullah Hamad                                                                                                                                                                                              |
| Number Of Team Members | 3                                                                                                                                                                                                           |
| Max Participants       | 3                                                                                                                                                                                                           |

A 'Back' button is located at the bottom right of the form.

Figure 32The team is complete

The screenshot shows the 'Participant Profile' page for Abdullah Hamad. The left sidebar is identical to the previous figure, with 'Profile' now highlighted. The main content area displays the user's profile and statistics:

| Section                              | Value | Action                        |
|--------------------------------------|-------|-------------------------------|
| <b>Abdullah Hamad</b><br>Participant |       | <a href="#">Edit Profile</a>  |
| <b>Hackathons Participated</b>       | 4     |                               |
| <b>Teams Joined</b>                  | 2     | <a href="#">View Teams</a>    |
| <b>Project Submitted</b>             | 2     | <a href="#">View Projects</a> |

Figure 33Participant Profile

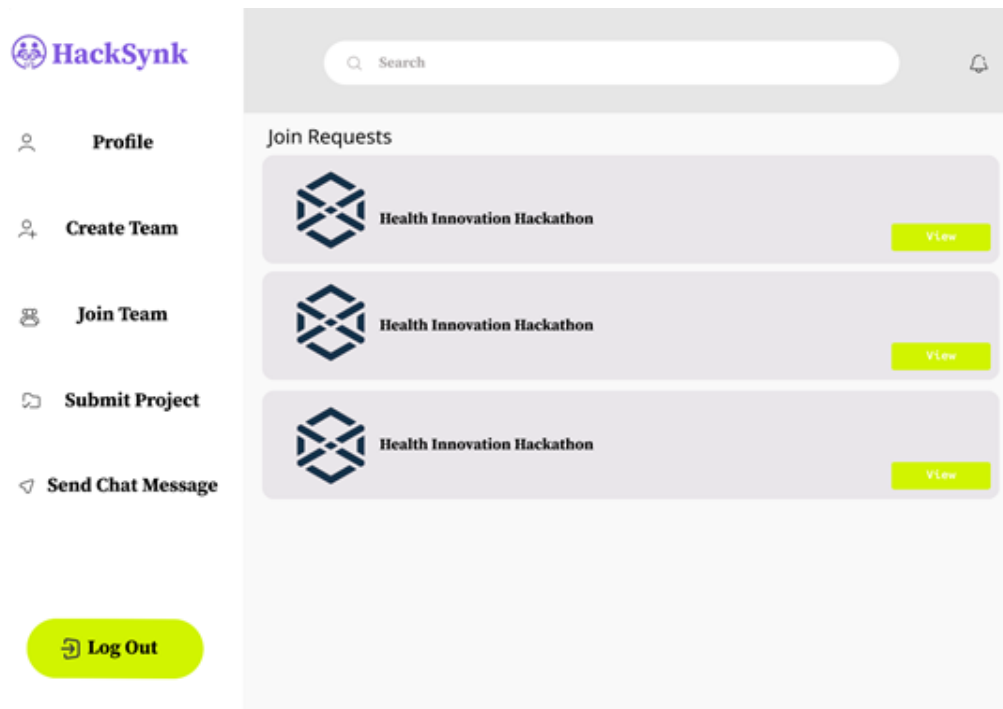


Figure 34Join Requests

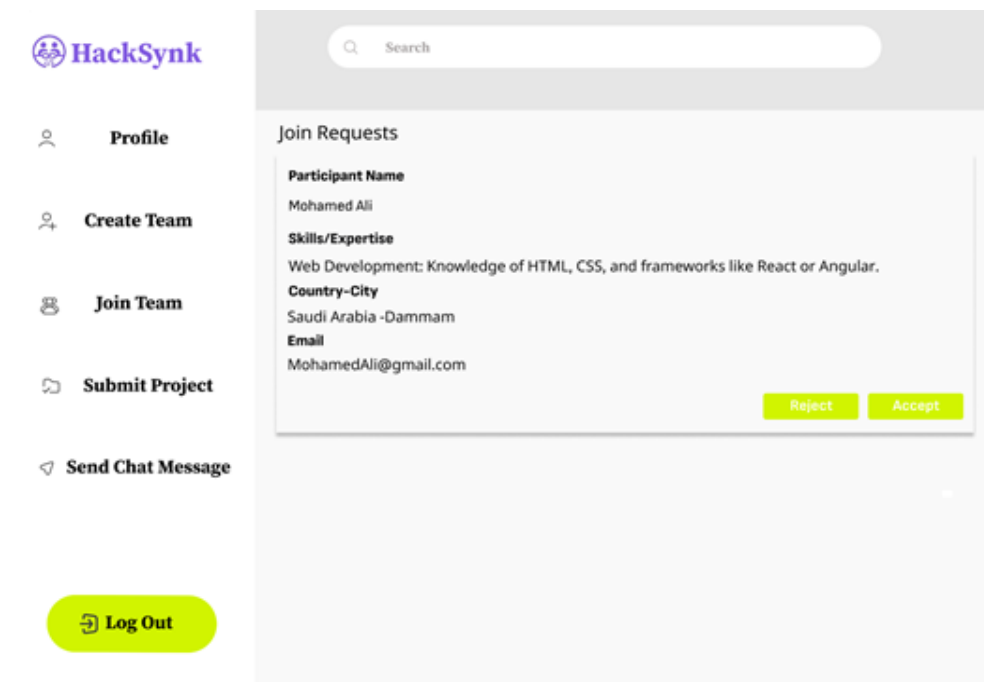


Figure 35View Join Requests

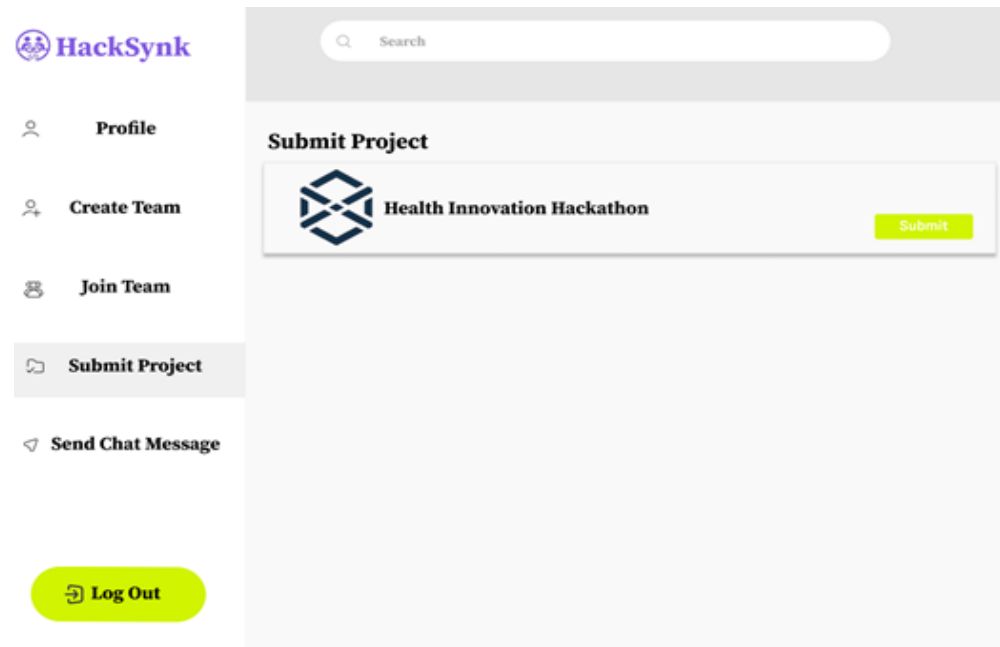


Figure 36Participant Submit Project-Leader

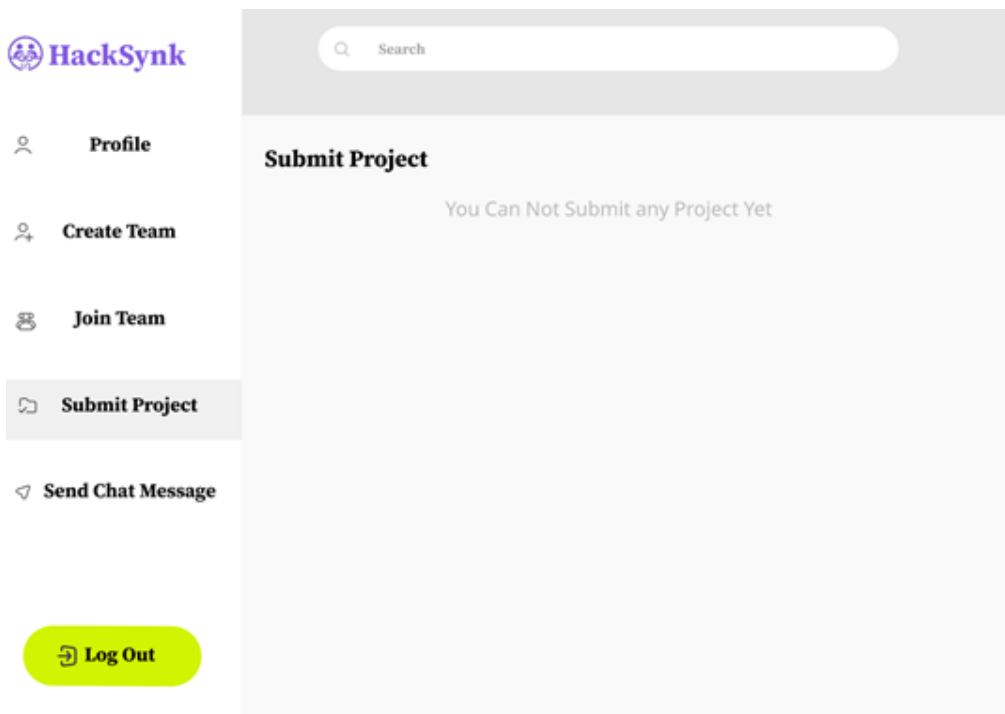


Figure 37Participant Submit Project-Not Leader

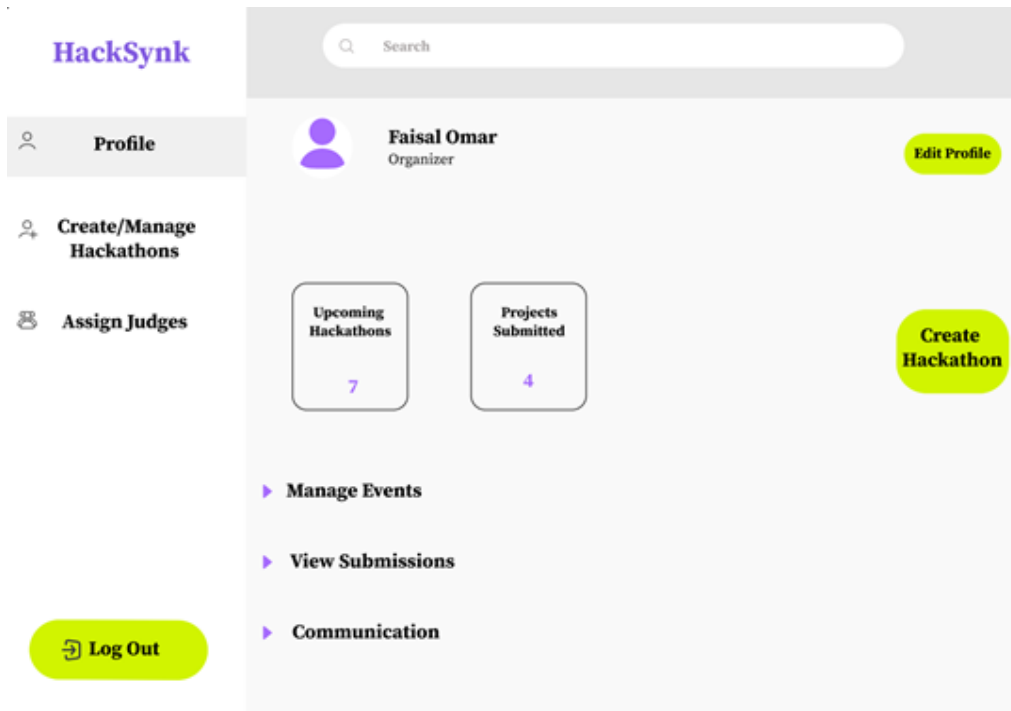


Figure 38 Organizer Profile

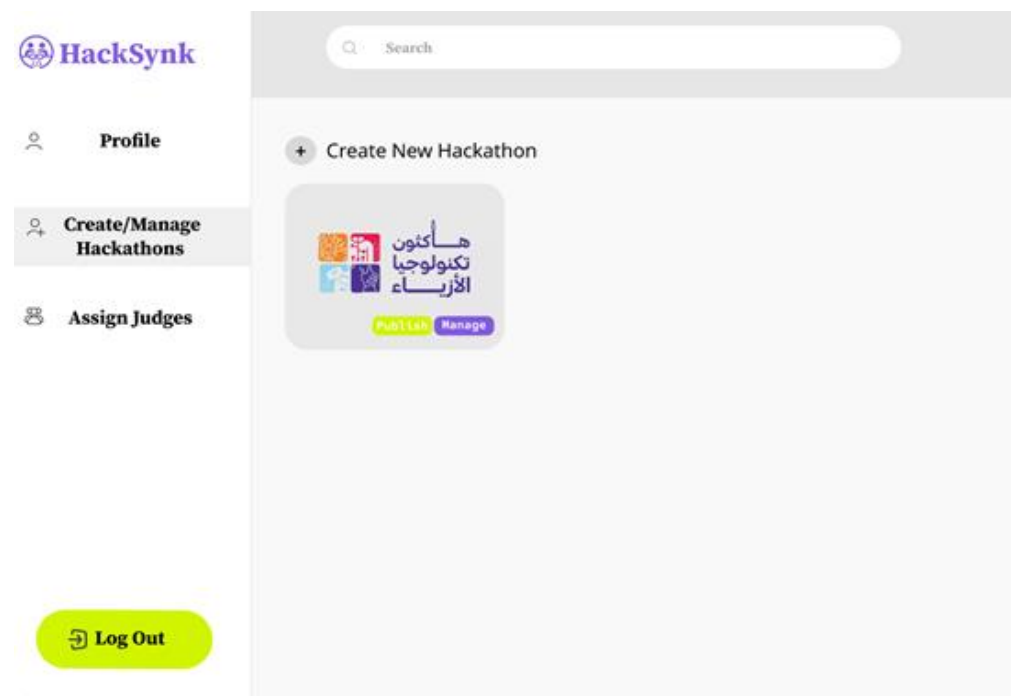


Figure 39 Create/Manage Hackathons

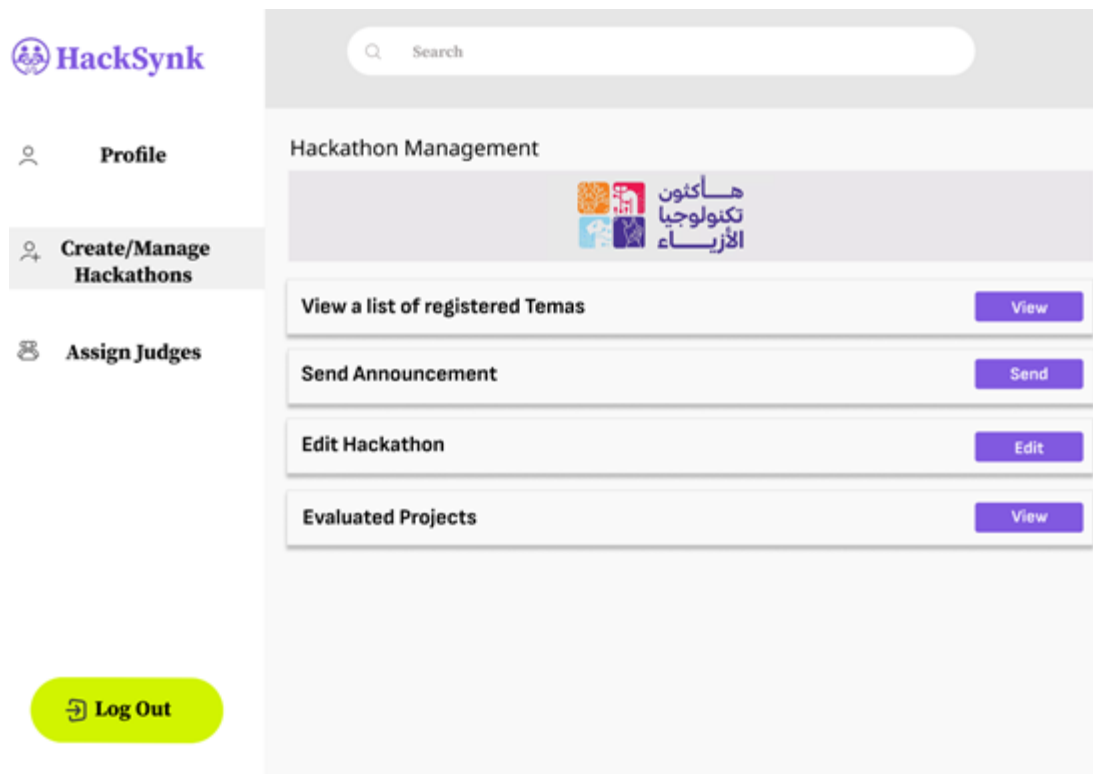


Figure 40Hackathon Management

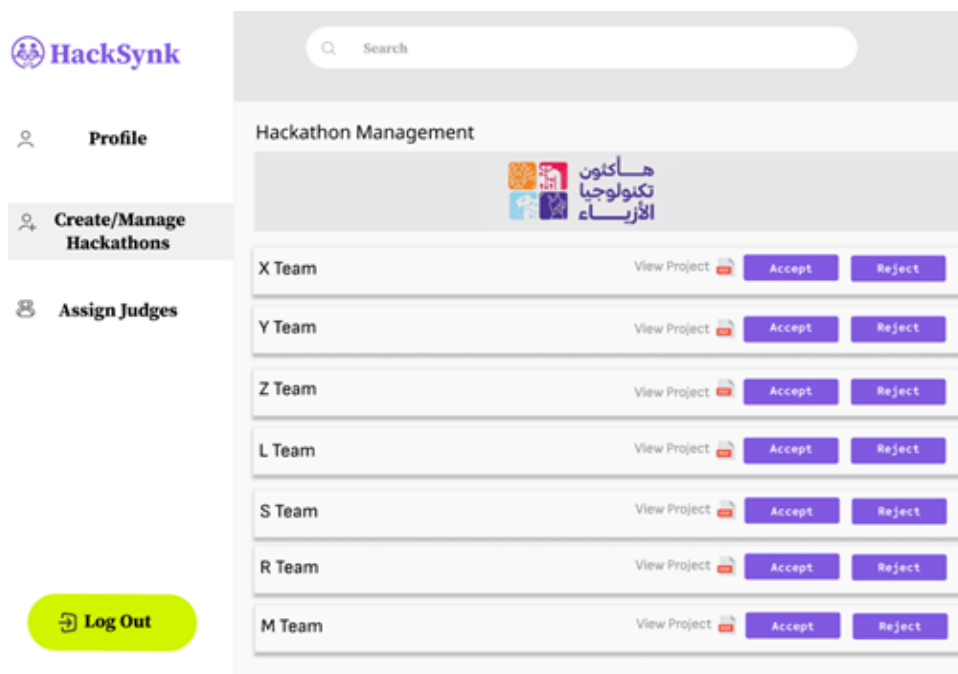


Figure 41View a list of registered Teams

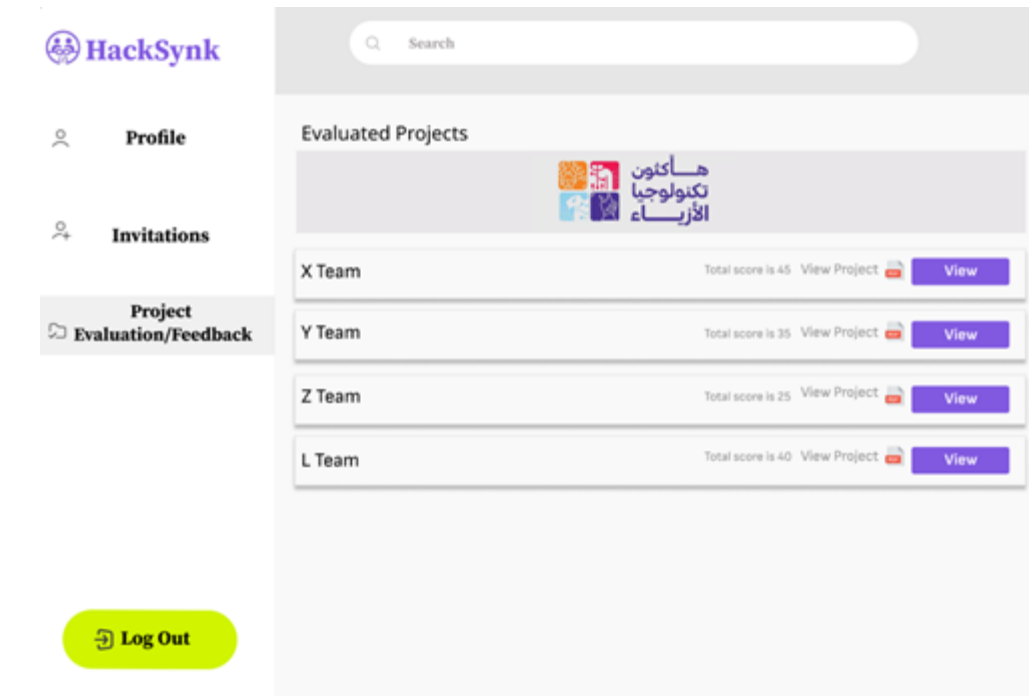


Figure 42View a list of registered Teams

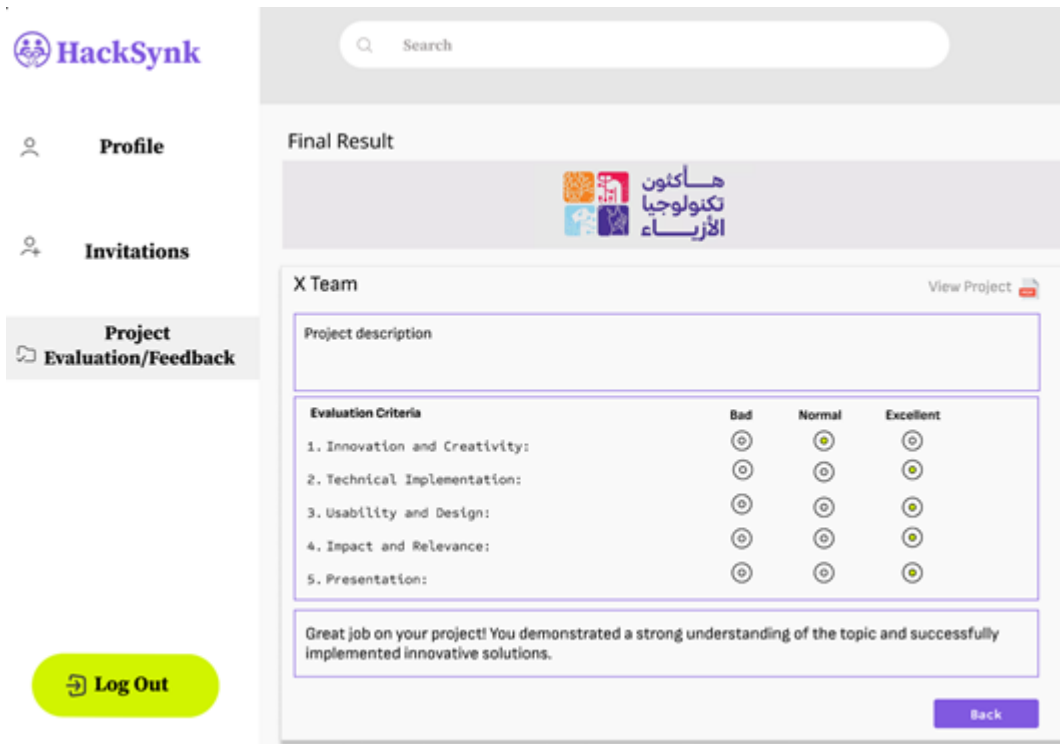


Figure 43View Final Result

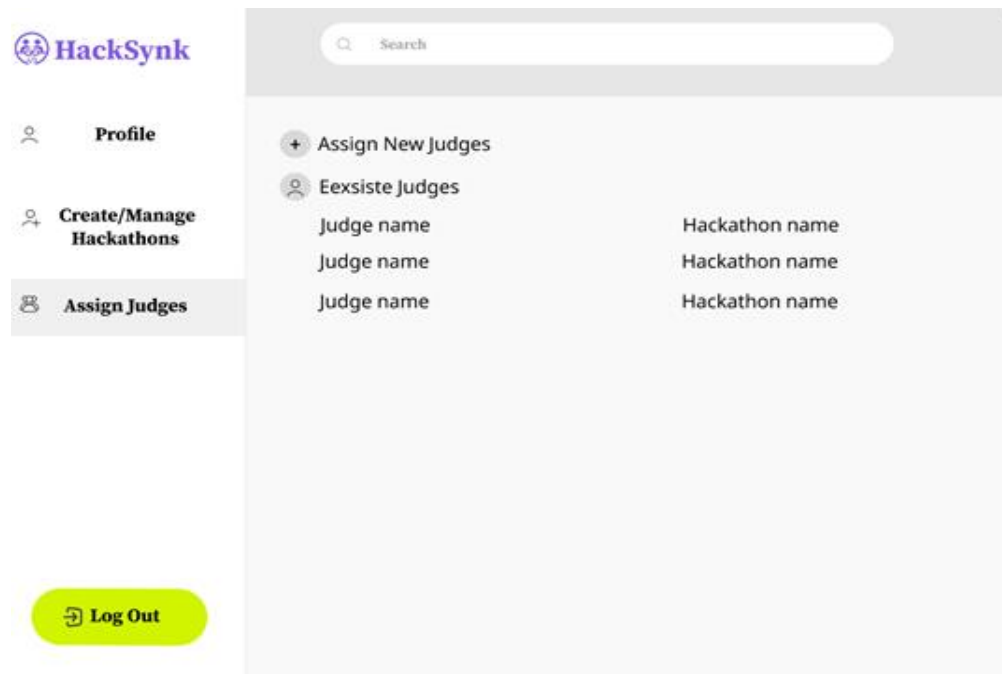


Figure 44Assign Judges

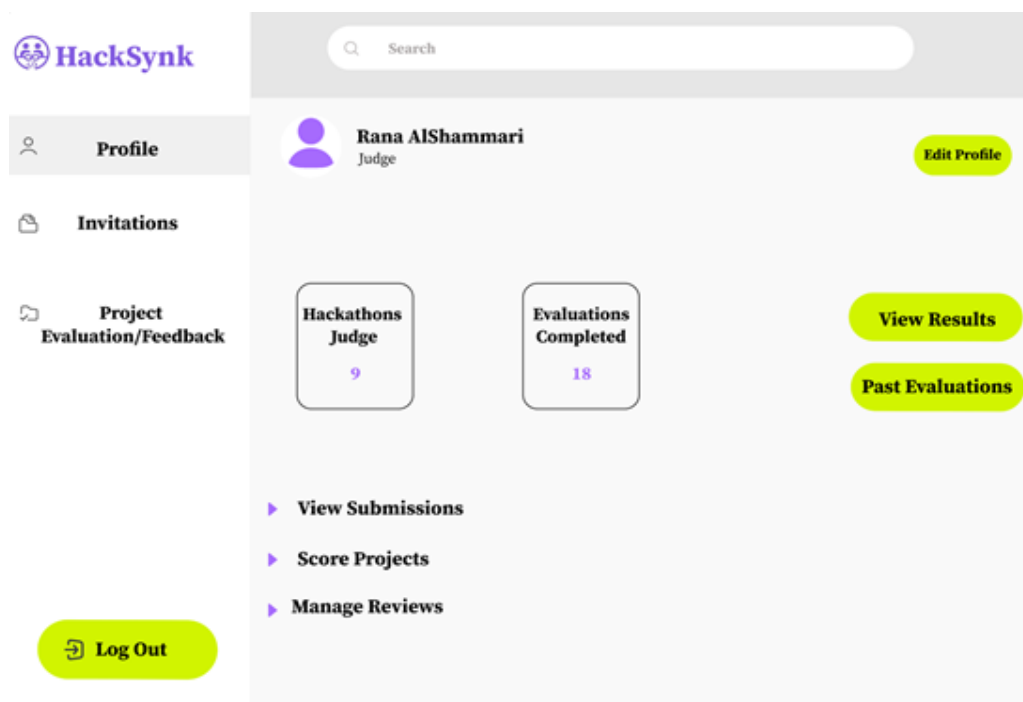


Figure 45Judge Profile



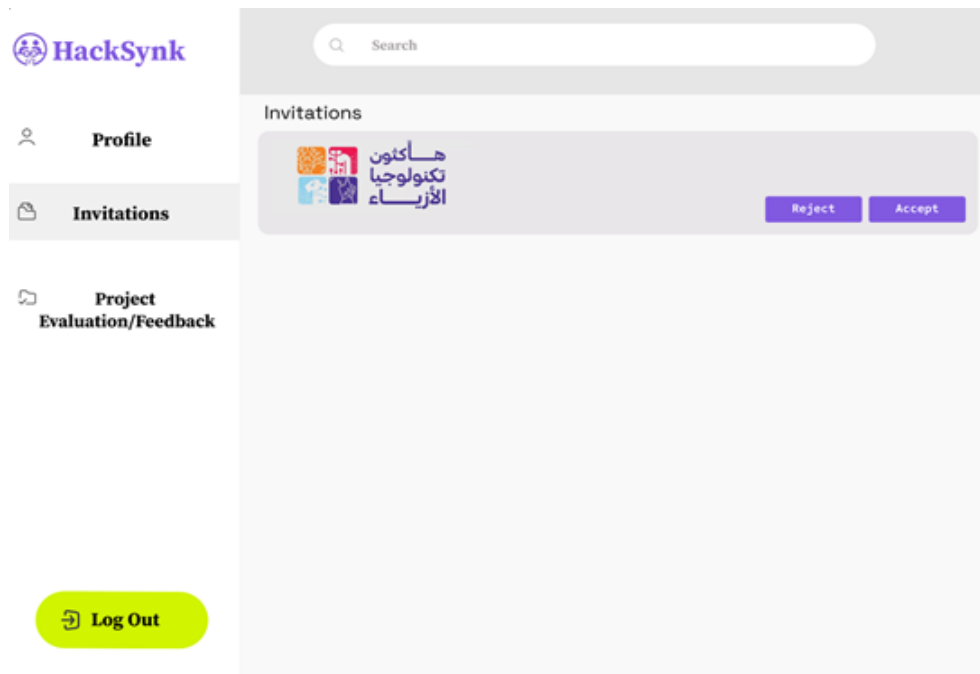


Figure 46Judge Invitation

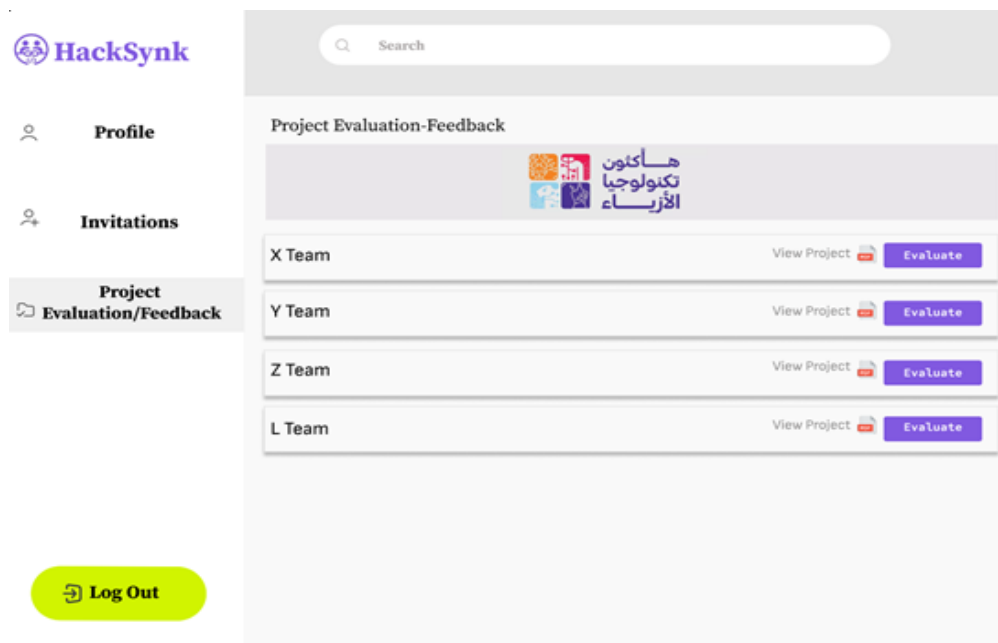


Figure 47Project Evaluation/Feedback

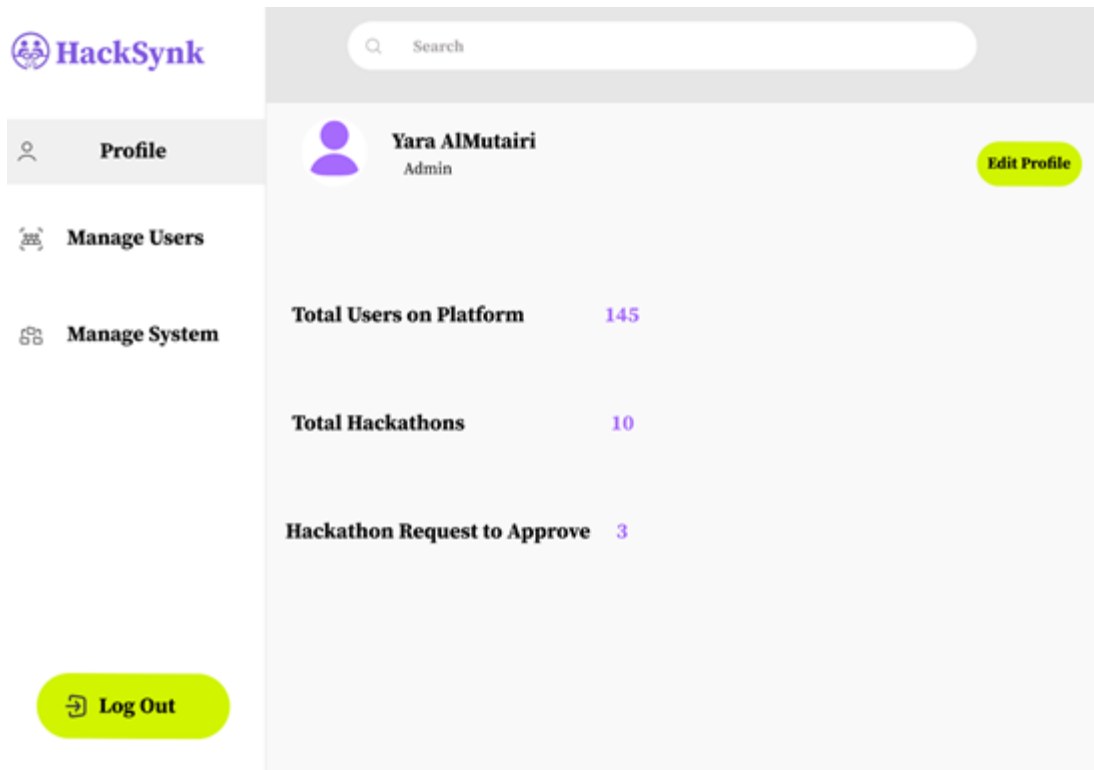


Figure 48 Admin Profile

### 3.3 Entity Relationship Diagram

Based on the ERD diagram from top to bottom, the HackSynk system is built around four distinct user roles: participants, organizers, judges, and admins. Each user type has its own authentication information including ID, full name, username (unique), email (unique), password, and creation timestamp.

Participants form the core user base and contain detailed profile information such as city/country, educational qualification, specialization, university, graduation year, certificates, years of experience, and account status. Each participant can have multiple participant\_skills, where each skill includes a skill name and skill level (Beginner, Intermediate, or Advanced).

Organizers create and manage hackathons. Each organizer has additional information including organization name and job title/position. Hackathons are comprehensive events containing a name, description, start and end dates, location, rules, prizes, image path,

minimum and maximum team limits, results announcement status, and detailed timeline information including submission periods, judging periods, and result announcement dates. Each hackathon is created by an organizer and can contain multiple teams.

Judges are assigned to hackathons through the `judge_hackathons` junction table, which tracks the assignment status (pending, accepted, or rejected) and assignment timestamp. Judges can also be specifically assigned to teams through `judge_team_assignments` for targeted evaluation. Each judge has a professional title and account status.

Teams are formed within hackathons and include a name, description, maximum participants limit, required skills, and are led by a participant (team leader). Teams can have multiple `team_members` (participants who have joined), and participants can submit `team_join_requests` with status tracking (pending, approved, or rejected). Each team also has its own `team_chats` for group communication.

Projects are submitted by teams and include a title, description, demo link, and are led by a participant. Each project can have multiple `project_files` (supporting documents, images, videos, presentations, etc.) uploaded by participants. Projects are evaluated by judges through `project_evaluations`, which contain comprehensive scoring across eleven criteria: innovation, implementation quality, design/usability, impact, problem relevance/clarity, functionality/features completeness, feasibility/scalability, team collaboration/presentation quality, technical complexity, and testing/reliability. Each criterion is scored from 1 to 10, with a calculated total score and optional feedback text. Projects can be ranked through `project_rankings`, which include ranking position, status (draft, confirmed, or final), and notes.

The system also includes a comprehensive communication infrastructure. Conversations are established between two participants for direct messaging, and each conversation contains multiple messages with support for text, file, and image message types, read status tracking, and timestamps. Similarly, `team_chats` contain `team_chat_messages` for team-based group communication with the same message type capabilities.

Finally, the system includes a notifications table that supports all user types (participants, organizers, judges, and admins) through a polymorphic relationship using `user_id` and `user_type` fields. Notifications track message content, notification type, read status, and creation timestamp.

The entire system is designed to support the complete hackathon lifecycle from creation and team formation through project submission, evaluation, ranking, and result announcement, with integrated communication and notification systems throughout.

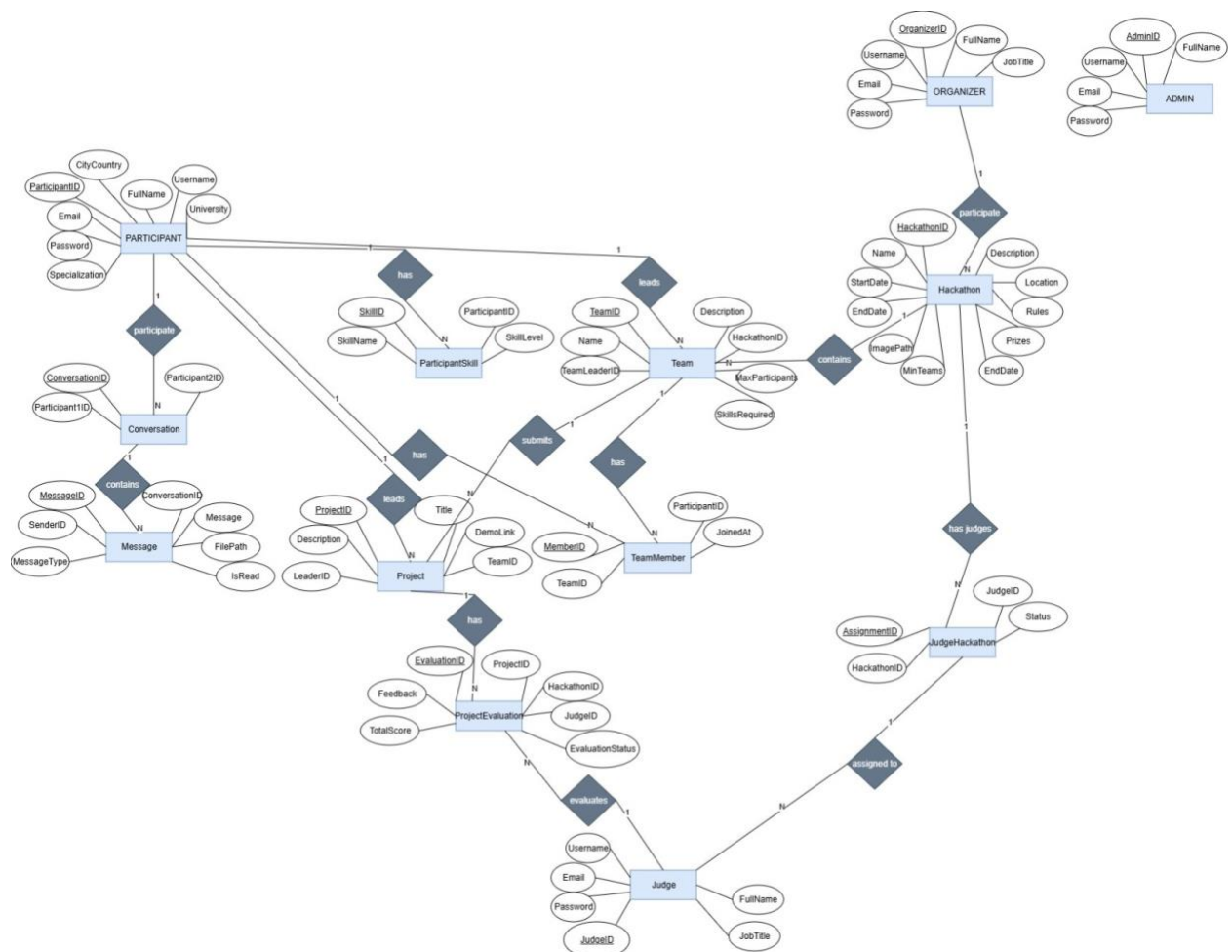


Figure 49 Entity Relationship Diagram

### 3.4 Normalization

The Logical Database Design for the HackSynk system presents a fully normalized database schema in Third Normal Form (3NF), comprising 19 interconnected entities that collectively support the complete hackathon management lifecycle from event creation to result announcement. The normalized structure is organized into five functional domains: (1) User Management with four role-based entities (Participants, Organizers, Judges, Admins) each with distinct attributes and relationships; (2) Hackathon Management including Hackathons, Teams, Team Members, and Team Join Requests with foreign key relationships to Participants and Hackathons; (3) Project Management with Projects, Project Files, Project Evaluations featuring 11 evaluation criteria, and Project Rankings; (4) Communication systems with Conversations and Messages for direct messaging, plus Team Chats for group communication; and (5) Supporting entities including Participant Skills, Judge Hackathons junction table, Judge Team Assignments, and a polymorphic Notifications table. The design enforces referential integrity through 30+ foreign key relationships with appropriate CASCADE and SET NULL actions, eliminates data redundancy through normalization, implements unique constraints on critical combinations (usernames, emails, team-participant pairs), and includes comprehensive validation through CHECK constraints on scores, dates, and status fields, ensuring data consistency and optimal query performance while maintaining production-ready standards for implementation in any modern relational database management system.



Figure 50 Normalization

### 3.5 Logical database design

The Logical Database Design for the HackSynk system represents a comprehensive, normalized database schema consisting of 20 interconnected tables organized into five functional domains: User Management (participants, organizers, judges, admins), Hackathon Management (hackathons, teams, team members, join requests), Project Management (projects, project files,

evaluations, rankings), Communication (conversations, messages, team chats), and Supporting Systems (participant skills, judge assignments, notifications). The design adheres to Third Normal Form (3NF) principles, ensuring data integrity, eliminating redundancy, and maintaining optimal query performance through strategic normalization and comprehensive constraint definitions.

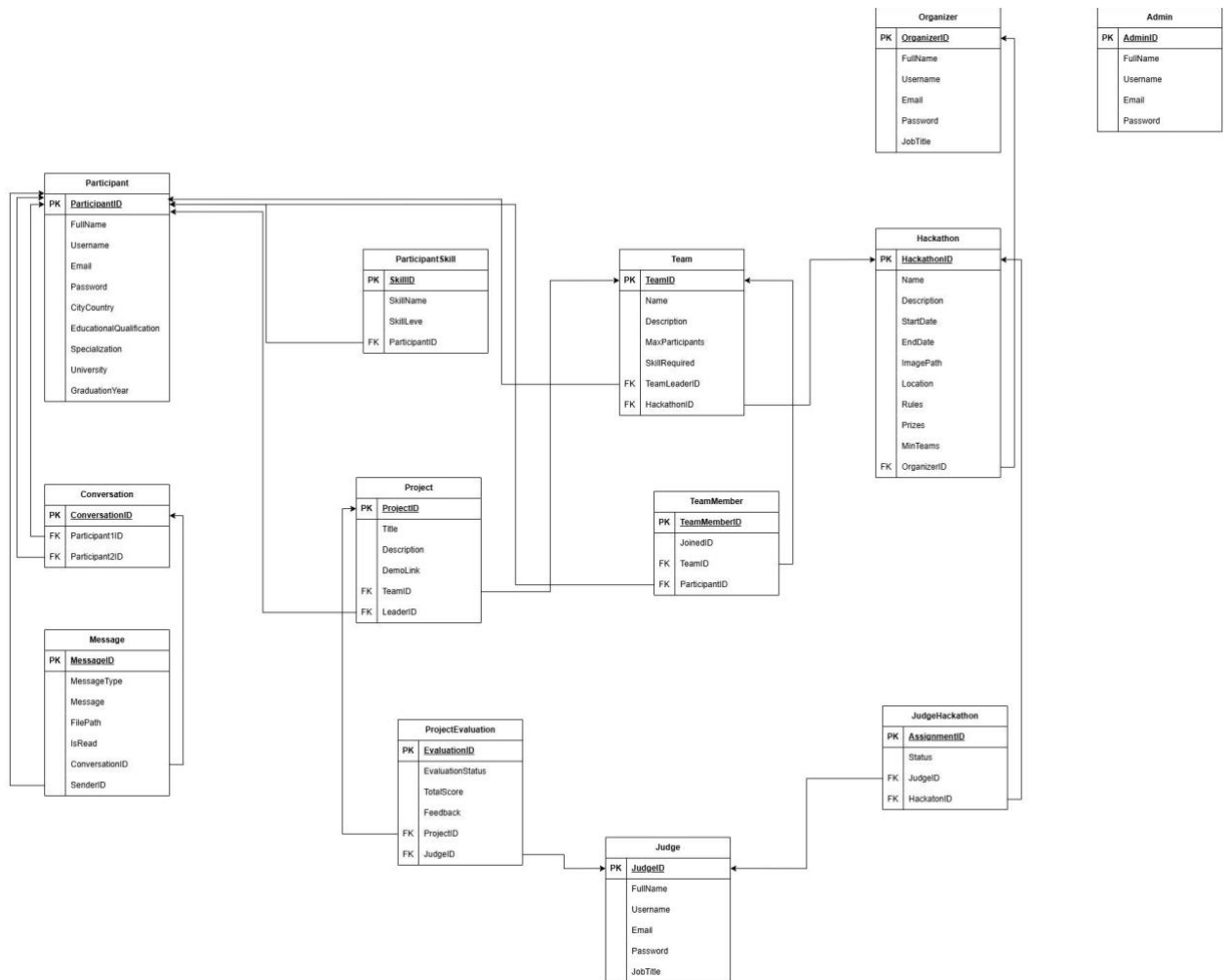


Figure 51 Logical Database Design

### 3.6 Physical database design

```
CREATE TABLE `admins` (
```

```

`id` int NOT NULL AUTO_INCREMENT,
`full_name` varchar(100) NOT NULL,
`username` varchar(50) NOT NULL,
`email` varchar(100) NOT NULL,
`password` varchar(255) NOT NULL,
`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (`id`),
UNIQUE KEY `username` (`username`),
UNIQUE KEY `email` (`email`)
)

```

| #                        | Name                | Type         | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|---------------------|--------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b>         | int(11)      |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>full_name</b>  | varchar(100) | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>username</b>   | varchar(50)  | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>email</b>      | varchar(100) | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>password</b>   | varchar(255) | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>created_at</b> | timestamp    |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 52Admins table

```

CREATE TABLE `participants` (
  `id` int NOT NULL AUTO_INCREMENT,
  `full_name` varchar(100) NOT NULL,
  `username` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(255) NOT NULL,
  `city_country` varchar(100) DEFAULT NULL,
  `educational_qualification` varchar(100) DEFAULT NULL,
  `specialization` varchar(100) DEFAULT NULL,
  `university` varchar(200) DEFAULT NULL,
  `graduation_year` year DEFAULT NULL,
  `certificates` text,

```



```

`years_experience` int DEFAULT NULL,
`status` enum('active','disabled') DEFAULT 'active',
`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (`id`),
UNIQUE KEY `username` (`username`),
UNIQUE KEY `email` (`email`)
)

```

| #                        | Name                               | Type                      | Collation          | Attributes | Null | Default             | Comments | Extra          | Action                                                           |
|--------------------------|------------------------------------|---------------------------|--------------------|------------|------|---------------------|----------|----------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 <b>id</b> 🔑                      | int(11)                   |                    |            | No   | None                |          | AUTO_INCREMENT | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 <b>full_name</b>                 | varchar(100)              | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 <b>username</b> 🔑                | varchar(50)               | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 <b>email</b> 🔑                   | varchar(100)              | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 5 <b>password</b>                  | varchar(255)              | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 6 <b>city_country</b>              | varchar(100)              | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 7 <b>educational_qualification</b> | varchar(100)              | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 8 <b>specialization</b>            | varchar(100)              | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 9 <b>university</b>                | varchar(200)              | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 10 <b>graduation_year</b>          | year(4)                   |                    |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 11 <b>certificates</b>             | text                      | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 12 <b>years_experience</b>         | int(11)                   |                    |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 13 <b>created_at</b>               | timestamp                 |                    |            | No   | current_timestamp() |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 14 <b>status</b>                   | enum('active','disabled') | utf8mb4_general_ci |            | Yes  | active              |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 53Participants table

```

CREATE TABLE `participant_skills` (
  `id` int NOT NULL AUTO_INCREMENT,
  `participant_id` int NOT NULL,
  `skill_name` varchar(100) NOT NULL,
  `skill_level` enum('Beginner','Intermediate','Advanced') NOT NULL,
  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `unique_participant_skill` (`participant_id`,`skill_name`),
  KEY `fk_participant_skills_participant` (`participant_id`),
  CONSTRAINT `fk_participant_skills_participant` FOREIGN KEY (`participant_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE

```

)

| #                        | Name             | Type                                         | Collation          | Attributes | Null | Default             | Comments | Extra          | Action                                                           |
|--------------------------|------------------|----------------------------------------------|--------------------|------------|------|---------------------|----------|----------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 id             | int(11)                                      |                    |            | No   | None                |          | AUTO_INCREMENT | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 participant_id | int(11)                                      |                    |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 skill_name     | varchar(100)                                 | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 skill_level    | enum('Beginner', 'Intermediate', 'Advanced') | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 5 created_at     | timestamp                                    |                    |            | No   | current_timestamp() |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 54Participant\_skills table

```
CREATE TABLE `organizers` (
  `id` int NOT NULL AUTO_INCREMENT,
  `full_name` varchar(100) NOT NULL,
  `username` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(255) NOT NULL,
  `organization_name` varchar(100) DEFAULT NULL,
  `job_title_position` varchar(100) DEFAULT NULL,
  `status` enum('active','disabled') DEFAULT 'active',
  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `username` (`username`),
  UNIQUE KEY `email` (`email`)
)
```

| #                        | Name                 | Type                       | Collation          | Attributes | Null | Default             | Comments | Extra          | Action                                                           |
|--------------------------|----------------------|----------------------------|--------------------|------------|------|---------------------|----------|----------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 id                 | int(11)                    |                    |            | No   | None                |          | AUTO_INCREMENT | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 full_name          | varchar(100)               | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 username           | varchar(50)                | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 email              | varchar(100)               | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 5 password           | varchar(255)               | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 6 organization_name  | varchar(100)               | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 7 job_title_position | varchar(100)               | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 8 created_at         | timestamp                  |                    |            | No   | current_timestamp() |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 9 status             | enum('active', 'disabled') | utf8mb4_general_ci |            | Yes  | active              |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 55organizers table

```

CREATE TABLE `judges` (
  `id` int NOT NULL AUTO_INCREMENT,
  `full_name` varchar(100) NOT NULL,
  `username` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(255) NOT NULL,
  `professional_title` varchar(100) DEFAULT NULL,
  `status` enum('active','disabled') DEFAULT 'active',
  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `username` (`username`),
  UNIQUE KEY `email` (`email`)
)

```

| #                        | Name                        | Type                      | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|-----------------------------|---------------------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b> 🔑               | int(11)                   |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>full_name</b>          | varchar(100)              | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>username</b> 🔑         | varchar(50)               | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>email</b> 🔑            | varchar(100)              | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>password</b>           | varchar(255)              | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>professional_title</b> | varchar(100)              | utf8mb4_general_ci |            | Yes  | NULL                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 7 <b>created_at</b>         | timestamp                 |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |
| <input type="checkbox"/> | 8 <b>status</b>             | enum('active','disabled') | utf8mb4_general_ci |            | Yes  | active              |          |                | Change  Drop  More |

Figure 56Judges table

```

CREATE TABLE `hackathons` (
  `id` int NOT NULL AUTO_INCREMENT,
  `name` varchar(200) NOT NULL,
  `description` text,
  `start_date` date NOT NULL,
  `end_date` date NOT NULL,
  `location` varchar(200) DEFAULT NULL,
  `rules` text,

```

```

`prizes` text,
`image_path` varchar(500) DEFAULT NULL,
`min_teams` int DEFAULT 1,
`max_teams` int DEFAULT 50,
`results_announced` tinyint(1) DEFAULT 0,
`results_announced_at` timestamp NULL DEFAULT NULL,
`submission_start` datetime DEFAULT NULL,
`submission_end` datetime DEFAULT NULL,
`judging_start` datetime DEFAULT NULL,
`judging_end` datetime DEFAULT NULL,
`result_announcement_at` datetime DEFAULT NULL,
`created_by` int DEFAULT NULL,
`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
`updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
PRIMARY KEY (`id`),
KEY `fk_hackathons_organizer` (`created_by`),
CONSTRAINT `fk_hackathons_organizer` FOREIGN KEY (`created_by`) REFERENCES
`organizers` (`id`) ON DELETE SET NULL
)

```

| #                        | Name                      | Type         | Collation          | Attributes | Null | Default             | Comments                      | Extra          | Action             |
|--------------------------|---------------------------|--------------|--------------------|------------|------|---------------------|-------------------------------|----------------|--------------------|
| <input type="checkbox"/> | 1 id                      | int(11)      |                    |            | No   | None                |                               | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 name                    | varchar(200) | utf8mb4_general_ci |            | No   | None                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 3 description             | text         | utf8mb4_general_ci |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 4 start_date              | date         |                    |            | No   | None                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 5 end_date                | date         |                    |            | No   | None                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 6 location                | varchar(200) | utf8mb4_general_ci |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 7 rules                   | text         | utf8mb4_general_ci |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 8 prizes                  | text         | utf8mb4_general_ci |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 9 image_path              | varchar(500) | utf8mb4_general_ci |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 10 created_by             | int(11)      |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 11 created_at             | timestamp    |                    |            | No   | current_timestamp() |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 12 updated_at             | timestamp    |                    |            | No   | current_timestamp() | ON UPDATE CURRENT_TIMESTAMP() |                | Change  Drop  More |
| <input type="checkbox"/> | 13 results_announced      | tinyint(1)   |                    |            | Yes  | 0                   |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 14 results_announced_at   | timestamp    |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 15 min_teams              | int(11)      |                    |            | Yes  | 1                   |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 16 max_teams              | int(11)      |                    |            | Yes  | 50                  |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 17 submission_start       | datetime     |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 18 submission_end         | datetime     |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 19 judging_start          | datetime     |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 20 judging_end            | datetime     |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |
| <input type="checkbox"/> | 21 result_announcement_at | datetime     |                    |            | Yes  | NULL                |                               |                | Change  Drop  More |

Figure 57hackathons table

```

CREATE TABLE `judge_hackathons` (
  `id` int NOT NULL AUTO_INCREMENT,
  `judge_id` int NOT NULL,
  `hackathon_id` int NOT NULL,
  `status` enum('pending','accepted','rejected') DEFAULT 'pending',
  `assigned_at` timestamp DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `unique_judge_hackathon` (`judge_id`,`hackathon_id`),
  KEY `fk_judge_hackathons_judge` (`judge_id`),
  KEY `fk_judge_hackathons_hackathon` (`hackathon_id`),
  CONSTRAINT `fk_judge_hackathons_judge` FOREIGN KEY (`judge_id`) REFERENCES
`judges` (`id`) ON DELETE CASCADE,
  CONSTRAINT `fk_judge_hackathons_hackathon` FOREIGN KEY (`hackathon_id`)
REFERENCES `hackathons` (`id`) ON DELETE CASCADE
)

```

| #                        | Name           | Type                                    | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|----------------|-----------------------------------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 id           | int(11)                                 |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 judge_id     | int(11)                                 |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 hackathon_id | int(11)                                 |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 status       | enum('pending', 'accepted', 'rejected') | utf8mb4_general_ci |            | Yes  | pending             |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 assigned_at  | timestamp                               |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 58Judge\_hackathons table

```

CREATE TABLE `teams` (
  `id` int NOT NULL AUTO_INCREMENT,
  `name` varchar(200) NOT NULL,
  `description` text,
  `hackathon_id` int NOT NULL,
  `max_participants` int NOT NULL DEFAULT 5,
  `skills_required` text,
  `team_leader_id` int NOT NULL,
  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,
  `updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `fk_teams_hackathon` (`hackathon_id`),
  KEY `fk_teams_leader` (`team_leader_id`),
  CONSTRAINT `fk_teams_hackathon` FOREIGN KEY (`hackathon_id`) REFERENCES
`hackathons` (`id`) ON DELETE CASCADE,
  CONSTRAINT `fk_teams_leader` FOREIGN KEY (`team_leader_id`) REFERENCES
`participants` (`id`) ON DELETE CASCADE
)

```

| #                        | Name               | Type         | Collation          | Attributes | Null | Default             | Comments | Extra                         | Action             |
|--------------------------|--------------------|--------------|--------------------|------------|------|---------------------|----------|-------------------------------|--------------------|
| <input type="checkbox"/> | 1 id               | int(11)      |                    |            | No   | None                |          | AUTO_INCREMENT                | Change  Drop  More |
| <input type="checkbox"/> | 2 name             | varchar(200) | utf8mb4_general_ci |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 3 description      | text         | utf8mb4_general_ci |            | Yes  | NULL                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 4 hackathon_id     | int(11)      |                    |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 5 max_participants | int(11)      |                    |            | No   | 5                   |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 6 skills_required  | text         | utf8mb4_general_ci |            | Yes  | NULL                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 7 team_leader_id   | int(11)      |                    |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 8 created_at       | timestamp    |                    |            | No   | current_timestamp() |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 9 updated_at       | timestamp    |                    |            | No   | current_timestamp() |          | ON UPDATE CURRENT_TIMESTAMP() | Change  Drop  More |

Figure 59 Teams table

```

CREATE TABLE `team_members` (

  `id` int NOT NULL AUTO_INCREMENT,

  `team_id` int NOT NULL,

  `participant_id` int NOT NULL,

  `joined_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  PRIMARY KEY (`id`),

  UNIQUE KEY `unique_team_participant` (`team_id`,`participant_id`),

  KEY `fk_team_members_team` (`team_id`),

  KEY `fk_team_members_participant` (`participant_id`),

  CONSTRAINT `fk_team_members_team` FOREIGN KEY (`team_id`) REFERENCES
`teams` (`id`) ON DELETE CASCADE,

  CONSTRAINT `fk_team_members_participant` FOREIGN KEY (`participant_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE

)

```

| #                        | Name             | Type      | Collation | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|------------------|-----------|-----------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 id             | int(11)   |           |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 team_id        | int(11)   |           |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 participant_id | int(11)   |           |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 joined_at      | timestamp |           |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 60team\_members table

```

CREATE TABLE `team_join_requests` (

  `id` int NOT NULL AUTO_INCREMENT,

  `team_id` int NOT NULL,

  `participant_id` int NOT NULL,

  `status` enum('pending','approved','rejected') DEFAULT 'pending',

  `requested_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  `responded_at` timestamp NULL DEFAULT NULL,

  PRIMARY KEY (`id`),

  UNIQUE KEY `unique_team_participant_request` (`team_id`,`participant_id`),

  KEY `fk_team_join_requests_team` (`team_id`),

  KEY `fk_team_join_requests_participant` (`participant_id`),

  CONSTRAINT `fk_team_join_requests_team` FOREIGN KEY (`team_id`) REFERENCES
`teams` (`id`) ON DELETE CASCADE,

  CONSTRAINT `fk_team_join_requests_participant` FOREIGN KEY (`participant_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE

)

```



| #                        | Name                    | Type                                    | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|-------------------------|-----------------------------------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b>             | int(11)                                 |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>team_id</b>        | int(11)                                 |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>participant_id</b> | int(11)                                 |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>status</b>         | enum('pending', 'approved', 'rejected') | utf8mb4_general_ci |            | Yes  | pending             |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>requested_at</b>   | timestamp                               |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>responded_at</b>   | timestamp                               |                    |            | Yes  | NULL                |          |                | Change  Drop  More |

Figure 61 Team\_join\_requests table

```

CREATE TABLE `projects` (

  `id` int NOT NULL AUTO_INCREMENT,

  `title` varchar(200) NOT NULL,

  `description` text,

  `demo_link` varchar(500) DEFAULT NULL,

  `team_id` int NOT NULL,

  `leader_id` int NOT NULL,

  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  `updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

  PRIMARY KEY (`id`),

  KEY `fk_projects_team` (`team_id`),

  KEY `fk_projects_leader` (`leader_id`),

  CONSTRAINT `fk_projects_team` FOREIGN KEY (`team_id`) REFERENCES `teams` (`id`)
ON DELETE CASCADE,

```

```

CONSTRAINT `fk_projects_leader` FOREIGN KEY (`leader_id`) REFERENCES
`participants` (`id`) ON DELETE CASCADE
)

```

| #                        | Name                 | Type         | Collation          | Attributes | Null | Default             | Comments | Extra                         | Action             |
|--------------------------|----------------------|--------------|--------------------|------------|------|---------------------|----------|-------------------------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b>          | int(11)      |                    |            | No   | None                |          | AUTO_INCREMENT                | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>title</b>       | varchar(200) | utf8mb4_general_ci |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>description</b> | text         | utf8mb4_general_ci |            | Yes  | NULL                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>demo_link</b>   | varchar(500) | utf8mb4_general_ci |            | Yes  | NULL                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>team_id</b>     | int(11)      |                    |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>leader_id</b>   | int(11)      |                    |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 7 <b>created_at</b>  | timestamp    |                    |            | No   | current_timestamp() |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 8 <b>updated_at</b>  | timestamp    |                    |            | No   | current_timestamp() |          | ON UPDATE CURRENT_TIMESTAMP() | Change  Drop  More |

Figure 62 Projects table

```

CREATE TABLE `project_files` (

  `id` int NOT NULL AUTO_INCREMENT,

  `project_id` int NOT NULL,

  `file_name` varchar(255) NOT NULL,

  `original_name` varchar(255) NOT NULL,

  `file_path` varchar(500) NOT NULL,

  `file_type` enum('pdf','image','video','archive','presentation','other') NOT NULL,

  `file_size` int NOT NULL,

  `mime_type` varchar(100) NOT NULL,

  `uploaded_by` int NOT NULL,

  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  PRIMARY KEY (`id`),

```

KEY `fk\_project\_files\_project` (`project\_id`),

KEY `fk\_project\_files\_uploaded\_by` (`uploaded\_by`),

CONSTRAINT `fk\_project\_files\_project` FOREIGN KEY (`project\_id`) REFERENCES  
`projects` (`id`) ON DELETE CASCADE,

CONSTRAINT `fk\_project\_files\_uploaded\_by` FOREIGN KEY (`uploaded\_by`)  
REFERENCES `participants` (`id`) ON DELETE CASCADE

)

| #                        | Name                   | Type                                              | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|------------------------|---------------------------------------------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b>            | int(11)                                           |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>project_id</b>    | int(11)                                           |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>file_name</b>     | varchar(255)                                      | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>original_name</b> | varchar(255)                                      | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>file_path</b>     | varchar(500)                                      | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>file_type</b>     | enum('pdf','image','video','archive','presenta... | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 7 <b>file_size</b>     | int(11)                                           |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 8 <b>mime_type</b>     | varchar(100)                                      | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 9 <b>uploaded_by</b>   | int(11)                                           |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 10 <b>created_at</b>   | timestamp                                         |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 63Project\_files table

CREATE TABLE `conversations` (

`id` int NOT NULL AUTO\_INCREMENT,

`participant1\_id` int NOT NULL,

`participant2\_id` int NOT NULL,

`created\_at` timestamp DEFAULT CURRENT\_TIMESTAMP,

`updated\_at` timestamp DEFAULT CURRENT\_TIMESTAMP ON UPDATE  
CURRENT\_TIMESTAMP,

PRIMARY KEY (`id`),

UNIQUE KEY `unique\_conversation` (`participant1\_id`,`participant2\_id`),

```

KEY `fk_conversations_participant1` (`participant1_id`),

KEY `fk_conversations_participant2` (`participant2_id`),

CONSTRAINT `fk_conversations_participant1` FOREIGN KEY (`participant1_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE,

CONSTRAINT `fk_conversations_participant2` FOREIGN KEY (`participant2_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE

)

```

| #                        | Name              | Type      | Collation | Attributes | Null | Default             | Comments | Extra                         | Action             |
|--------------------------|-------------------|-----------|-----------|------------|------|---------------------|----------|-------------------------------|--------------------|
| <input type="checkbox"/> | 1 id              | int(11)   |           |            | No   | None                |          | AUTO_INCREMENT                | Change  Drop  More |
| <input type="checkbox"/> | 2 participant1_id | int(11)   |           |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 3 participant2_id | int(11)   |           |            | No   | None                |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 4 created_at      | timestamp |           |            | No   | current_timestamp() |          |                               | Change  Drop  More |
| <input type="checkbox"/> | 5 updated_at      | timestamp |           |            | No   | current_timestamp() |          | ON UPDATE CURRENT_TIMESTAMP() | Change  Drop  More |

Figure 64 Conversations table

```

CREATE TABLE `messages` (

`id` int NOT NULL AUTO_INCREMENT,

`conversation_id` int NOT NULL,

`sender_id` int NOT NULL,

`message` text NOT NULL,

`message_type` enum('text','file','image') DEFAULT 'text',

`file_path` varchar(500) DEFAULT NULL,

`is_read` tinyint(1) DEFAULT 0,

`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

PRIMARY KEY (`id`),

```

```

KEY `fk_messages_conversation` (`conversation_id`),

KEY `fk_messages_sender` (`sender_id`),

CONSTRAINT `fk_messages_conversation` FOREIGN KEY (`conversation_id`)
REFERENCES `conversations` (`id`) ON DELETE CASCADE,

CONSTRAINT `fk_messages_sender` FOREIGN KEY (`sender_id`) REFERENCES
`participants` (`id`) ON DELETE CASCADE
)

```

| #                        | Name                     | Type                          | Collation          | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|--------------------------|-------------------------------|--------------------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b>              | int(11)                       |                    |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>conversation_id</b> | int(11)                       |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>sender_id</b>       | int(11)                       |                    |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>message</b>         | text                          | utf8mb4_general_ci |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>message_type</b>    | enum('text', 'file', 'image') | utf8mb4_general_ci |            | Yes  | text                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6 <b>file_path</b>       | varchar(500)                  | utf8mb4_general_ci |            | Yes  | NULL                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 7 <b>is_read</b>         | tinyint(1)                    |                    |            | Yes  | 0                   |          |                | Change  Drop  More |
| <input type="checkbox"/> | 8 <b>created_at</b>      | timestamp                     |                    |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 65Messages table

Messages table

```

CREATE TABLE `team_chats` (

`id` int NOT NULL AUTO_INCREMENT,

`team_id` int NOT NULL,

`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

`updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

PRIMARY KEY (`id`),

UNIQUE KEY `unique_team_chat` (`team_id`),

```

```

KEY `fk_team_chats_team` (`team_id`),

CONSTRAINT `fk_team_chats_team` FOREIGN KEY (`team_id`) REFERENCES `teams`
(`id`) ON DELETE CASCADE

)

```

| #                        | Name         | Type      | Collation | Attributes | Null | Default             | Comments | Extra                         | Action                                                           |
|--------------------------|--------------|-----------|-----------|------------|------|---------------------|----------|-------------------------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 id         | int(11)   |           |            | No   | None                |          | AUTO_INCREMENT                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 team_id    | int(11)   |           |            | No   | None                |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 created_at | timestamp |           |            | No   | current_timestamp() |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 updated_at | timestamp |           |            | No   | current_timestamp() |          | ON UPDATE CURRENT_TIMESTAMP() | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 66Team\_chats table

```

CREATE TABLE `team_chat_messages` (

`id` int NOT NULL AUTO_INCREMENT,

`team_chat_id` int NOT NULL,

`sender_id` int NOT NULL,

`message` text NOT NULL,

`message_type` enum('text','file','image') DEFAULT 'text',

`file_path` varchar(500) DEFAULT NULL,

`is_read` tinyint(1) DEFAULT 0,

`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

PRIMARY KEY (`id`),

KEY `fk_team_chat_messages_chat` (`team_chat_id`),

KEY `fk_team_chat_messages_sender` (`sender_id`),

```

```
CONSTRAINT `fk_team_chat_messages_chat` FOREIGN KEY (`team_chat_id`)
REFERENCES `team_chats` (`id`) ON DELETE CASCADE,
```

```
CONSTRAINT `fk_team_chat_messages_sender` FOREIGN KEY (`sender_id`)
REFERENCES `participants` (`id`) ON DELETE CASCADE
```

)

| #                        | Name           | Type                          | Collation          | Attributes | Null | Default             | Comments | Extra          | Action                                                           |
|--------------------------|----------------|-------------------------------|--------------------|------------|------|---------------------|----------|----------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 id           | int(11)                       |                    |            | No   | None                |          | AUTO_INCREMENT | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 team_chat_id | int(11)                       |                    |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 sender_id    | int(11)                       |                    |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 message      | text                          | utf8mb4_general_ci |            | No   | None                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 5 message_type | enum('text', 'file', 'image') | utf8mb4_general_ci |            | Yes  | text                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 6 file_path    | varchar(500)                  | utf8mb4_general_ci |            | Yes  | NULL                |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 7 is_read      | tinyint(1)                    |                    |            | Yes  | 0                   |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 8 created_at   | timestamp                     |                    |            | No   | current_timestamp() |          |                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 67Team\_chat\_messages table

```
CREATE TABLE `project_evaluations` (
  `id` int NOT NULL AUTO_INCREMENT,
  `project_id` int NOT NULL,
  `hackathon_id` int NOT NULL,
  `judge_id` int NOT NULL,
  `innovation` int NOT NULL,
  `implementation_quality` int NOT NULL,
  `design_usability` int NOT NULL,
  `impact` int NOT NULL,
  `problem_relevance_clarity` int NOT NULL,
  `functionality_features_completeness` int NOT NULL,
```

```

`feasibility_scalability` int NOT NULL,

`team_collaboration_presentation_quality` int NOT NULL,

`technical_complexity` int NOT NULL,

`testing_reliability` int NOT NULL,

`total_score` int GENERATED ALWAYS AS ((`innovation` + `implementation_quality` +
`design_usability` + `impact` + coalesce(`problem_relevance_clarity`,0) +
coalesce(`functionality_features_completeness`,0) + coalesce(`feasibility_scalability`,0) +
coalesce(`team_collaboration_presentation_quality`,0) + coalesce(`technical_complexity`,0) +
coalesce(`testing_reliability`,0))) STORED,

`feedback` text,

`evaluation_status` enum('pending','completed') DEFAULT 'completed',

`created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

`updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

PRIMARY KEY (`id`),

UNIQUE KEY `unique_judge_project_evaluation` (`judge_id`,`project_id`),

KEY `fk_project_evaluations_project` (`project_id`),

KEY `fk_project_evaluations_hackathon` (`hackathon_id`),

KEY `fk_project_evaluations_judge` (`judge_id`),

CONSTRAINT `fk_project_evaluations_project` FOREIGN KEY (`project_id`)
REFERENCES `projects` (`id`) ON DELETE CASCADE,

CONSTRAINT `fk_project_evaluations_hackathon` FOREIGN KEY (`hackathon_id`)
REFERENCES `hackathons` (`id`) ON DELETE CASCADE,

```



CONSTRAINT `fk\_project\_evaluations\_judge` FOREIGN KEY (`judge\_id`) REFERENCES  
 `judges` (`id`) ON DELETE CASCADE,

CONSTRAINT `chk\_innovation` CHECK ((`innovation` >= 1 and `innovation` <= 10)),

CONSTRAINT `chk\_implementation\_quality` CHECK ((`implementation\_quality` >= 1 and  
 `implementation\_quality` <= 10)),

CONSTRAINT `chk\_design\_usability` CHECK ((`design\_usability` >= 1 and  
 `design\_usability` <= 10)),

CONSTRAINT `chk\_impact` CHECK ((`impact` >= 1 and `impact` <= 10)),

CONSTRAINT `chk\_problem\_relevance\_clarity` CHECK ((`problem\_relevance\_clarity` >= 1  
 and `problem\_relevance\_clarity` <= 10)),

CONSTRAINT `chk\_functionality\_features\_completeness` CHECK  
 ((`functionality\_features\_completeness` >= 1 and `functionality\_features\_completeness` <= 10)),

CONSTRAINT `chk\_feasibility\_scalability` CHECK ((`feasibility\_scalability` >= 1 and  
 `feasibility\_scalability` <= 10)),

CONSTRAINT `chk\_team\_collaboration\_presentation\_quality` CHECK  
 ((`team\_collaboration\_presentation\_quality` >= 1 and `team\_collaboration\_presentation\_quality`  
 <= 10)),

CONSTRAINT `chk\_technical\_complexity` CHECK ((`technical\_complexity` >= 1 and  
 `technical\_complexity` <= 10)),

CONSTRAINT `chk\_testing\_reliability` CHECK ((`testing\_reliability` >= 1 and  
 `testing\_reliability` <= 10))

| #                        | Name                                       | Type                        | Collation          | Attributes | Null | Default             | Comments                      | Extra            | Action             |
|--------------------------|--------------------------------------------|-----------------------------|--------------------|------------|------|---------------------|-------------------------------|------------------|--------------------|
| <input type="checkbox"/> | 1 id                                       | int(11)                     |                    |            | No   | None                |                               | AUTO_INCREMENT   | Change  Drop  More |
| <input type="checkbox"/> | 2 project_id                               | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 3 hackathon_id                             | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 4 judge_id                                 | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 5 innovation                               | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 6 implementation_quality                   | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 7 design_usability                         | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 8 impact                                   | int(11)                     |                    |            | No   | None                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 9 feedback                                 | text                        | utf8mb4_general_ci |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 10 evaluation_status                       | enum('pending','completed') | utf8mb4_general_ci |            | Yes  | completed           |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 11 created_at                              | timestamp                   |                    |            | No   | current_timestamp() |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 12 updated_at                              | timestamp                   |                    |            | No   | current_timestamp() | ON UPDATE CURRENT_TIMESTAMP() |                  | Change  Drop  More |
| <input type="checkbox"/> | 13 problem_relevance_clarity               | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 14 functionality_features_completeness     | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 15 feasibility_scalability                 | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 16 team_collaboration_presentation_quality | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 17 technical_complexity                    | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 18 testing_reliability                     | int(11)                     |                    |            | Yes  | NULL                |                               |                  | Change  Drop  More |
| <input type="checkbox"/> | 19 total_score                             | int(11)                     |                    |            | Yes  | NULL                |                               | STORED GENERATED | Change  Drop  More |

Figure 68Project\_evaluations table

```

CREATE TABLE `project_rankings` (

  `id` int NOT NULL AUTO_INCREMENT,

  `project_id` int NOT NULL,

  `ranking_position` int NOT NULL,

  `ranking_status` enum('draft','confirmed','final') DEFAULT 'draft',

  `notes` text,

  `created_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  `updated_at` timestamp DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

  PRIMARY KEY (`id`),

  UNIQUE KEY `unique_project_ranking` (`project_id`),

  UNIQUE KEY `unique_position_per_hackathon` (`ranking_position`,`project_id`),

  KEY `fk_project_rankings_project` (`project_id`),

```

```
CONSTRAINT `fk_project_rankings_project` FOREIGN KEY (`project_id`) REFERENCES
`projects` (`id`) ON DELETE CASCADE,
```

```
CONSTRAINT `chk_ranking_position` CHECK ((`ranking_position` >= 1))

)
```

| #                        | Name               | Type                                | Collation          | Attributes | Null | Default             | Comments | Extra                         | Action                                                           |
|--------------------------|--------------------|-------------------------------------|--------------------|------------|------|---------------------|----------|-------------------------------|------------------------------------------------------------------|
| <input type="checkbox"/> | 1 id               | int(11)                             |                    |            | No   | None                |          | AUTO_INCREMENT                | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 2 project_id       | int(11)                             |                    |            | No   | None                |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 3 ranking_position | int(11)                             |                    |            | No   | None                |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 4 ranking_status   | enum('draft', 'confirmed', 'final') | utf8mb4_general_ci |            | Yes  | draft               |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 5 notes            | text                                | utf8mb4_general_ci |            | Yes  | NULL                |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 6 created_at       | timestamp                           |                    |            | No   | current_timestamp() |          |                               | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |
| <input type="checkbox"/> | 7 updated_at       | timestamp                           |                    |            | No   | current_timestamp() |          | ON UPDATE CURRENT_TIMESTAMP() | <a href="#">Change</a> <a href="#">Drop</a> <a href="#">More</a> |

Figure 69Project\_rankings table

```
CREATE TABLE `judge_team_assignments` (

  `id` int NOT NULL AUTO_INCREMENT,

  `judge_id` int NOT NULL,

  `team_id` int NOT NULL,

  `hackathon_id` int NOT NULL,

  `assigned_at` timestamp DEFAULT CURRENT_TIMESTAMP,

  PRIMARY KEY (`id`),

  UNIQUE KEY `unique_assignment` (`judge_id`,`team_id`,`hackathon_id`),

  KEY `fk_judge_team_assignments_judge` (`judge_id`),

  KEY `fk_judge_team_assignments_team` (`team_id`),

  KEY `fk_judge_team_assignments_hackathon` (`hackathon_id`),

  CONSTRAINT `fk_judge_team_assignments_judge` FOREIGN KEY (`judge_id`)
REFERENCES `judges` (`id`) ON DELETE CASCADE,
```

```

    CONSTRAINT `fk_judge_team_assignments_team` FOREIGN KEY (`team_id`)
REFERENCES `teams` (`id`) ON DELETE CASCADE,

    CONSTRAINT `fk_judge_team_assignments_hackathon` FOREIGN KEY (`hackathon_id`)
REFERENCES `hackathons` (`id`) ON DELETE CASCADE
)

```

| #                        | Name                    | Type      | Collation | Attributes | Null | Default             | Comments | Extra          | Action             |
|--------------------------|-------------------------|-----------|-----------|------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 <b>id</b> 🔑           | int(11)   |           |            | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 <b>judge_id</b> 🔑     | int(11)   |           |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 <b>team_id</b> 🔑      | int(11)   |           |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 <b>hackathon_id</b> 🔑 | int(11)   |           |            | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5 <b>assigned_at</b>    | timestamp |           |            | No   | current_timestamp() |          |                | Change  Drop  More |

Figure 70 Judge\_team\_assignments table

```

CREATE TABLE `notifications` (

    `id` int NOT NULL AUTO_INCREMENT,

    `user_id` int NOT NULL,

    `user_type` enum('participant','organizer','judge','admin') NOT NULL,

    `message` text NOT NULL,

    `type` varchar(50) NOT NULL,

    `is_read` tinyint(1) DEFAULT 0,

    `created_at` datetime DEFAULT CURRENT_TIMESTAMP,

    PRIMARY KEY (`id`)

)

```

| #                        | Name | Type              | Collation                                          | Attributes         | Null | Default             | Comments | Extra          | Action             |
|--------------------------|------|-------------------|----------------------------------------------------|--------------------|------|---------------------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1    | <b>id</b>         | int(11)                                            |                    | No   | None                |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2    | <b>user_id</b>    | int(11)                                            |                    | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3    | <b>user_type</b>  | enum('participant', 'organizer', 'judge', 'admin') | utf8mb4_general_ci | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4    | <b>message</b>    | text                                               | utf8mb4_general_ci | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 5    | <b>type</b>       | varchar(50)                                        | utf8mb4_general_ci | No   | None                |          |                | Change  Drop  More |
| <input type="checkbox"/> | 6    | <b>is_read</b>    | tinyint(1)                                         |                    | Yes  | 0                   |          |                | Change  Drop  More |
| <input type="checkbox"/> | 7    | <b>created_at</b> | datetime                                           |                    | Yes  | current_timestamp() |          |                | Change  Drop  More |

Figure 71 Notifications table

```

CREATE TABLE `notifications` (
  `id` int NOT NULL AUTO_INCREMENT,
  `user_id` int NOT NULL,
  `user_type` enum('participant','organizer','judge','admin') NOT NULL,
  `message` text NOT NULL,
  `type` varchar(50) NOT NULL,
  `is_read` tinyint(1) DEFAULT 0,
  `read_at` datetime DEFAULT NULL,
  `related_entity` varchar(50) DEFAULT NULL,
  `related_id` int DEFAULT NULL,
  `created_at` datetime DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_notifications_user` (`user_id`,`user_type`),
  KEY `idx_notifications_is_read` (`is_read`),
  KEY `idx_notifications_created_at` (`created_at`),
  CONSTRAINT `chk_notification_type` CHECK (char_length(`type`) >= 2)
);

```

## 4. DEVELOPMENT

### 4.1 Tools/ technologies selected for implementation.

**HTML:** HTML formed the foundation for structuring and organizing the web pages of the HackSynk platform.

**Example:** Structuring the home page of HackSynk with a navigation menu, a header section featuring the platform's tagline "The Home for Hackathons", and various content sections such as "Active and Upcoming Hackathons", "About Us", and "Contact Us."

**CSS:** CSS was utilized to enhance the visual design and layout of HackSynk's web pages. It ensured the user interface was aesthetically pleasing and easy to navigate by styling the HTML components.

**Example:** CSS was used to ensure consistent fonts and colors across all pages for a unified look, including styling for hackathon cards, dashboard layouts, and chat interfaces.

**JavaScript:** JavaScript was integrated to provide interactive features and improve functionality in the HackSynk system. It facilitated input validation, managed form submissions, and dynamically displayed available data.

**Example:** JavaScript was used to filter hackathons based on their status "Upcoming", "Active", or "Ended", filter users by role (Participant, Organizer, Judge, Admin), and filter teams based on hackathon selection in the participant dashboard.

**PHP:** PHP served as the backend scripting language, handling the server-side operations of the HackSynk platform. It managed tasks such as processing user data, storing it in the database, and retrieving data.

**Example:** Processing user registration on HackSynk. When a new participant submits the registration form, PHP collects the form data (full name, username, email, password, city/country), validates it (e.g., checks if the username or email already exists, validates email format, ensures password meets minimum length requirements), and securely stores it in the database with hashed passwords. It also handles authentication and session management for different user roles (participants, organizers, judges, and administrators).

**PhpMyAdmin:** phpMyAdmin, a web-based tool, was employed to oversee and manage the MySQL databases used by HackSynk. It assisted in creating the database structure and organizing user and data effectively.

**Example:** Using phpMyAdmin to create a relational database for HackSynk. The database contains tables for users (participants, organizers, judges, admins), teams, hackathons, projects, evaluations, chat messages, and notifications. For instance, an admin might use phpMyAdmin to query the database and view all active hackathons or to add a new judge assignment to a hackathon.

**XAMPP:** XAMPP provided a local development environment, combining tools like Apache, MySQL, and PHP. It was instrumental in running and testing the HackSynk system locally, enabling developers to identify and resolve issues before deploying the application on a live server.

**Example:** Running the HackSynk platform locally on a developer's computer to test new features such as real-time chat functionality, project submission with file uploads, or judge evaluation system. XAMPP provides the necessary environment (Apache server and MySQL database) for the developer to simulate real-world scenarios and fix any bugs before deploying to a live server.

## 4.2 Web system navigation design

HackSynk is a comprehensive hackathon management platform that provides role-based navigation for four distinct user types: Participants who form teams and submit projects, Organizers who create and manage hackathon events, Judges who evaluate submissions using a 10-criteria scoring system, and Admins who oversee user management and system operations. The navigation structure follows the complete hackathon lifecycle from event discovery and team formation through project submission, evaluation, and results announcement, with integrated communication features including direct messaging, team chats, and system notifications. Each role has a dedicated dashboard with hierarchical navigation paths that reflect their specific responsibilities, enabling seamless interaction with hackathons, teams, projects, evaluations, and rankings while maintaining clear separation of concerns and workflow-based organization.





WEB SYSTEM NAVIGATION DIAGRAM

4.3 User interface design

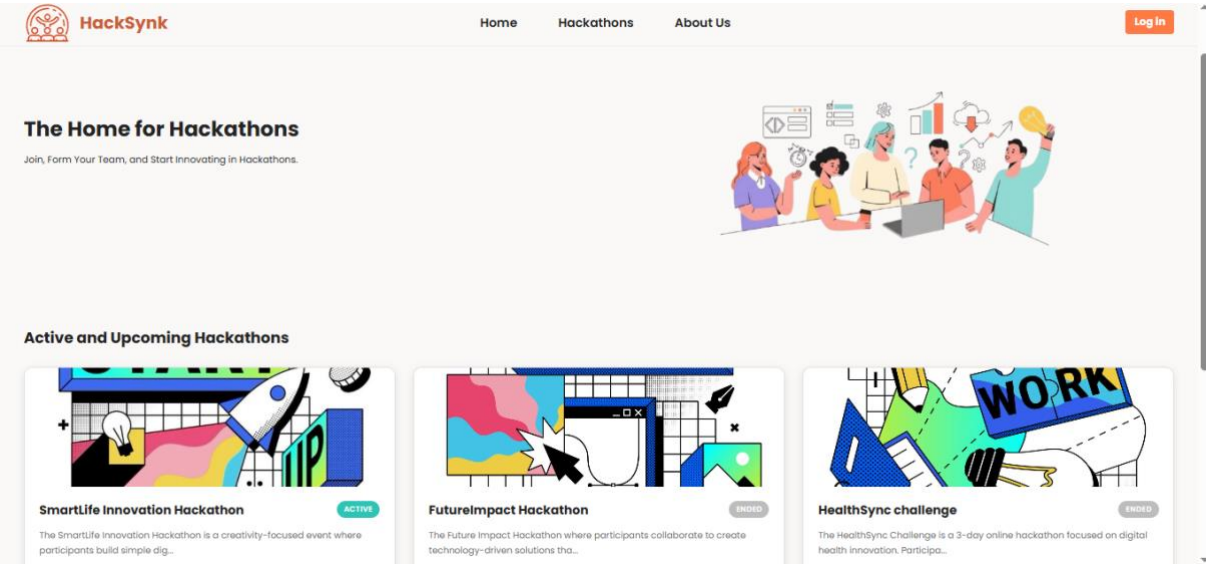


Figure 72Home Page

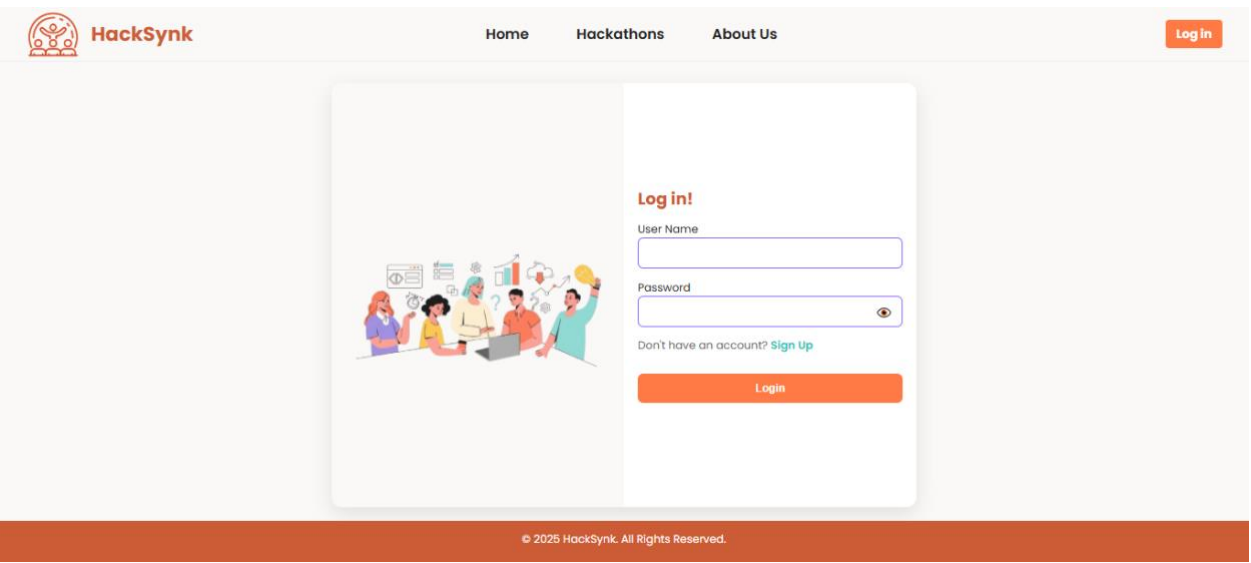


Figure 73Login Page

**HackSynk**

[Home](#)[Hackathons](#)[About Us](#)[Log In](#)



### Create Your Account

[Already have an account? \[Login\]\(#\)](#)

[Register](#)

Figure 74Participant Signup Page

**HackSynk**

[Home](#)[Hackathons](#)[About Us](#)[Log In](#)



### Create Your Account

[Already have an account? \[Login\]\(#\)](#)

[Register](#)

[Start](#)

Figure 75Organizer Signup Page



[Home](#)
[Hackathons](#)
[About Us](#)
[Log In](#)



### Create Your Account

Already have an account? [Login](#)

Register

Figure 76Judge Signup Page



[Home](#)
[Hackathons](#)
[About Us](#)

M

mariam  
Participant

Profile

Create Team

Join Team

Submit Project

Results

Team Chat

Chat Messages

Log Out

#### General Information

USERNAME:  
mariam1

CITY/COUNTRY:  
alshsa

YEARS OF EXPERIENCE:  
3 years

#### Education

EDUCATIONAL QUALIFICATION:  
Bachelor's Degree in Computer Science

SPECIALIZATION:  
Software Engineering, Artificial Intelligence

UNIVERSITY:  
Imam Abdulrahman Bin Faisal University

GRADUATION YEAR:  
2022

CERTIFICATES:

- AWS Cloud Practitioner Certification
- Machine Learning by Stanford (Coursera)
- Python for Data Science - IBM Skills Network
- Advanced JavaScript and React - Udemy

#### Skills

CSS

Intermediate

HTML

Intermediate

JavaScript

Intermediate

Node.js

Intermediate

PHP

Advanced

Python

Beginner

SQL

Intermediate

Figure 77Participant Profile

**Edit Profile**

Full Name \*  
mariam

Email \*  
mariam@gmail.com

Username \*  
mariam1

New Password (leave blank to keep current)

Confirm New Password

City/Country  
alaha

Educational Qualification  
Bachelor's Degree in Computer Science

Specialization  
Software Engineering, Artificial Intelligence

Figure 78 Edit Participant Profile

**HackSynk** Home Hackathons About Us

**Create Team**

Select Hackathon \*  
Choose a hackathon...

Team Name \*  
Enter your team name

Team Description  
Describe your team's goals and skills...

Maximum Participants \*  
5 members

Skills Required  
Enter skill name Beginner ADD SKILL

Reset CREATE TEAM

Log Out

Figure 79 Create Team

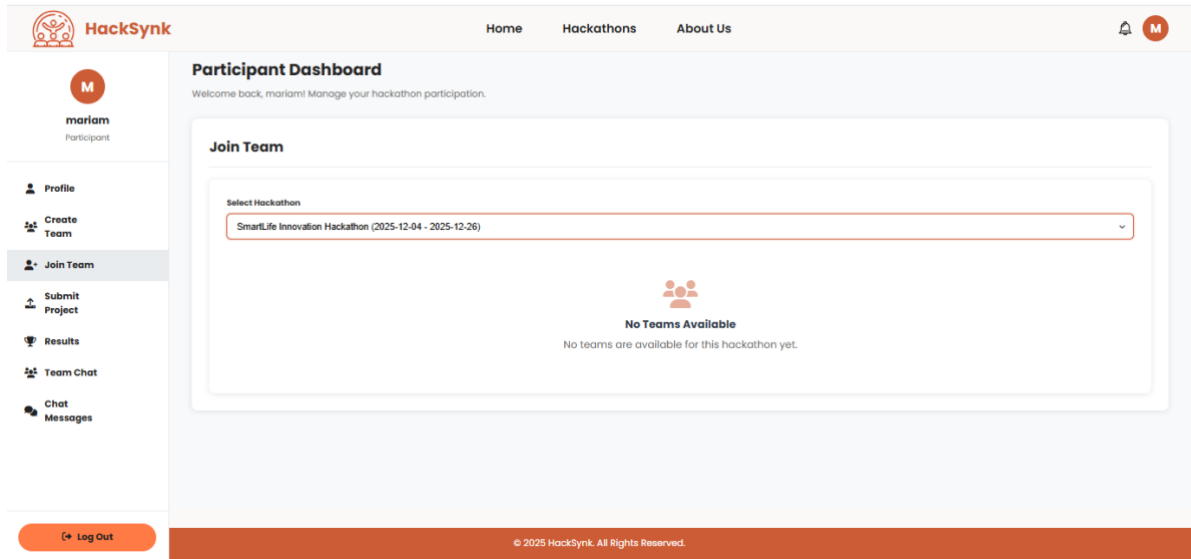


Figure 80Join Team

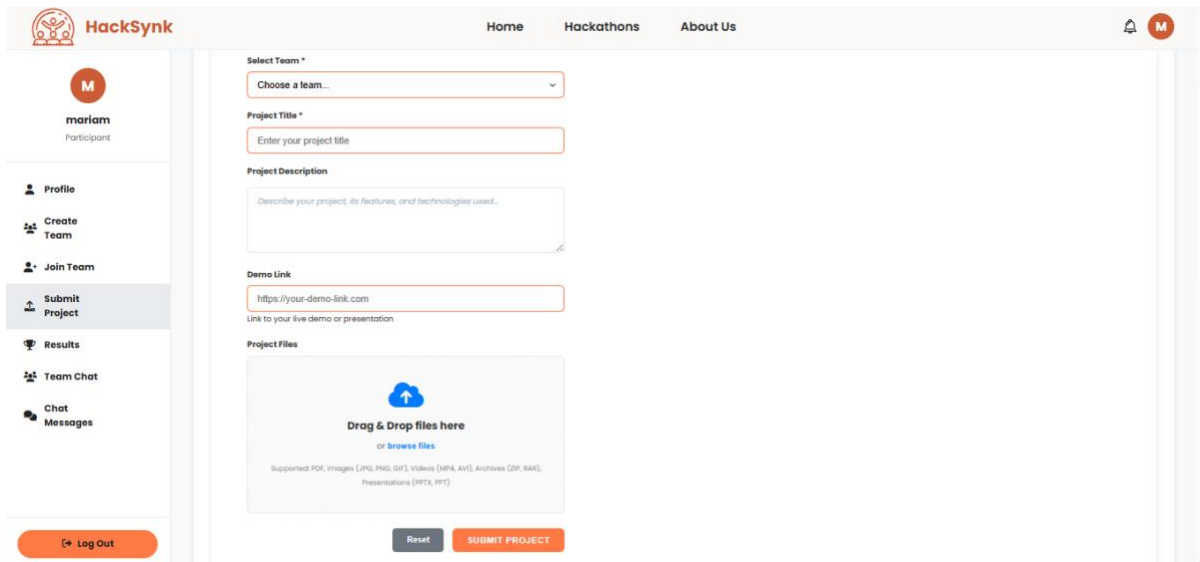


Figure 81Submit Project

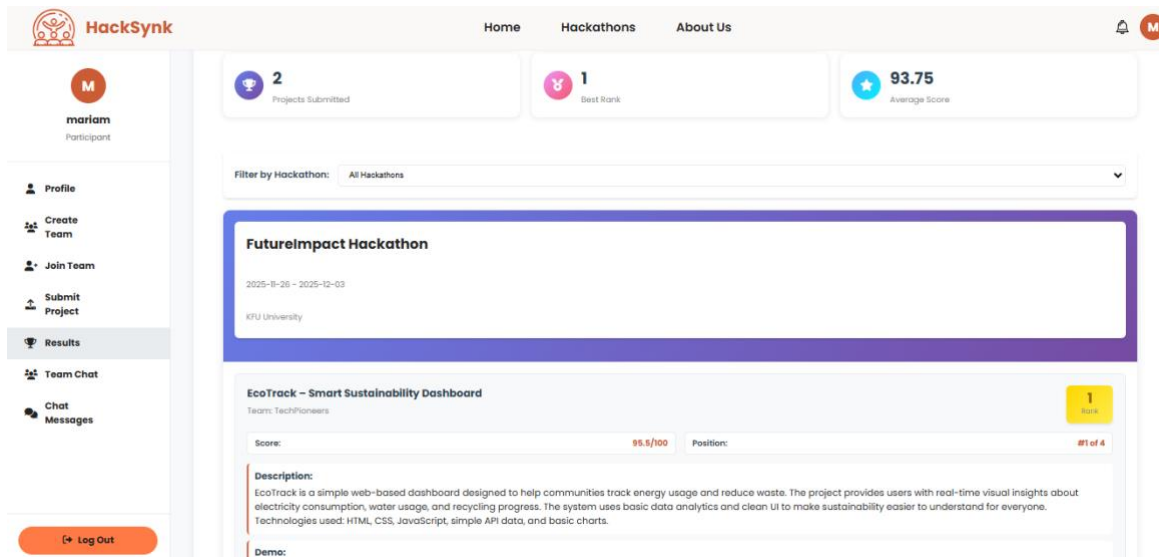


Figure 82Results Page

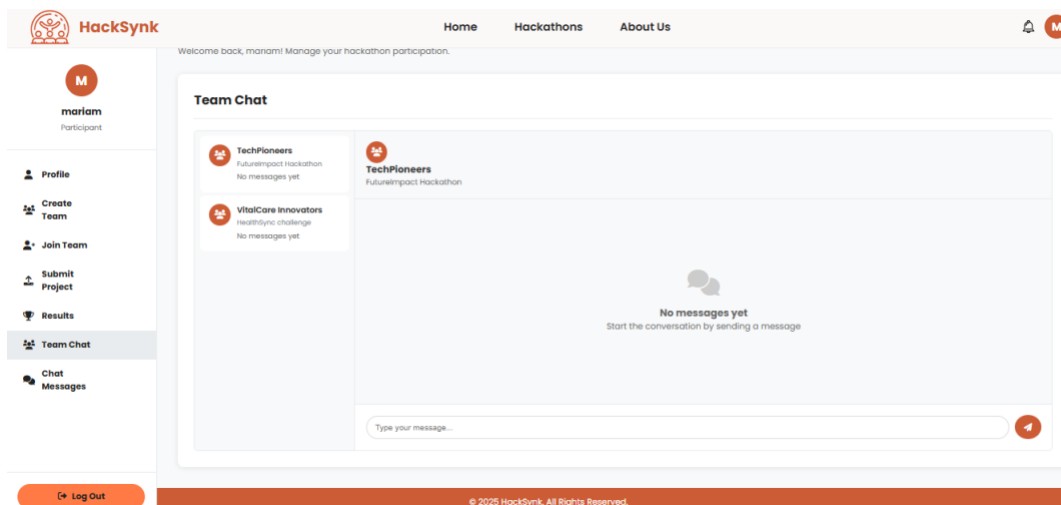


Figure 83Team Chat

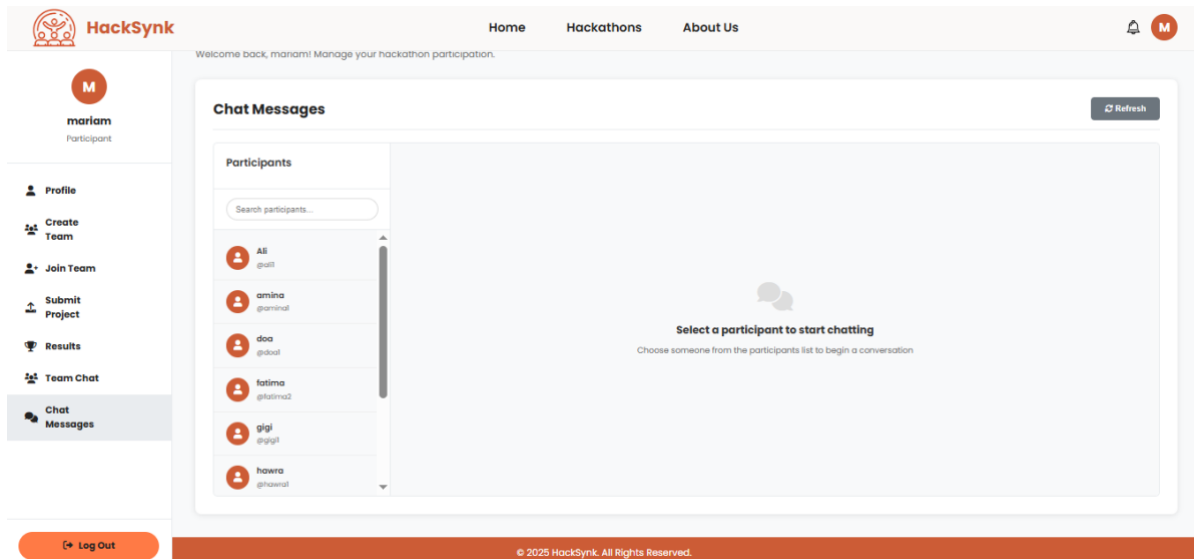


Figure 84Chat Messages

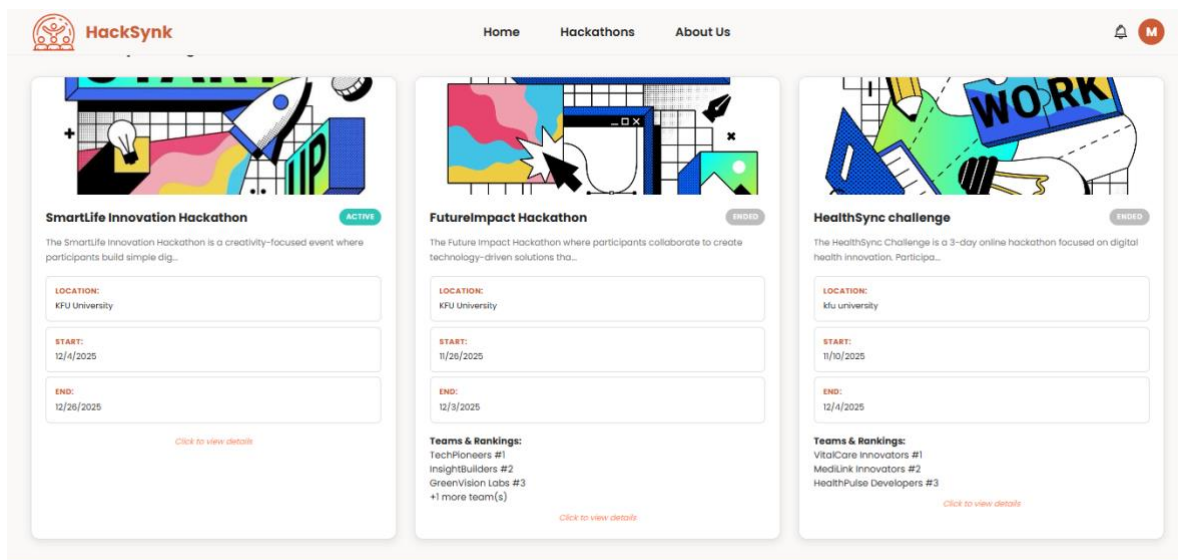


Figure 85Hackathons Page

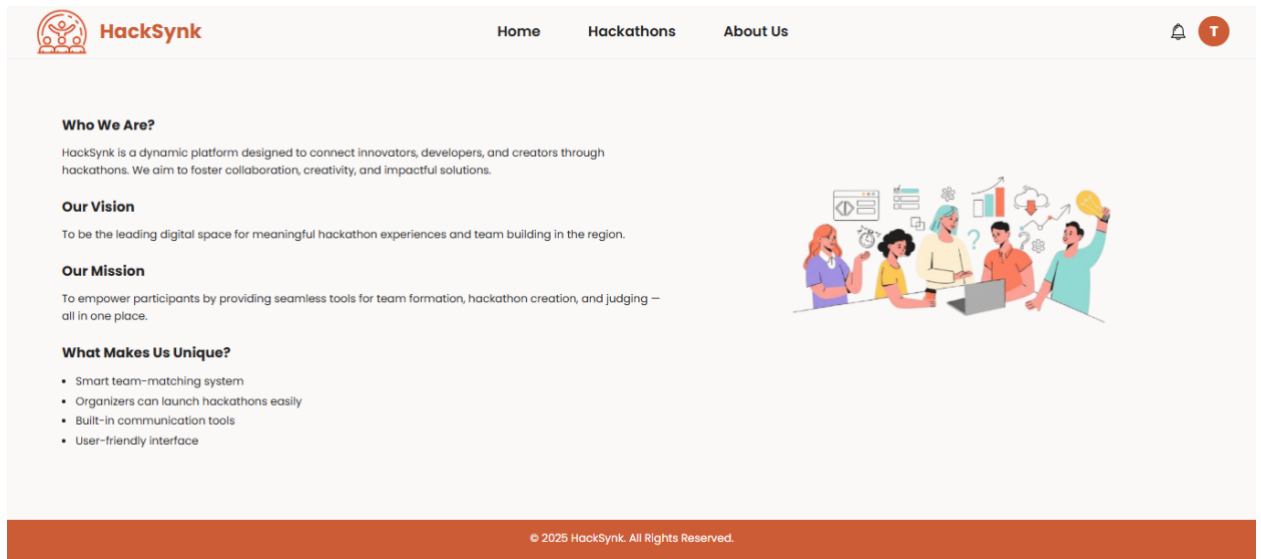


Figure 86About Us Page

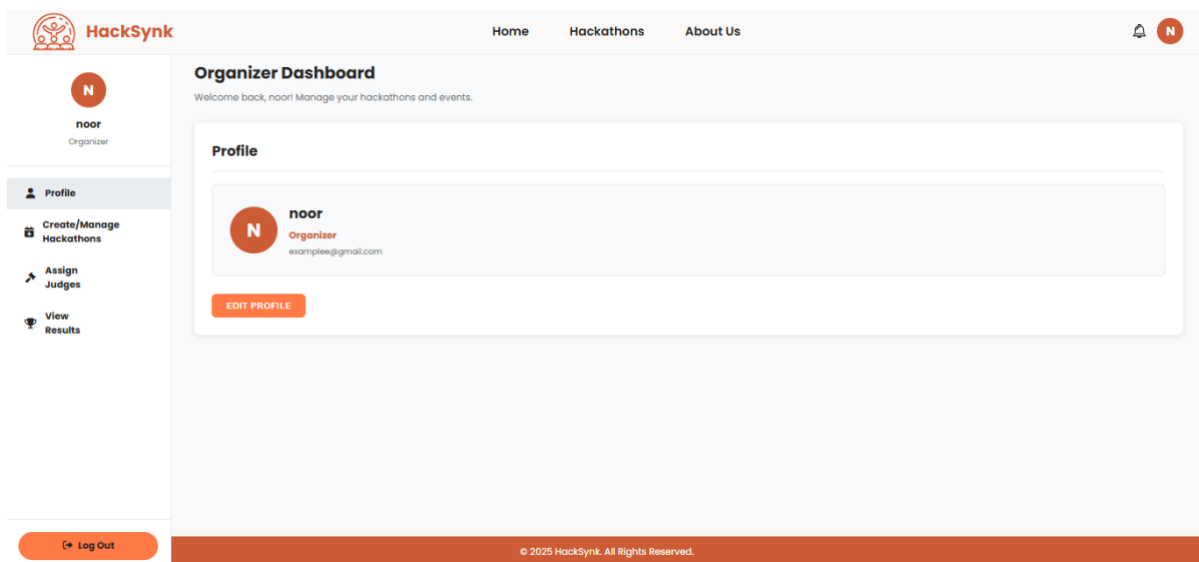


Figure 87 orgnizer profile



### Edit Profile

Full Name \*

noor

Email \*

example@gmail.com

Username \*

noor1

New Password (leave blank to keep current)

Confirm New Password

Organization Name

kfu

Job Title/Position

manager

Cancel UPDATE PROFILE

Figure 88Organizer Profile

HackSynk

[Home](#)
[Hackathons](#)
[About Us](#)

N

noor

Organizer

[Profile](#)
[Create/Manage Hackathons](#)
[Assign Judges](#)
[View Results](#)

[Log Out](#)

### Create/Manage Hackathons

[+ CREATE NEW HACKATHON](#)

#### SmartLife Innovation Hackathon

ONGOING

The SmartLife Innovation Hackathon is a creativity-focused event where participants build simple digital tools that...

KFU University

12/4/2025 - 12/26/2025

[View](#)
[EDIT](#)
[Delete](#)

#### FutureImpact Hackathon

COMPLETED

The Future Impact Hackathon where participants collaborate to create technology-driven solutions that improve...

KFU University

11/26/2025 - 12/3/2025

[View](#)
[EDIT](#)
[Delete](#)

#### HealthSync challenge

COMPLETED

The HealthSync Challenge is a 3-day online hackathon focused on digital health innovation. Participants will design and...

kfu university

11/10/2025 - 12/4/2025

[View](#)
[EDIT](#)
[Delete](#)

Figure 89Create/Manage Hackathons

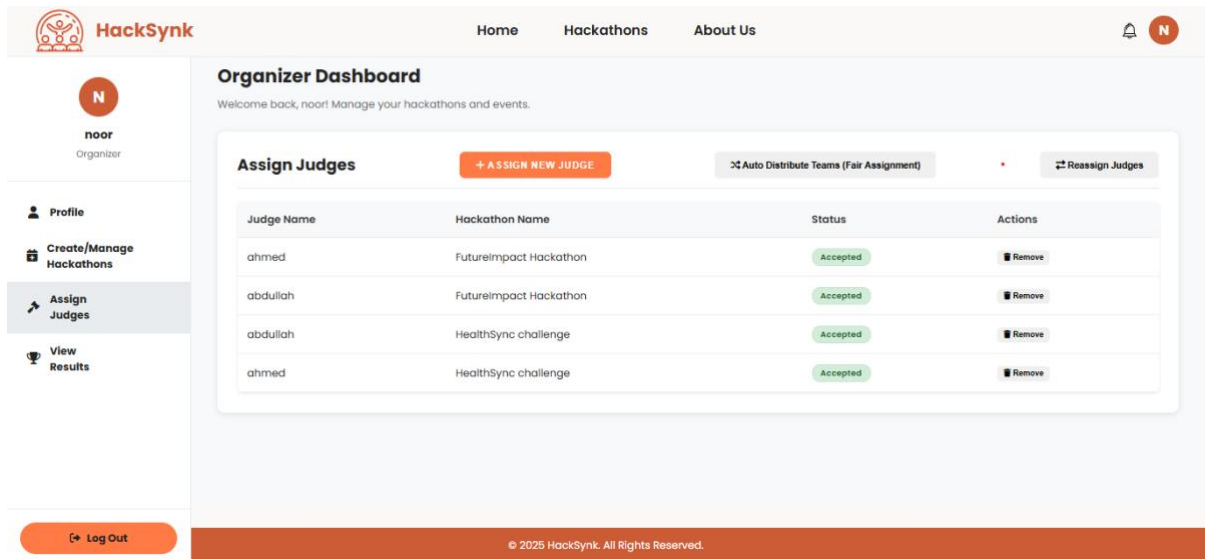


Figure 90Assign Judges

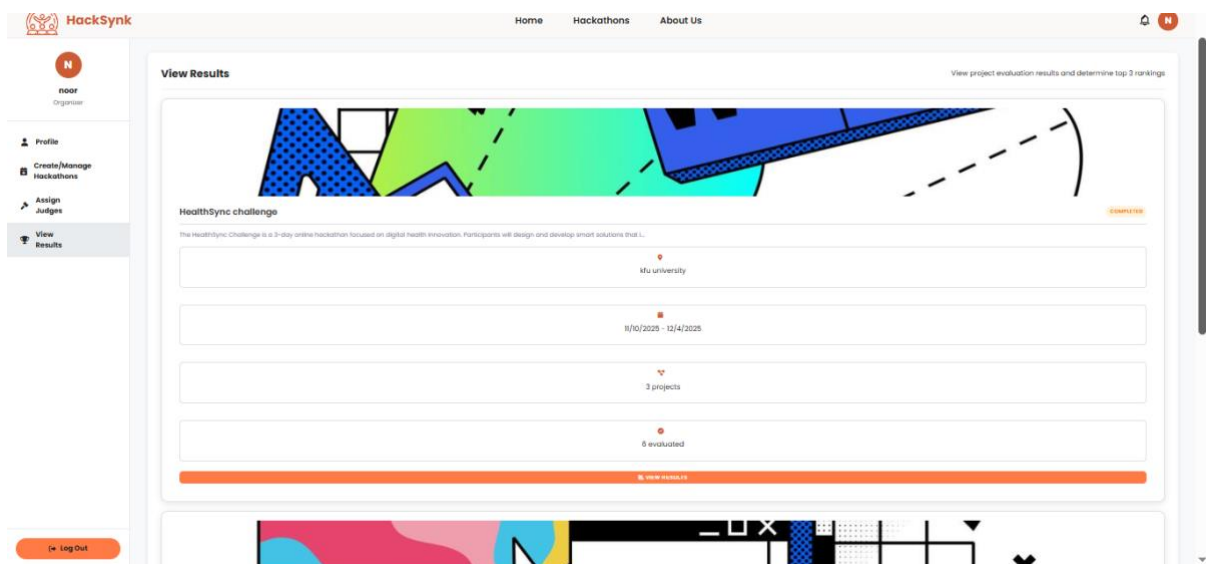


Figure 91View Results

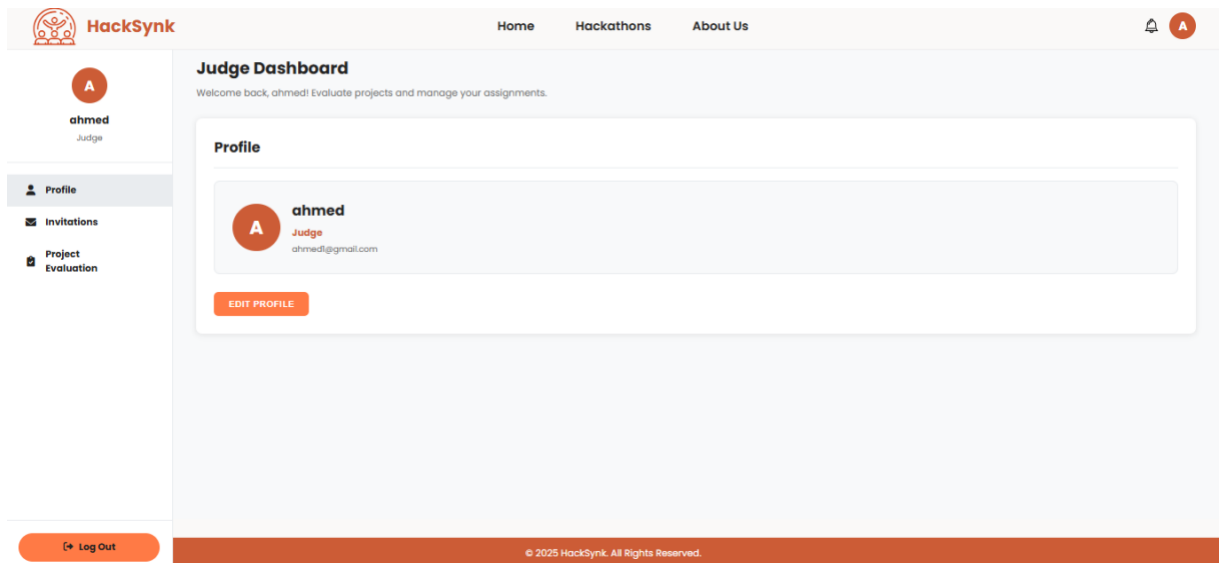


Figure 92 Judge Profile

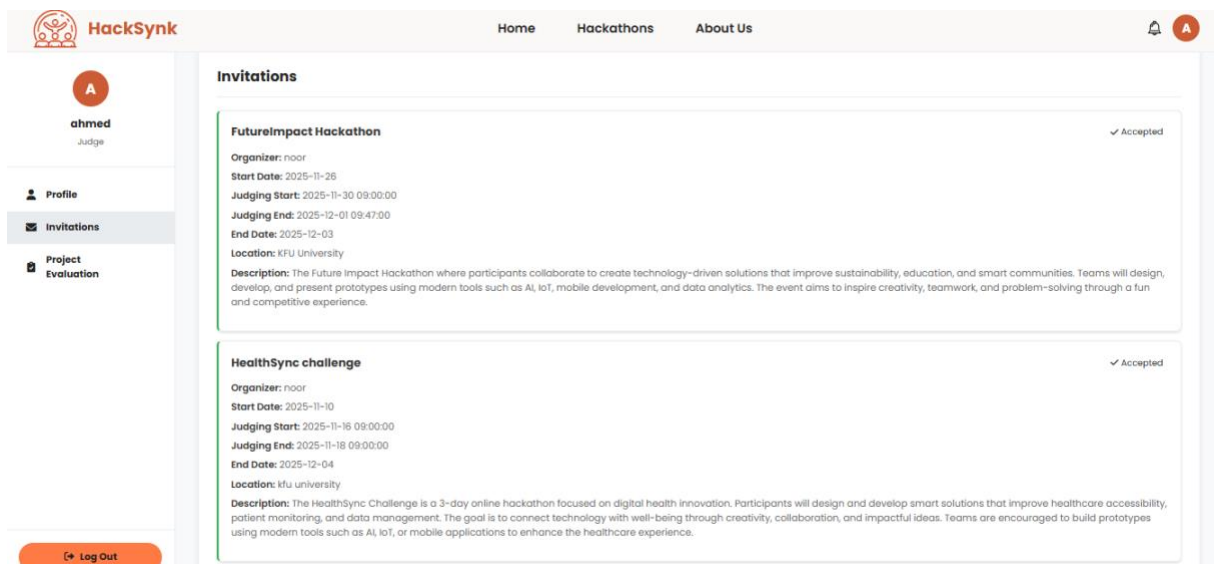


Figure 93 Invitations

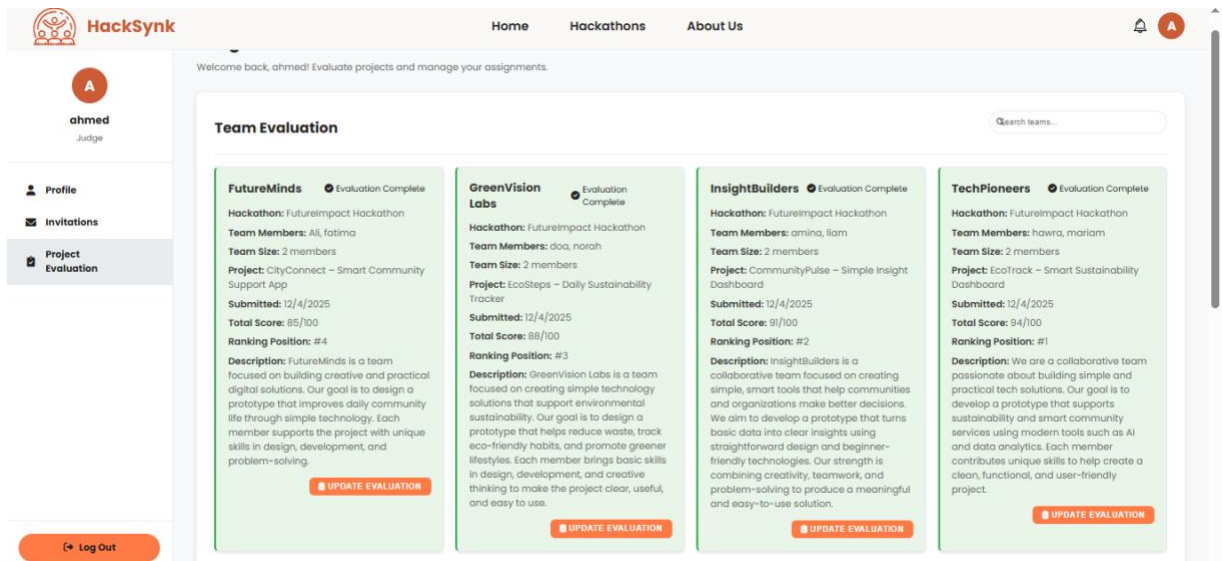


Figure 94Team Evaluation

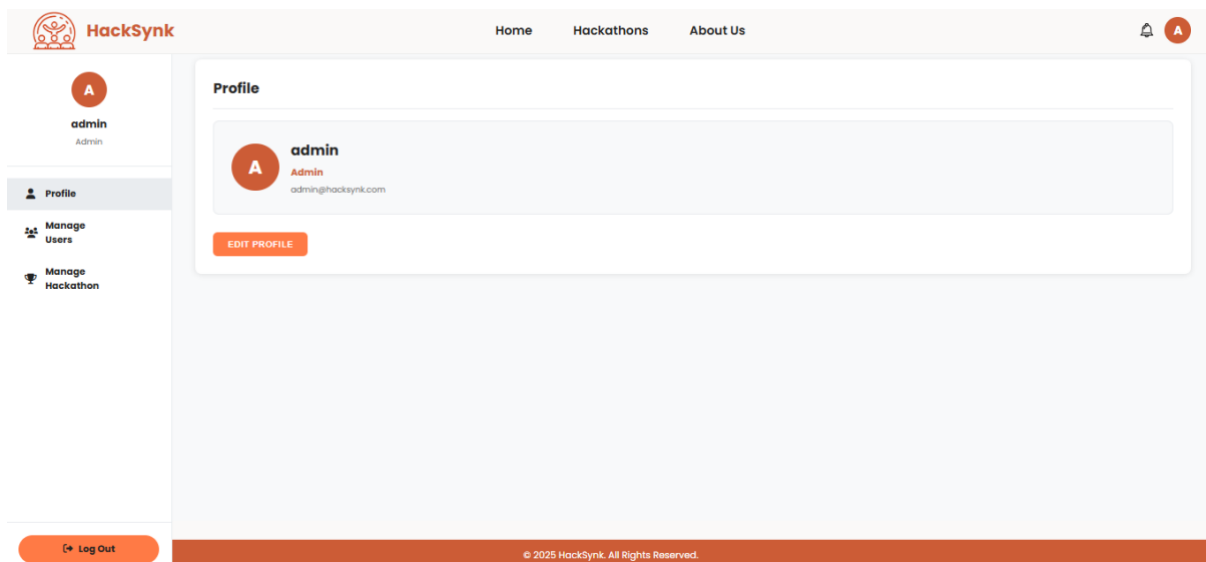


Figure 95Admin Profile

Home

Hackathons

About Us

admin

Admin

Profile

Manage Users

Manage Hackathon

Log Out

Manage Users

Search by name or email...

All Roles

Total Users: 13

+ ADD USER

| FULL NAME | EMAIL              | ROLE        | STATUS | ACTIONS |
|-----------|--------------------|-------------|--------|---------|
| Ali       | ali@gmail.com      | PARTICIPANT | ACTIVE |         |
| abdullah  | abdullah@gmail.com | JUDGE       | ACTIVE |         |
| ahmed     | ahmed@gmail.com    | JUDGE       | ACTIVE |         |
| amina     | amina@gmail.com    | PARTICIPANT | ACTIVE |         |
| doa       | doa@gmail.com      | PARTICIPANT | ACTIVE |         |
| fatma     | example@gmail.com  | PARTICIPANT | ACTIVE |         |
| gigi      | gigi@gmail.com     | PARTICIPANT | ACTIVE |         |
| hawa      | hawa@gmail.com     | PARTICIPANT | ACTIVE |         |
| iam       | iam@gmail.com      | PARTICIPANT | ACTIVE |         |
| malak     | malak@gmail.com    | PARTICIPANT | ACTIVE |         |
| mariam    | mariam@gmail.com   | PARTICIPANT | ACTIVE |         |

Figure 96Manage Users

Home

Hackathons

About Us

admin

Admin

Profile

Manage Users

Manage Hackathon

Log Out

Manage Hackathon

+ REFRESH

SmartLife Innovation Hackathon

ACTIVE

Description:

The SmartLife Innovation Hackathon is a creativity-focused event where participants build simple digital tools that make everyday life easier. Teams will design prototypes that help with productivity, time management, smart home ideas, or community convenience. The goal is to create small solutions that improve daily routines using user-friendly design and basic technology.

Location:

KFU University

Start Date:

12/4/2025

End Date:

12/26/2025

Teams:

0

EDIT DATES

FutureImpact Hackathon

PAST

Description:

The Future Impact Hackathon where participants collaborate to create technology-driven solutions that improve sustainability, education, and smart communities. Teams will design, develop, and present prototypes using modern tools such as AI, IoT, mobile development, and data analytics. The event aims to inspire creativity, teamwork, and problem-solving through a fun and competitive experience.

Figure 97Manage Hackathon

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## 5. TESTING AND CONVERSION

### 5.1 Test plan

#### *Test requirements*

#### **Use cases:**

- The organizer signs into the system, adds, edits, or deletes hackathons, assigns judges to hackathons, and manages hackathon results
- The judge signs into the system, views assigned projects, evaluates projects using multiple criteria, and provides feedback to participants
- The participant signs into the system, creates or joins teams, submits projects with file uploads, communicates via direct and team chat, and views evaluation results after announcement
- The admin signs into the system, manages user accounts (participants, organizers, judges), enables or disables user accounts, and monitors system statistics

#### **Functional Requirements:**

HackSynk's functional requirements are designed to ensure the platform provides a smooth and efficient experience for all user roles, including participants, organizers, judges, and administrators. The focus is on verifying the core functionalities that enable users to manage hackathons, teams, projects, and evaluations, ensuring the system operates as intended while meeting the needs of its diverse audience.

#### **1. Organizer Hackathon Management:**

- Ensure that organizers can log into the system successfully.
- Test the functionality of adding, editing, and deleting hackathons to ensure organizers can manage their events efficiently.

- Validate the judge assignment process to ensure organizers can assign judges to specific hackathons.
- Test the team distribution functionality to ensure teams are properly assigned to judges for evaluation.
- Verify the results management system, including setting project rankings and announcing results to participants.

## **2. Judge Project Evaluation:**

- Confirm that judges can successfully log in to the platform.
- Test the ability of judges to view assigned projects and teams for evaluation.
- Validate the evaluation process using multiple criteria (Innovation, Technical Implementation, Design/Usability, Impact, Problem Relevance, Functionality, Feasibility, Team Collaboration, Technical Complexity, Testing/Reliability) to ensure accurate project assessment.
- Verify that judges can provide written feedback to participants along with their evaluations.
- Test the judge invitation system to ensure judges can accept or reject hackathon invitations.

## **3. Participant Interaction with the Platform:**

- Ensure participants can log in and access the platform without issues.
- Test the functionality for browsing hackathons, creating teams, and joining existing teams (direct join or request to join).
- Verify that participants can submit projects with titles, descriptions, demo links, and file uploads (PDF format).
- Test the ability to update or delete projects before the submission deadline.

- Validate the real-time chat functionality, including direct messaging with other participants and team-based group chat.
- Verify that participants can view evaluation results and judge feedback after results are announced by organizers.

#### **4. Admin System Oversight:**

- Validate that administrators can log in and efficiently oversee the platform.
- Test the user management functionality to ensure admins can add, edit, and delete user accounts (participants, organizers, judges).
- Verify the ability to enable or disable user accounts to control platform access.
- Test the system overview functionality to monitor platform statistics and user activity.
- Validate that admins can view detailed user profiles and manage user roles effectively.

### **Non-Functional requirements:**

#### **1. Performance Requirements:**

- The system shall respond to user actions (e.g., logging in, submitting projects, sending chat messages) within 2 seconds under normal load conditions.
- The system shall handle up to 100 concurrent users without performance degradation.

#### **2. Scalability:**

The system shall be designed to scale horizontally to support increasing numbers of users, hackathons, teams, projects, and evaluations.

#### **3. Availability:**



The system shall have an uptime of 99.9%, ensuring minimal downtime for users during active hackathon events.

#### **4. Usability:**

- The system shall have an intuitive and user-friendly interface for all user roles (participants, organizers, judges, and admins).
- The system shall provide clear navigation and tooltips to guide users through its functionalities, including dashboard layouts, hackathon browsing, team management, and project submission processes.

#### **5. Maintainability:**

The system shall be developed using modular and reusable components to facilitate updates and maintenance, with clear separation between frontend (HTML/CSS/JavaScript) and backend (PHP) components.

#### **Testing Strategy:**

The testing strategy for the HackSynk platform is designed to ensure a comprehensive evaluation of its features and functionalities, tailored to meet the needs of participants, organizers, judges, and administrators. This structured approach ensures the platform delivers a seamless, efficient, and user-friendly experience.

##### **1. Core Feature Verification:**

The initial phase focuses on validating the platform's core features, such as user registration for different roles (participants, organizers, judges), login authentication, and navigation across different dashboards. These essential functions are tested for reliability across all user roles.

## **2. Role-Specific Functionality Testing:**

Subsequent testing targets specific functionalities for each role. Participants undergo tests for team formation, project submission, and chat functionality. Organizers are tested for hackathon management, judge assignment, and results management. Judges are tested for project evaluation and feedback submission. Administrators are tested for user management and system oversight capabilities.

## **3. User Experience Assessment:**

The platform's usability and interface are tested to confirm an intuitive experience for all users. This includes assessing navigation clarity across different dashboards (participant, organizer, judge, admin), ease of accessing features such as hackathon browsing, team management, and project submission, and responsiveness on both desktop and mobile devices.

## **4. Performance Evaluation:**

Special focus is placed on testing the platform's performance under various conditions, including high traffic during active hackathon periods and concurrent user activity (multiple participants submitting projects simultaneously, multiple judges evaluating projects, real-time chat messaging). This phase ensures HackSynk remains fast and responsive, even during peak usage.

## **5. Error Handling and Resolution:**

A key component of the strategy involves testing the platform's ability to handle errors gracefully. Scenarios like invalid inputs during registration or project submission, failed file uploads, unavailable hackathons, unauthorized access attempts, or network interruptions during chat messaging are evaluated to ensure users receive clear guidance and error resolution.

## **6. Data Security and Integrity Checks:**

Testing protocols emphasize the protection of user data and transactional accuracy. This includes validating secure authentication processes and password hashing mechanisms to prevent data breaches or unauthorized access. Additionally, tests verify that user roles are properly enforced, ensuring participants cannot access organizer functions, judges can only evaluate assigned projects, and data integrity is maintained across team formations, project submissions, and evaluations.

By adhering to this robust testing strategy, the HackSynk platform aims to deliver a dependable, efficient, and secure system that meets the expectations of all its users.

## 5.2 Unit testing

### USER REGISTRATION CASE

|                                |                                                                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name                 | Registration Success                                                                                                                                                                                                                   |
| Purpose                        | Verify successful participant, organizer, and judge registration                                                                                                                                                                       |
| Component Name                 | participant-register.php, organizer-register.php, and judge-register.php                                                                                                                                                               |
| Component Type (method/object) | \$_POST method                                                                                                                                                                                                                         |
| Test Description (Steps)       | <ol style="list-style-type: none"><li>1. Click on "Join our community" or navigate to signup page</li><li>2. Choose a role (Participant, Organizer, or Judge)</li><li>3. Fill out valid information on the registration form</li></ol> |

|                    |                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
|                    | 4. Click the "Register" button<br>5. Account is created                                                           |
| Input Provided     | FullName, UserName, Email, Password, PhoneNumber (optional), Address (optional), Role-specific fields             |
| Expected Outcome   | A new user account is created, and the user is redirected to the success page.                                    |
| Actual Outcome     | The account is created successfully and shows the following message (Registration successful! You can now login.) |
| Status (Pass/Fail) | Pass                                                                                                              |

Table 3:USER REGISTRATION CASE

#### HACKATHON CREATION TEST CASE

|                                |                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name                 | Create Hackathon Success                                                                                                                                                          |
| Purpose                        | Verify successful hackathon creation by organizer with all required fields                                                                                                        |
| Component Name                 | hackathons.php (create action)                                                                                                                                                    |
| Component Type (method/object) | \$_POST method                                                                                                                                                                    |
| Test Description (Steps)       | 1. Login as organizer<br>2. Navigate to organizer dashboard<br>3. Click "Create Hackathon"<br>4. Fill out all required fields: Name, Description, Start Date, End Date, Location, |

|                    |                                                                                                                                                                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Timeline dates</p> <p>5. Set min_teams and max_teams</p> <p>6. Click "Create" button</p>                                                                                                                                                                                                                                              |
| Input Provided     | <p>name: "Tech Innovation 2024", description: "Annual tech hackathon", start_date: "2024-06-01", end_date: "2024-06-03", location: "Riyadh", submission_start: "2024-06-01", submission_end: "2024-06-02", judging_start: "2024-06-02", judging_end: "2024-06-02", result_announcement_at: "2024-06-03", min_teams: 5, max_teams: 20</p> |
| Expected Outcome   | <p>Hackathon is created successfully and message displayed: "Hackathon created successfully"</p>                                                                                                                                                                                                                                         |
| Actual Outcome     | <p>Hackathon is created successfully and shows: "Hackathon created successfully"</p>                                                                                                                                                                                                                                                     |
| Status (Pass/Fail) | <p>Pass</p>                                                                                                                                                                                                                                                                                                                              |

Table 4:HACKATHON CREATION TEST CASE

#### TEAM CREATION TEST CASE

|                                |                                                |
|--------------------------------|------------------------------------------------|
| Test Case Name                 | Create Team Success                            |
| Purpose                        | Verify successful team creation by participant |
| Component Name                 | teams.php (create action)                      |
| Component Type (method/object) | \$_POST method                                 |

|                          |                                                                                                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Description (Steps) | <ol style="list-style-type: none"> <li>1. Login as participant</li> <li>2. Navigate to participant dashboard</li> <li>3. Select a hackathon</li> <li>4. Click "Create Team"</li> <li>5. Fill team name, description, max participants</li> <li>6. Click "Create" button</li> </ol> |
| Input Provided           | name: "Team Alpha", description: "Innovative solutions team", hackathon_id: 1, max_participants: 5                                                                                                                                                                                 |
| Expected Outcome         | Team is created successfully and message displayed: "Team created successfully"                                                                                                                                                                                                    |
| Actual Outcome           | Team is created successfully and shows: "Team created successfully"                                                                                                                                                                                                                |
| Status (Pass/Fail)       | Pass                                                                                                                                                                                                                                                                               |

Table 5:TEAM CREATION TEST CASE

#### PROJECT SUBMISSION TEST CASE

|                                |                                                                           |
|--------------------------------|---------------------------------------------------------------------------|
| Test Case Name                 | Submit Project Success                                                    |
| Purpose                        | Verify successful project submission by participant team                  |
| Component Name                 | projects.php (create action)                                              |
| Component Type (method/object) | \$_POST method                                                            |
| Test Description (Steps)       | <ol style="list-style-type: none"> <li>1. Login as participant</li> </ol> |

|                    |                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | 2. Navigate to "Submit Project" section<br>3. Select team<br>4. Fill project title, description, demo link<br>5. Upload project files (PDF)<br>6. Click "Submit Project" button |
| Input Provided     | team_id: 1, title: "AI-Powered Solution", description: "Innovative AI solution for healthcare", demo_link: "https://demo.example.com"                                           |
| Expected Outcome   | Project is submitted successfully and message displayed: "Project submitted successfully"                                                                                       |
| Actual Outcome     | Project submitted successfully and shows: "Project submitted successfully"                                                                                                      |
| Status (Pass/Fail) | Pass                                                                                                                                                                            |

Table 6:PROJECT SUBMISSION TEST CASE

### 5.3 Integration Testing

#### VIEW MY TEAMS

|                          |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| Test Case Name           | View my teams which are teams created or joined by the participant.                |
| Purpose                  | To enable a participant to view all their team details.                            |
| External Dependencies    | login.php, participant-dashboard.php                                               |
| Test Description (Steps) | 1. Login to the system as a participant.<br>2. Get access to a participant account |

|                    |                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | 3. Go to participant dashboard page<br>4. Navigate to "My Teams" section<br>5. Click on "View Teams".<br>6. All teams created or joined by the participant are displayed. |
| Expected Outcome   | All teams created or joined by the participant are displayed.                                                                                                             |
| Actual Outcome     | All teams are displayed correctly.                                                                                                                                        |
| Status (Pass/Fail) | Pass                                                                                                                                                                      |
| Comments           | -                                                                                                                                                                         |

Table 7:VIEW MY TEAMS

#### VIEW MY PROJECTS

|                          |                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name           | View my projects which are projects submitted by the participant's teams.                                                                                                                                                                                                |
| Purpose                  | To enable a participant to view all their project details.                                                                                                                                                                                                               |
| External Dependencies    | login.php, participant-dashboard.php                                                                                                                                                                                                                                     |
| Test Description (Steps) | 1. Login to the system as a participant.<br>2. Get access to a participant account<br>3. Go to participant dashboard page<br>4. Navigate to "My Projects" section<br>5. Click on "View Projects".<br>6. All projects submitted by the participant's teams are displayed. |



|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| Expected Outcome   | All projects submitted by the participant's teams are displayed. |
| Actual Outcome     | All projects are displayed correctly.                            |
| Status (Pass/Fail) | Pass                                                             |
| Comments           | -                                                                |

Table 8:VIEW MY PROJECTS

### VIEW MY HACKATHONS

|                          |                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name           | View my hackathons which are hackathons created by the organizer.                                                                                                                                                                                                                                                                              |
| Purpose                  | To enable an organizer to view all their hackathon details.                                                                                                                                                                                                                                                                                    |
| External Dependencies    | login.php, organizer-dashboard.php                                                                                                                                                                                                                                                                                                             |
| Test Description (Steps) | <ol style="list-style-type: none"> <li>1. Login to the system as an organizer.</li> <li>2. Get access to an organizer account</li> <li>3. Go to organizer dashboard page</li> <li>4. Navigate to "My Hackathons" section</li> <li>5. Click on "View Hackathons".</li> <li>6. All hackathons created by the organizer are displayed.</li> </ol> |
| Expected Outcome         | All hackathons created by the organizer are displayed.                                                                                                                                                                                                                                                                                         |
| Actual Outcome           | All hackathons are displayed correctly.                                                                                                                                                                                                                                                                                                        |
| Status (Pass/Fail)       | Pass                                                                                                                                                                                                                                                                                                                                           |

|          |   |
|----------|---|
| Comments | - |
|----------|---|

Table 9:VIEW MY HACKATHONS

#### VIEW ASSIGNED HACKATHONS

|                          |                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name           | View assigned hackathons which are hackathons assigned to the judge.                                                                                                                                                                                                                                                                         |
| Purpose                  | To enable a judge to view all hackathons they are assigned to evaluate.                                                                                                                                                                                                                                                                      |
| External Dependencies    | login.php, judge-dashboard.php                                                                                                                                                                                                                                                                                                               |
| Test Description (Steps) | <ol style="list-style-type: none"> <li>1. Login to the system as a judge.</li> <li>2. Get access to a judge account</li> <li>3. Go to judge dashboard page</li> <li>4. Navigate to "Assigned Hackathons" section</li> <li>5. Click on "View Assigned Hackathons".</li> <li>6. All hackathons assigned to the judge are displayed.</li> </ol> |
| Expected Outcome         | All hackathons assigned to the judge are displayed.                                                                                                                                                                                                                                                                                          |
| Actual Outcome           | All assigned hackathons are displayed correctly.                                                                                                                                                                                                                                                                                             |
| Status (Pass/Fail)       | Pass                                                                                                                                                                                                                                                                                                                                         |
| Comments                 | -                                                                                                                                                                                                                                                                                                                                            |

Table 10:VIEW ASSIGNED HACKATHONS

## 5.4 System Testing

### ADMIN TASKS

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Case Name           | Admin's main tasks through the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Purpose                  | <p>To ensure that the admin:</p> <ul style="list-style-type: none"><li>• Can access the system and perform actions</li><li>• Can view the data of users (participants, organizers, judges), hackathons, teams, and projects</li><li>• Can enable and disable user accounts</li><li>• Can view all hackathons in the system</li><li>• Can view all teams in the system</li><li>• Can view all projects in the system</li><li>• Can manage system settings</li></ul>                                                                                                                                                                                                                               |
| Test Description (Steps) | <ul style="list-style-type: none"><li>• Log in as Admin.</li><li>• Navigate to "User Management" section.</li><li>• View all participants, organizers, and judges</li><li>• Enable and disable user accounts</li></ul> <hr/> <ul style="list-style-type: none"><li>• Log in as Admin.</li><li>• Navigate to "Hackathon Overview" section.</li><li>• View all hackathons in the system</li></ul> <hr/> <ul style="list-style-type: none"><li>• Log in as Admin.</li><li>• Navigate to "Hackathon Overview" section.</li><li>• View all teams in the system</li></ul> <hr/> <ul style="list-style-type: none"><li>• Log in as Admin.</li><li>• Navigate to "Hackathon Overview" section.</li></ul> |

|                    |                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <ul style="list-style-type: none"> <li>• View all projects in the system</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as Admin.</li> <li>• Navigate to "System Settings" section.</li> <li>• Manage system configuration</li> </ul> |
| Expected Outcome   | To successfully take all actions in the admin profile.                                                                                                                                                                                                  |
| Actual Outcome     | The admin successfully takes all actions in the admin profile.                                                                                                                                                                                          |
| Status (Pass/Fail) | Pass                                                                                                                                                                                                                                                    |
| Remarks            | -                                                                                                                                                                                                                                                       |

Table 11:ADMIN TASKS

#### PARTICIPANT TASKS

| Test Case Name | Participant's main tasks through the system                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose        | <p>To ensure that the participant:</p> <ul style="list-style-type: none"> <li>• Can access the system and perform actions</li> <li>• Can view and edit their profile</li> <li>• Can manage their skills</li> <li>• Can browse and view hackathons</li> <li>• Can create teams and join teams</li> <li>• Can manage team join requests</li> <li>• Can submit projects and upload project files</li> <li>• Can edit and delete projects</li> <li>• Can send and receive direct messages</li> </ul> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <ul style="list-style-type: none"> <li>• Can participate in team chats</li> <li>• Can view notifications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Test Description (Steps) | <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "My Profile" section.</li> <li>• View and edit profile information</li> <li>• Add, edit, and remove skills</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "Browse Hackathons" section.</li> <li>• View hackathon details and join hackathons</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "My Teams" section.</li> <li>• Create new teams</li> <li>• Browse and join existing teams</li> <li>• Approve or reject team join requests (as team leader)</li> <li>• Leave teams</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "My Projects" section.</li> <li>• Submit new projects</li> <li>• Upload project files (PDF, images, videos, presentations, archives)</li> <li>• Edit and delete projects (before deadline)</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "Messages" section.</li> <li>• Send and receive direct messages</li> </ul> |

|                    |                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <ul style="list-style-type: none"> <li>• Participate in team chats</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Participant.</li> <li>• Navigate to "Notifications" section.</li> <li>• View all system notifications</li> </ul> |
| Expected Outcome   | The participant successfully performs all actions related to their profile, teams, projects, and communication.                                                                                                                                           |
| Actual Outcome     | The participant successfully performs all actions as expected.                                                                                                                                                                                            |
| Status (Pass/Fail) | Pass                                                                                                                                                                                                                                                      |
| Remarks            | -                                                                                                                                                                                                                                                         |

Table 12:PARTICIPANT TASKS

#### ORGANIZER TASKS

| Test Case Name | Organizer's main tasks through the system                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose        | <p>To ensure that the organizer:</p> <ul style="list-style-type: none"> <li>• Can access the system and perform actions</li> <li>• Can view and edit their profile</li> <li>• Can create hackathons with all details</li> <li>• Can edit and update hackathons</li> <li>• Can set hackathon timeline (submission period, judging period, result announcement)</li> <li>• Can assign judges to hackathons</li> <li>• Can view assigned judges</li> </ul> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <ul style="list-style-type: none"> <li>• Can manage judge invitations</li> <li>• Can view all project evaluations</li> <li>• Can set project rankings</li> <li>• Can announce hackathon results</li> <li>• Can view notifications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Test Description (Steps) | <ul style="list-style-type: none"> <li>• Log in as an Organizer.</li> <li>• Navigate to "My Profile" section.</li> <li>• View and edit profile information</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as an Organizer.</li> <li>• Navigate to "My Hackathons" section.</li> <li>• Create new hackathons</li> <li>• Set hackathon details (name, description, dates, location)</li> <li>• Upload hackathon images</li> <li>• Define rules and guidelines</li> <li>• Set prize information</li> <li>• Edit existing hackathons</li> <li>• Set timeline (submission start/end, judging start/end, result announcement)</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as an Organizer.</li> <li>• Navigate to "Judge Management" section.</li> <li>• Assign judges to hackathons</li> <li>• View assigned judges</li> <li>• Manage judge invitations (send invitations, track acceptances)</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as an Organizer.</li> </ul> |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <ul style="list-style-type: none"> <li>• Navigate to "Results Management" section.</li> <li>• View all project evaluations</li> <li>• Set project rankings based on evaluations</li> <li>• Announce hackathon results</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as an Organizer.</li> <li>• Navigate to "Notifications" section.</li> <li>• View all system notifications</li> </ul> |
| Expected Outcome   | The organizer successfully performs all actions related to hackathon creation, judge management, and results management.                                                                                                                                                                                                                                                                                    |
| Actual Outcome     | The organizer successfully performs all actions as expected.                                                                                                                                                                                                                                                                                                                                                |
| Status (Pass/Fail) | Pass                                                                                                                                                                                                                                                                                                                                                                                                        |
| Remarks            | -                                                                                                                                                                                                                                                                                                                                                                                                           |

Table 13:ORGANIZER TASKS

#### JUDGE TASKS

| Test Case Name | Judge's main tasks through the system                                                                                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose        | <p>To ensure that the judge:</p> <ul style="list-style-type: none"> <li>• Can access the system and perform actions</li> <li>• Can view and edit their profile</li> <li>• Can view assigned hackathons</li> <li>• Can view hackathon details</li> <li>• Can view assigned teams for evaluation</li> </ul> |



|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <ul style="list-style-type: none"> <li>• Can evaluate projects using 10 criteria scoring system</li> <li>• Can provide feedback on projects</li> <li>• Can view past evaluations</li> <li>• Can view notifications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Test Description (Steps) | <ul style="list-style-type: none"> <li>• Log in as a Judge.</li> <li>• Navigate to "My Profile" section.</li> <li>• View and edit profile information</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Judge.</li> <li>• Navigate to "Assigned Hackathons" section.</li> <li>• View all assigned hackathons</li> <li>• View hackathon details</li> <li>• View assigned teams for evaluation</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Judge.</li> <li>• Navigate to "Project Evaluations" section.</li> <li>• Evaluate projects using 10 criteria (innovation, implementation quality, design usability, impact, problem relevance clarity, functionality features completeness, feasibility scalability, team collaboration presentation quality, technical complexity, testing reliability)</li> <li>• Provide written feedback on projects</li> <li>• View past evaluations</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Log in as a Judge.</li> <li>• Navigate to "Notifications" section.</li> </ul> |

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                    | • View all system notifications                                                                                                  |
| Expected Outcome   | The judge successfully performs all actions related to viewing assigned hackathons, evaluating projects, and providing feedback. |
| Actual Outcome     | The judge successfully performs all actions as expected.                                                                         |
| Status (Pass/Fail) | Pass                                                                                                                             |

Table 14: JUDGE TASKS

## 5.5 Conversion plan

The conversion plan for the HackSynk platform entails a strategic initiative to evolve into an innovative online platform that bridges the gap between participants, organizers, and judges in a manner that aligns with Saudi Arabia's Vision 2030 and its digitization policy. The main objective of this is to develop a seamless, user-friendly environment where participants can browse and join hackathons, form teams, submit projects, and collaborate effectively, while organizers can create and manage hackathon events, assign judges, and oversee results, and judges can evaluate projects and provide constructive feedback.

This transition is driven by a growing digital economy and the greater need for accessible innovation platforms in Saudi Arabia. Such a new system smoothened operations all the way from managing hackathon creation, team formation, project submission, judge assignment, to evaluation processes, and even participant interactions through real-time communication.

The main stakeholders include participants, organizers, judges, and administrators. Each has a custom functionality: accessibility and collaboration tools for the participants, efficient event management tools for organizers to develop and grow their hackathon businesses,

comprehensive evaluation tools for judges to assess projects fairly, and performance oversight to allow administrators to optimize systems and maintain platform stability.

These are expected to impact cultural, economic, and operational dimensions. By embracing technological advancements, HackSynk will have the ability to allow participants to explore a wide variety of hackathons, form teams seamlessly, and showcase their innovative projects, organizers to manage hackathon events efficiently and track participant engagement, judges to evaluate projects using standardized criteria and provide valuable feedback, and administrators to monitor and optimize the entire platform ecosystem. This transformation aims at positioning HackSynk to be the cornerstone of Saudi Arabia's innovation, technology, and digital ecosystem, fostering collaboration, creativity, and impactful solutions that contribute to the nation's technological advancement.

## **6. SYSTEM DOCUMENTATION**

### **6.1 User manual**

#### *Organizer Features*

As an organizer, you can:

#### **1. Sign In:**

- Navigate to the website's homepage.
- Click on the "Login" button.
- Select "Organizer" as your role (or the system will detect your role automatically).
- Enter your username and password, then click "Sign In."

#### **2. Create Hackathons:**

- Once logged in, go to the "Organizer Dashboard."
- Click on the "Create/Manage Hackathons" tab in the sidebar.
- Click the "Create Hackathon" button or "Add Hackathon" option.
- Fill in the hackathon details including:
  - Hackathon name and description
  - Start date and end date
  - Location
  - Submission deadline
  - Judging period dates
  - Result announcement date
  - Rules and guidelines
  - Prize information
  - Upload hackathon image

- Minimum and maximum team sizes
- Click "Submit" or "Create" to save your hackathon.

### **3. Edit Hackathons:**

- Go to the "Create/Manage Hackathons" section in your dashboard.
- Find the hackathon you wish to edit from the list.
- Click the "Edit" button next to the hackathon.
- Modify the required details (name, dates, rules, prizes, etc.).
- Click "Save Changes" to update the hackathon.

### **4. Delete Hackathons:**

- Locate the hackathon you want to delete in the "Create/Manage Hackathons" section.
- Click "Delete" next to the hackathon.
- Confirm the deletion when prompted.

### **5. View Hackathons:**

- Navigate to the "Create/Manage Hackathons" page to see all your created hackathons.
- View hackathon statistics including number of participants, teams, and submitted projects.

### **6. Assign Judges:**

- Go to the "Assign Judges" section in your dashboard.
- Select a hackathon from the dropdown list.
- View available judges or search for judges.
- Click "Assign Judge" to add a judge to the selected hackathon.
- Judges will receive invitations that they can accept or reject.

### **7. Distribute Teams to Judges:**

- In the "Assign Judges" section, you can automatically or manually distribute teams to judges for evaluation.

- Click "Auto Distribute" to automatically assign teams, or manually assign specific teams to judges.

- This ensures balanced workload distribution among judges.

## **8. View and Manage Results:**

- Navigate to the "View Results" section in your dashboard.

- Select a hackathon to view all project evaluations and scores.

- Review evaluation scores from all judges.

- Set final project rankings based on evaluation results.

- Click "Announce Results" to make results visible to participants.

## *Judge Features*

As a judge, you can:

### **1. Sign In:**

- Navigate to the website's homepage.

- Click on the "Login" button.

- Select "Judge" as your role (or the system will detect your role automatically).

- Enter your username and password, then click "Sign In."

### **2. View Invitations:**

- Once logged in, go to the "Judge Dashboard."

- Click on the "Invitations" tab in the sidebar.

- View all hackathon invitations sent by organizers.

- Invitations will show their status: "Pending", "Accepted", or "Rejected".

### **3. Accept or Reject Invitations:**

- In the "Invitations" section, review pending hackathon invitations.

- Click "Accept" to join a hackathon as a judge, or "Reject" to decline the invitation.
- You will receive a confirmation message after responding to an invitation.

#### **4. View Assigned Projects:**

- Go to the "Project Evaluation" section in your dashboard.
- View all teams and projects assigned to you for evaluation.
- Click on a project to view its details including:
  - Project title and description
  - Demo link (if provided)
  - Uploaded project files (PDF format)
  - Team members information

#### **5. Evaluate Projects:**

- In the "Project Evaluation" section, select a project to evaluate.
- Fill out the evaluation form with scores (1-10) for each criterion:
  - Innovation and Creativity
  - Technical Implementation
  - Design and Usability
  - Impact and Relevance
  - Problem Relevance
  - Functionality
  - Feasibility
  - Team Collaboration
  - Technical Complexity
  - Testing and Reliability

- Provide written feedback in the feedback section.
- Click "Submit Evaluation" to save your evaluation.
- You can update your evaluation before the judging period ends.

## **6. View Evaluation Status:**

- Track which projects you have evaluated and which are still pending.
- View your evaluation history and submitted scores.

### *Participant Features*

As a participant, you can:

#### **1. Sign In:**

- Navigate to the website's homepage.
- Click on the "Login" button.
- Select "Participant" as your role (or the system will detect your role automatically).
- Enter your username and password, then click "Sign In."

#### **2. Browse Hackathons:**

- From the homepage or "Hackathons" page, browse available hackathons.
- Click on a hackathon to view its details including:
  - Description, dates, and location
  - Rules and guidelines
  - Prize information
  - Number of teams and participants
- Filter hackathons by status: "Upcoming", "Active", or "Ended".

#### **3. Create Team:**

- Go to the "Participant Dashboard."



- Click on the "Create Team" tab in the sidebar.
- Select a hackathon from the dropdown.
- Enter team name and description.
- Click "Create Team" to form your team.
- As the team leader, you can manage team membership and approve join requests.

#### **4. Join Team:**

- Go to the "Join Team" section in your dashboard.
- Browse available teams for a specific hackathon.
- Click "Join Team" if the team is open, or "Request to Join" if approval is required.
- Wait for the team leader to approve your request.
- You will receive a notification when your request is approved or rejected.

#### **5. Manage Team Join Requests (as Team Leader):**

- If you are a team leader, go to the "Create Team" section.
- View pending join requests for your teams.
- Click "Approve" to accept a request or "Reject" to decline it.

#### **6. Submit Project:**

- As a team leader, go to the "Submit Project" section in your dashboard.
- Select the hackathon and team for which you want to submit a project.
- Fill in project details:
  - Project title
  - Description
  - Demo link (optional)

- Upload project files (PDF format).
- Click "Submit Project" to save your submission.
- You can update or delete your project before the submission deadline.

#### **7. View Results:**

- Navigate to the "Results" section in your dashboard.
- View evaluation results and rankings after organizers announce them.
- See your project's scores, average scores, and judge feedback.
- View project rankings for the hackathon.

#### **8. Team Chat:**

- Go to the "Team Chat" section in your dashboard.
- Select a team to view and participate in team-based group chat.
- Send messages to all team members for collaboration and coordination.
- View message history and mark messages as read.

#### **9. Direct Messages:**

- Click on the "Chat Messages" tab in your dashboard.
- View your conversations with other participants.
- Start a new conversation by selecting a participant.
- Send and receive direct messages for private communication.
- View message status (read/unread).

#### **10. Manage Profile:**

- Go to the "Profile" section in your dashboard.
- Click "Edit Profile" to update your information.
- Add or update your skills, education, experience, and certificates.

- Save changes to update your profile.

### *Admin Features*

As an administrator, you can:

#### **1. Sign In:**

- Navigate to the admin login page or the main login page.
- Enter your admin username and password.
- Click "Sign In" to access the admin dashboard.

#### **2. Manage Users:**

- Go to the "Manage Users" section in the admin dashboard.
- View all users (participants, organizers, and judges) in the system.
- Use the search bar to find specific users by name or email.
- Filter users by role using the role filter dropdown.

#### **3. Add Users:**

- In the "Manage Users" section, click "Add User" button.
- Fill in user details including:
  - Full name
  - Username
  - Email
  - Password
  - Role (Participant, Organizer, or Judge)
- Click "Submit" to create the user account.

#### **4. Edit Users:**

- Find the user you want to edit in the "Manage Users" section.

- Click the "Edit" button next to the user.
- Update user information (name, email, password, etc.).
- Click "Save Changes" to update the user account.

### **5. Delete Users:**

- Locate the user you want to delete in the "Manage Users" section.
- Click "Delete" next to the user.
- Confirm the deletion when prompted.

### **6. Enable or Disable User Accounts:**

- In the "Manage Users" section, find the user account.
- Click "Disable" to deactivate a user account, or "Enable" to reactivate it.
- Disabled users cannot log in to the system.
- Confirm the action when prompted.

### **7. View System Statistics:**

- Access the dashboard overview to view system-wide statistics.
- Monitor total number of users, hackathons, teams, and projects.
- View platform activity and engagement metrics.

### **8. Manage Hackathons (Oversight):**

- Go to the "Manage Hackathon" section in the admin dashboard.
- View all hackathons created by organizers.
- Monitor hackathon activity and participant engagement.
- Oversee platform content and ensure system stability.

## **6.2 Technical support manuals**

The Technical Support Manual provides practical solutions to common HackSynk platform technical issues as an easy fix and ensure user experience continuity. Below, some of the

problems may be pointed out with respective solutions based on the functional aspect of the HackSynk platform.

#### ***A. Site Not Accessible***

● Issue: There is no internet

⚙ Solution: Your device should use stable internet access, and after that restart your router to take effect.

● Issue: Platform didn't load.

⚙ Solution: Refresh the page or switch to a different network. Try using a different device to access the platform if the issue continues.

● Issue: The server is unavailable, or the website is undergoing maintenance.

⚙ Solution: Check the HackSynk homepage or email notifications for maintenance updates or server status.

#### ***B. Pages take too much time to load***

● Issue: Poor internet connection.

⚙ Solution: Switch to a wired connection or get closer to your router to get a better signal.

● Issue: Incompatibility with your browser.

⚙ Solution: Update your browser to the latest version or switch to a browser optimized for HackSynk, such as Google Chrome or Firefox.

● Issue: High traffic on the platform.

✿ Solution: Try accessing the platform during off-peak hours for faster load times.

### ***C. Server Not Found***

● Issue: Internet connectivity issues.

✿ Solution: Restart your router and ensure your device is properly connected to the network.

● Issue: Device-specific issues.

✿ Solution: Restart your device and try accessing the platform again.

● Issue: Cached data is outdated.

✿ Solution: Clear your browser cache and cookies, then refresh the platform.

### ***D. Account Problems***

● Issue: Cannot create a new account.

✿ Solution: All the required fields should be correctly filled in, and your email address should be valid. If the registration is not successful, please try with another email address.

● Issue: Cannot log in to your account.

✿ Solution: Verify that you are using the correct username and password. Ensure that your account has not been disabled by an administrator. If you forgot your password, contact the system administrator for assistance.

● Issue: Account is disabled.

✿ Solution: If your account has been disabled, contact the system administrator to reactivate your account. Disabled accounts cannot access the platform.

### ***E. Project Submission Issues***

● Issue: Cannot upload project files.

✿ Solution: Ensure that your files are in PDF format and do not exceed the maximum file size limit. Check your internet connection and try uploading again. If the issue persists, try uploading files one at a time.

● Issue: Project submission deadline has passed.

✿ Solution: Once the submission deadline has passed, you cannot submit or modify your project. Contact the hackathon organizer if you have extenuating circumstances.

● Issue: Cannot edit or delete submitted project.

✿ Solution: You can only edit or delete your project before the submission deadline. If your project has already been evaluated by judges, it cannot be modified.

### ***F. Team Management Issues***

● Issue: Cannot join a team.

✿ Solution: Ensure that the team has not reached its maximum member limit. If the team requires approval, wait for the team leader to approve your request. Check that you are not already a member of another team for the same hackathon.

● Issue: Cannot create a team.

☼ Solution: Verify that you are registered for the hackathon. Ensure that you have not already created or joined a team for this hackathon, as participants can only be part of one team per hackathon.

### ***G. Evaluation Issues***

● Issue: Cannot submit evaluation.

☼ Solution: Ensure that you have filled in all required evaluation criteria (scores from 1-10 for each criterion). Verify that you are within the judging period set by the organizer. Check your internet connection and try submitting again.

● Issue: Cannot view assigned projects.

☼ Solution: Ensure that you have accepted the hackathon invitation from the organizer. Verify that teams have been distributed to you by the organizer. Contact the organizer if you believe projects should be assigned to you.

### ***H. Chat and Messaging Issues***

● Issue: Messages are not sending.

☼ Solution: Check your internet connection. Refresh the page and try sending the message again. Ensure that you are connected to the conversation or team chat.

● Issue: Cannot see new messages.

☼ Solution: Refresh the chat page to load new messages. Check that you are viewing the correct conversation or team chat. Clear your browser cache if messages are not updating.



By addressing these issues effectively, the HackSynk platform ensures reliability and smoothness for all users.

## **6.3 Online Help and FAQs**

### **What is HackSynk?**

HackSynk is a comprehensive hackathon management platform designed for participants, organizers, judges, and administrators to manage hackathon events, team formation, project submission, and evaluation efficiently. It connects innovators, developers, and creators, enabling them to collaborate, compete, and showcase their projects seamlessly.

### **How can I register on HackSynk?**

1. Visit the HackSynk homepage and click "Sign Up" or "Register."
2. Choose your role: Participant, Organizer, or Judge.
3. Fill in the registration form with your details (Full Name, Username, Email, Password, City/Country, etc.).
4. Click "Sign Up" or "Register" to complete the process.
5. You can now log in and access your role-specific dashboard.

### **What roles does the system have?**

HackSynk supports the following roles:

- ☐ Participant: Browses hackathons, creates or joins teams, submits projects, communicates via chat, and views evaluation results.
- ☐ Organizer: Creates and manages hackathons, assigns judges, distributes teams for evaluation, sets project rankings, and announces results.
- ☐ Judge: Receives hackathon invitations, evaluates projects using multiple criteria, provides feedback to participants, and manages assigned projects.

❑ Admin: Oversees the platform, manages user accounts (participants, organizers, judges), enables/disables accounts, and monitors system statistics.

### **Is my data safe on HackSynk?**

Yes, HackSynk prioritizes your data security. We use advanced encryption techniques, secure password hashing, secure authentication, and regular security audits to ensure your information is protected. User roles are properly enforced to prevent unauthorized access.

### **How do I join a hackathon?**

1. Browse available hackathons from the homepage or "Hackathons" page.
2. Click on a hackathon to view its details.
3. Create a team or join an existing team for that hackathon.
4. Once you are part of a team, you can participate in the hackathon.

### **How do I create a team?**

1. Log in to your Participant Dashboard.
2. Click on "Create Team" in the sidebar.
3. Select the hackathon you want to participate in.
4. Enter your team name and description.
5. Click "Create Team" to form your team.
6. As the team leader, you can approve or reject join requests from other participants.

### **How do I submit a project?**

1. As a team leader, go to the "Submit Project" section in your Participant Dashboard.
2. Select the hackathon and team for which you want to submit a project.

3. Fill in project details (title, description, demo link).
4. Upload project files in PDF format.
5. Click "Submit Project" to save your submission.
6. You can update or delete your project before the submission deadline.

### **How does project evaluation work?**

1. Organizers assign judges to hackathons and distribute teams to judges.
2. Judges receive invitations and can accept or reject them.
3. Judges view assigned projects and evaluate them using 10 criteria (scored 1-10 each).
4. Judges provide written feedback along with their scores.
5. Organizers review all evaluations, set final rankings, and announce results.
6. Participants can view their evaluation results and feedback after results are announced.

### **Can I communicate with other participants?**

Yes, HackSynk provides two types of communication:

- ☐ Direct Messages: Send private 1-on-1 messages with other participants.
- ☐ Team Chat: Participate in group chat with your team members for collaboration and coordination.

### **What should I do if I encounter an issue?**

If you face any issues, you can:

- ☐ Check the Technical Support Manual (Section 6.2) for common problems and solutions.
- ☐ Contact the system administrator if your account is disabled or you need assistance.

☐ For hackathon-specific issues, contact the hackathon organizer.

**Can I switch my role after registering?**

No, roles are fixed upon registration. However, you can create a new account with a different role if you need to access different functionalities.

**Can I be part of multiple teams?**

No, participants can only be part of one team per hackathon. However, you can participate in multiple different hackathons with different teams.

**Can I edit or delete my submitted project?**

Yes, you can edit or delete your project before the submission deadline. Once the deadline has passed or your project has been evaluated by judges, you cannot modify it.

**How do judges receive project assignments?**

1. Organizers assign judges to specific hackathons.
2. Judges receive invitations in their dashboard.
3. After accepting an invitation, organizers distribute teams to judges.
4. Judges can then view and evaluate their assigned projects.

**Can I see evaluation results before they are announced?**

No, evaluation results are only visible to participants after the organizer announces them. This ensures fairness and prevents premature disclosure of results.

**What happens if I miss the project submission deadline?**

Once the submission deadline has passed, you cannot submit or modify your project. Contact the hackathon organizer if you have extenuating circumstances.

**Can organizers modify hackathons after creation?**

Yes, organizers can edit hackathon details (dates, rules, prizes, etc.) at any time. However, certain changes may affect ongoing registrations or submissions.

**How do I know if my team join request was approved?**

You will receive a notification when your team join request is approved or rejected by the team leader. You can also check the status in your dashboard.

**Can I leave a team after joining?**

Yes, you can leave a team at any time. However, if you are the team leader, you may need to transfer leadership or delete the team if you leave.

**What file formats are supported for project submissions?**

Currently, HackSynk supports PDF format for project file uploads. Ensure your files are in PDF format and do not exceed the maximum file size limit.

**Can I view hackathon results from previous events?**

Yes, once results are announced by organizers, you can view them in the "Results" section of your Participant Dashboard. Results include scores, rankings, and judge feedback.

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